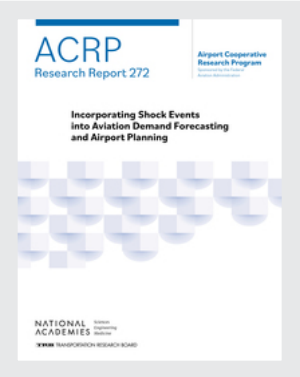
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ACRP Research Report 272

Airport Cooperative Research Program

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Incorporating Shock Events into Aviation Demand Forecasting and Airport Planning

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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 272

Incorporating Shock Events into Aviation Demand Forecasting and Airport Planning

Ian Kincaid
Christopher Greer
InterVISTAS Consulting USA LLC
Washington, DC

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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

Research problem statements for ACRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the AOC to formulate the research program by identifying the highest priority projects and defining funding levels and expected products.

Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

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Dr. Michael Tretheway of Inter*VISTAS* was the original Principal Investigator. Sadly, Dr. Tretheway passed away during the project. The role of Principal Investigator was taken on by Ian Kincaid of Inter*VISTAS*, and Christopher Greer of Inter*VISTAS* was Project Manager. Other project researchers involved were Kathryn Tooley, Jody Kositsky, Nelly Alandou, Mark Taylor, and Josh Cohn of Inter*VISTAS*; Doug Banez of Hubpoint Strategic Advisors; Lois Kramer of KRAMER aerotek; Max Henrion of Lumina Decision Systems; and David Vose of RSA Archer. Eden Muldal of Inter*VISTAS* served as the Administrative Officer for the project.

The project team would like to sincerely thank the airports that contributed to the guide through interviews and review of the draft materials.

This guide is dedicated to the memory of Dr. Michael Tretheway.

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FOREWORD

By Matthew J. Griffin
Staff Officer
Transportation Research Board

ACRP Research Report 272: Incorporating Shock Events into Aviation Demand Forecasting and Airport Planning presents a methodology to help airport operators identify relevant shock events, understand their potential impact on airport business, and develop strategies to better withstand and respond to them. The report will be of interest to airports of all sizes and the planners and consultants who support them.

Traditional airport forecasting and planning are largely based on long-term trends, assume gradual and incremental change, and generally do not reflect the possibility of unpredicted shock events or black swan events—unexpected, high-impact events that are without precedent. Shock events are characterized by their rarity and potential for severe impacts and long-lasting consequences. While this report is meant to help airports understand and address significant national or global shock events, methodologies developed from this research may be used by airports to respond to regional and local challenges as well.

Under ACRP Project 03-60, "Incorporating Shock Events into Aviation Demand Forecasting and Airport Planning," Inter*VISTAS* Consulting USA LLC was asked to produce a guide and methodology to help airports understand and account for potential shock events in forecasts and planning. These objectives aim to support the preparation of analyses to better forecast and plan for such events and support airports that are reacting to and recovering from them. The research team drew on experiences from the COVID-19 pandemic and other shock events within both the airport industry and other sectors to develop a five-step, systematic methodology to address shock events in airport forecasting, management, and planning. Inter*VISTAS* led the research with support from Lumina Decision Systems; Hubpoint Strategic Advisors, LLC; KRAMER aerotek, Inc.; and RSA Archer.

The report and its Appendix B, a summary of relevant literature reviewed, are available on the National Academies Press website (nap.nationalacademies.org) by searching for *ACRP Research Report 272: Incorporating Shock Events into Aviation Demand Forecasting and Airport Planning.*

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CHAPTER 1

Introduction

Purpose of This Guide

The COVID-19 pandemic has made clear the vulnerability of the aviation industry (along with many other sectors of the economy) to shock events. As documented in this guide, government measures to control the COVID-19 pandemic resulted in sharp declines in air traffic (except for air cargo in some instances) followed by a prolonged and uncertain period of traffic recovery. While the pandemic is one of the largest events to impact the aviation industry, other previous shock events have also exerted profound impacts on the aviation industry. Most recessions are not so unusual as to be totally unexpected, but the scale of the 2008–2009 Great Recession was greater than anything experienced since 1929 and resulted in a significant downturn in traffic as well as changes to the structure and development of the aviation market (e.g., airline consolidation). The terrorist attacks of September 11, 2001 (9/11) impacted traffic development, industry structure, and airport facility requirements (security screening, etc.). Other terrorist events, while not impacting traffic significantly, led to changes in airport facilities and processes, such as hold-baggage screening; reconciliation following the 1988 Lockerbie bombing disaster; and the banning of liquids following the 2006 terrorist liquid-explosive plot.

Typically, the long-term forecasting, management, and planning of airports are based largely on a continuation of "business as usual" and generally do not give much focus to the possibility of shock events, even though they have led to material changes in airport management and design. One reason is that predicting the nature, timing, and scale of such events is almost impossible; many are "black swan" events—unexpected, high-impact events that are without precedent. While airport forecasting and planning have tended to focus on gradual and incremental change—rising traffic levels as the economy grows—it is arguable that infrequent, one-off events have had as much impact on airport development as long-term, gradual growth. Clearly, the future is uncertain, and airport forecasting and planning need a way to recognize and address the risks and uncertainties facing the development of airports.

The purpose of this guide is not to provide a means of predicting the exact timing and scale of shock events as this is largely impossible in any practical sense. Rather, the guide provides a methodology to aid airports in identifying and describing relevant shock events, understanding their potential impact on various aspects of the airport business, and developing approaches and strategies to enable the airport to better withstand and respond to shock events if they occur. The objective is to contribute to a more resilient airport business model by assisting the airport to better respond to shock events robustly. While not all shock events can be identified or foreseen, the intent is that an airport would be better positioned for these eventualities generally and can benefit from past expertise and experience drawn from the COVID-19 pandemic and other shock events from within both the airport industry and other sectors.

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2 Incorporating Shock Events into Aviation Demand Forecasting and Airport Planning

How to Use This Guide

The guide is designed to be relevant and accessible to a wide range of users. The findings and methodology apply to those involved in air traffic forecasting and airport planning, whether internal to the airport organization or external consultants. However, one of the key findings from the research is the need for full organizational engagement in addressing shock events. Many of these events have had significant implications for the finances, operation, management, and strategic development of airports. Therefore, the guide will be of interest to a wide range of practitioners and stakeholders involved in the airport sector.

The guide is organized into four main parts plus a concluding section and appendices.

Part I discusses the nature and impact of shock events that have affected the aviation sector starting with 1970 and leading up to an examination of the impact of the COVID-19 pandemic on U.S. aviation and airports. This includes the national impacts on airport traffic, operations, and finances as well as the response of airport management and government. Findings are also provided from 11 case studies at airports that were affected by the pandemic in different ways. In addition, Part I looks at other shock events impacting the aviation industry, such as the SARS (severe acute respiratory syndrome) pandemic, 9/11, Hurricane Katrina, the 2008–2009 Great Recession, and examples of airline exit and entry. While history is an imperfect guide to the future, this part provides lessons learned from previous shock events.

Part II provides an overview of existing approaches for identifying, understanding, addressing, and mitigating shock events, both in the aviation industry and in other industries and sectors. It provides a baseline understanding of approaches developed to handle shock events specifically or uncertainties more generally (which can be applied to shock events). This is supplemented with more detailed information in the appendices.

Part III outlines a systematic methodology to understand and address shock events in airport forecasting, management, and planning, drawing on findings from the research in Parts I and II as well as prior experience in risk identification and risk management exercises in a range of economic sectors. It builds on established methodologies but is tailored to the circumstances and requirements of the airport sector. The methodology is general enough to be used for airports of all types and sizes and includes guidelines for simplifying the methodology for small airports with limited time and resources.

Part IV illustrates the methodology by applying it to example airports of different sizes and traffic mixes and facing different market and economic conditions.

Part V provides some overall conclusions from the research and identifies areas for further or related research.

It is not necessary for the reader to review the entire guide to understand the methodology provided. Some users may prefer to skip the historical and background information in Parts I and II and focus instead on the methodology in Part III and its application in Part IV, referring back to Parts I and II for additional information where needed.

How the Research Was Conducted

The ACRP research team was made up of members with extensive experience in airport forecasting, planning, and finances as well as specialists in risk analysis and modeling. The team's combined knowledge and experience helped shape the project research and findings. The key research activities are summarized below:

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- The impact of the COVID-19 pandemic on aviation and airports was researched using various traffic and financial data sources (such as U.S. DOT, FAA, individual airport statistics and annual reports, and subscription data), published articles, and contemporary news stories. Remote interviews were conducted with 11 case study airports including personnel involved in planning, forecasting, air service development, and finance.
- Previous shock events were researched using a combination of published traffic data, airport annual reports, academic articles, and contemporary news stories.
- To examine existing approaches for identifying, understanding, addressing, and mitigating shock events, the research team conducted an extensive literature review. The literature review examined a range of industries and academic disciplines, including the aviation and transportation industry, risk and decision analysis, management theory, the financial and insurance industries, as well as other network industries such as telecommunications and energy generation and distribution.

Based on this research and drawing on the ACRP research team's collected experience, a systematic methodology was developed and refined that could be incorporated into existing airport practices that would aid in the identification, understanding, and management of potential shock events. This methodology was tested and modified through application to two example airports: Birmingham-Shuttlesworth International Airport and San Francisco International Airport. The application of the methodology was carried out by the ACRP research team, with draft versions sent to the respective airports and their feedback incorporated into the final write-up.

Related Materials

The guide touches on many related areas, including airport forecasting, planning, financial management, emergency response, and addressing uncertainty. While the guide is designed to be stand-alone, there are related ACRP publications that may benefit the reader. Most notably, *ACRP Report 76: Addressing Uncertainty about Future Airport Activity Levels in Airport Decision Making* (Kincaid et al. 2012), which, while not covering shock events in any detail, does cover the issue of risk and uncertainty in airport forecasting and planning. Other related ACRP publications that may be of interest include

- *ACRP Synthesis 2: Airport Aviation Activity Forecasting* (Spitz and Golaszewsi 2007);
- *ACRP Report 18: Passenger Air Service Development Techniques* (Martin 2009);
- *ACRP Report 20: Strategic Planning in the Airport Industry* (Ricondo & Associates et al. 2009);
- *ACRP Report 25: Airport Passenger Terminal Planning and Design, Volume 1 Guidebook* (Landrum & Brown et al. 2010);
- *ACRP Report 48: Impact of Jet Fuel Price Uncertainty on Airport Planning and Development* (Spitz and Berardino 2011);
- *ACRP Report 74: Application of Enterprise Risk Management at Airports* (Marsh Risk Consulting 2012); and
- *ACRP Synthesis 115: Practices in Airport Emergency Plans* (Murphy et al. 2021).

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PART I

Shock Events and the Aviation Industry

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CHAPTER 2

Overview of Shock Events

The COVID-19 pandemic led to the largest downturn in global commercial aviation in modern history. However, the aviation industry has been affected by pandemics before, such as SARS (2003), Swine Flu (2009), and MERS (Middle East Respiratory Syndrome) (2012), albeit nowhere on the scale of COVID-19. The industry as a whole and individual airports have also been affected by economic shocks and recessions, air market changes and airline failures, terrorist attacks (or attempts), and wars. In fact, these shock events occur with considerable frequency, as illustrated in Figure 1.

This timeline shows shock events potentially relevant to the U.S. aviation industry since 1970 in four categories: economic, airline market, pandemics and natural disasters, and wars and terrorist attacks. Some of these events had little material effect on the industry as a whole, while others were highly significant. In general, the industry has a long-term track record of recovering from these events and reverting to growth, although not always recovering to the previous trend. Furthermore, while the impact on the entire industry may be modest, the impacts on individual airports can be profound and long-lived: dehubbing, the loss of a major carrier, entry of ultra-low-cost carriers (ULCCs), new passenger processing requirements, etc. The impacts are not necessarily uniform, affecting some airports severely and others only slightly.

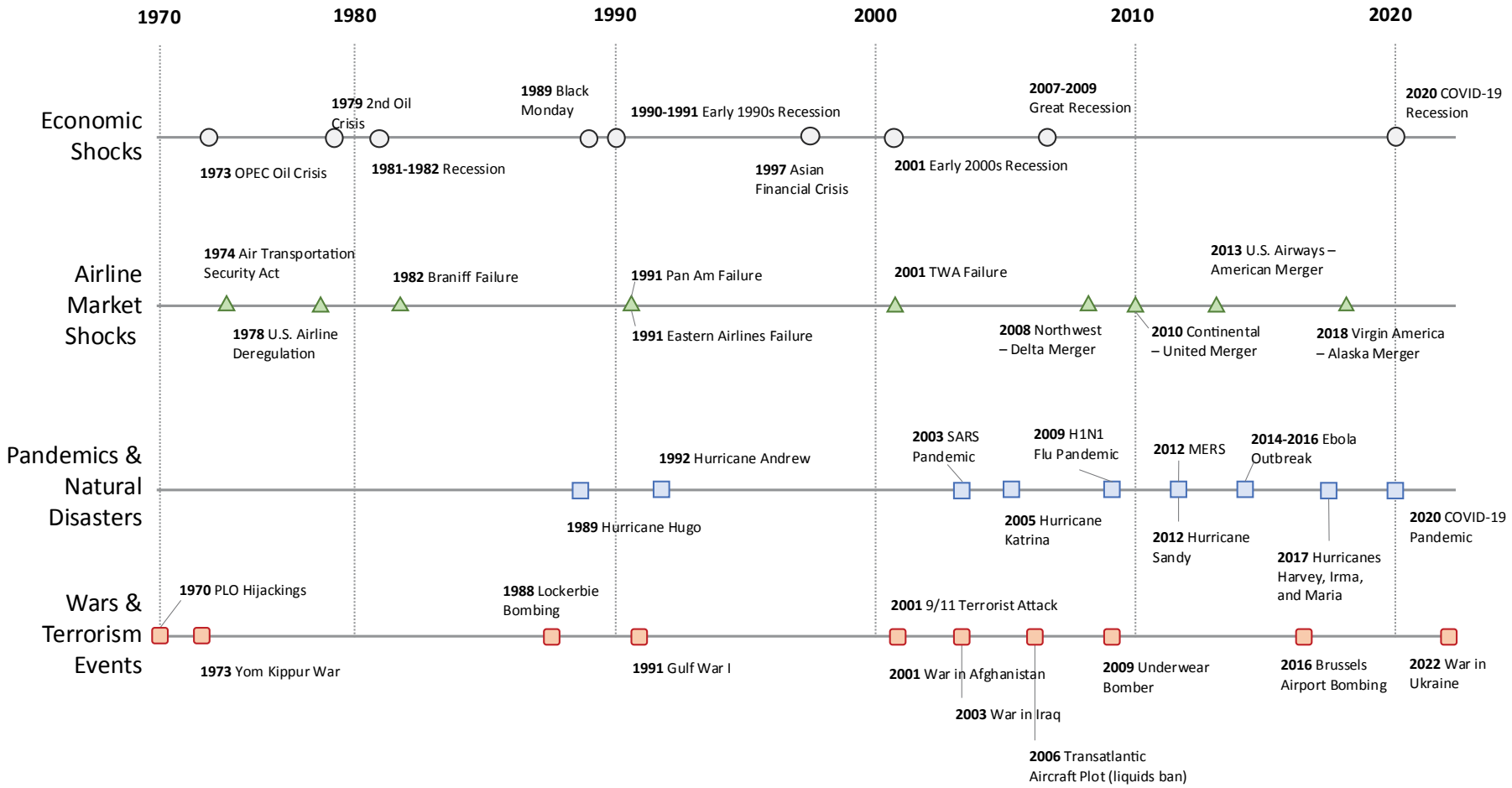
Defining Shock Events

It is useful to have a clear idea of what is meant by "shock event." In this guide, the term "shock event" has generally been used, but other terminology is used elsewhere. As documented in Part II, terms such as "black swan events," "grey swan events," "rare events," "disrupters," and "high-impact, low-probability events" are also used. Shock event is used in this document as it provides a general idea of the concept without any implied assumptions around the level of prior knowledge (black versus grey swan) or likelihood of occurring (low probability). Based on research conducted for this guide and the team's prior experience, the characteristics of a shock event can be summarized as follows.

- **Uncertainty exists around whether the event will occur and when.**
  The probability of the event is not necessarily low, but neither is it a certainty. Given the history of airline bankruptcies and mergers, the failure or exit of a key airline at an airport is not an extremely unlikely event or black swan, but if and when it might occur remains fairly uncertain. Where an event is a certainty or near certainty it should be built into the planning and management of the airport (e.g., poor weather conditions in the winter, the hurricane season disrupting operations in southeastern U.S. states).

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NOTE: OPEC = Organization of the Petroleum Exporting Countries; PLO = Palestine Liberation Organization.

Figure 1. Timeline of shock events in the U.S. aviation industry since 1970.

- **When the event does occur, it has the potential to impact the airport in a significant or strategic way (positively or negatively).**

  The event may have profound implications for the traffic, management, finances, and planning of the airport. This differentiates it from operational events such as a ground collision or snowstorms that disrupt operations for a fairly short time but do not lead to significant changes in the airport business or strategy.
- **The event itself can vary in its timeline.**

  Some events unfold over minutes or hours, such as terrorist attacks; others develop over weeks, months, or years, such as the COVID-19 pandemic. Regardless, the resulting impacts of such events are generally long-lived and can transform the path of the business.

To consider it another way, almost all airports base their medium- and long-term planning on forecasts that predict largely gradual and relatively consistent growth in air traffic volumes (it is extremely rare for an airport traffic forecast to predict long-term declining volumes). This is based on the extensive historical experience in the United States and elsewhere of economic growth and rising incomes leading to increasing air traffic volumes, often in combination with competitive airline markets and technological improvement. It is expected that forecasts will still be generated in this manner for the most part. However, consideration can also be given to potential shock events that could cause traffic development to diverge considerably from the long-term trend, perhaps permanently, as illustrated in Figure 2 showing total traffic volumes (but other aspects such as traffic mix can also be considered).

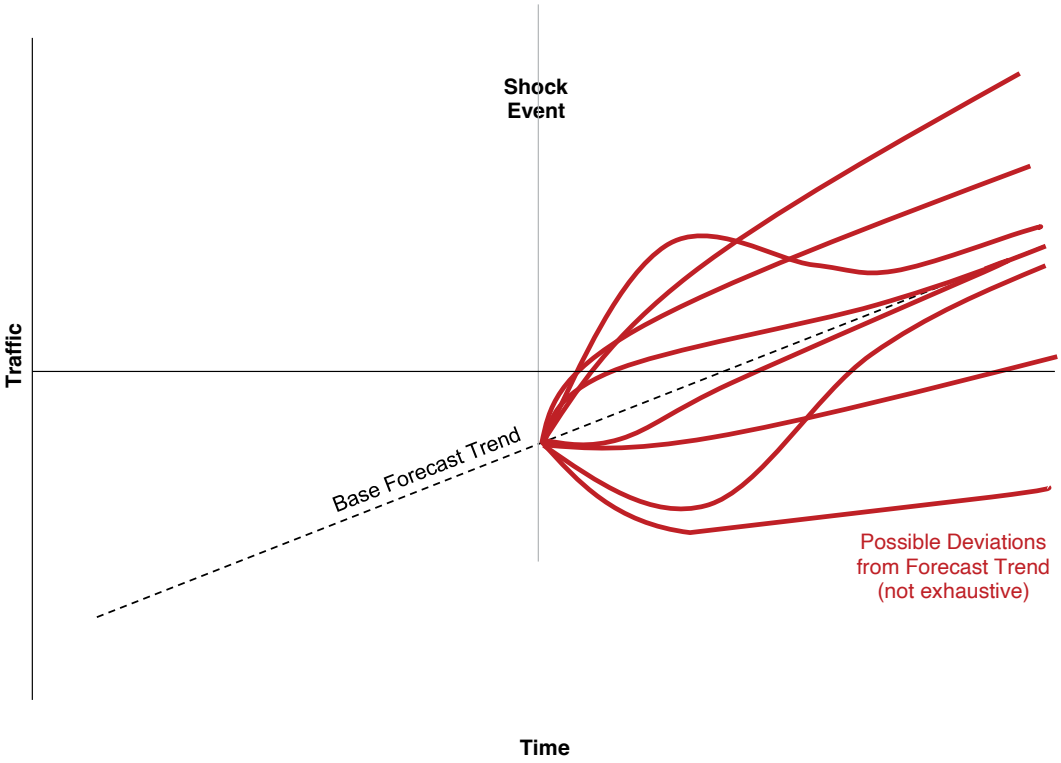


Figure 2. Possible impacts of shock events on the long-term trend (illustrative).

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CHAPTER 3

Impacts of the COVID-19 Pandemic on the U.S. Aviation System

Timeline of the Pandemic and Government Regulation

The key milestones in the development of the COVID-19 pandemic, the development of vaccines, and the government's (U.S. federal and state) response to the pandemic are summarized in Table 1. Milestones that directly affected travel and the travel industry are highlighted in **green**.

Figure 3 shows the development of the COVID-19 pandemic in the United States in terms of the number of new cases reported daily. Daily cases reached around 31,000 (weekly average) in early April 2020 and then flattened with the imposition of stay-at-home orders. With the removal of these orders and relaxation of other restrictions in summer 2020, cases started to rise again. Cases rose rapidly in winter 2020/2021, reaching 245,000 average daily cases per week due to the population being in closer proximity indoors and the emergence of more contagious variants of the virus, notably the Delta variant (the first U.S. case was recorded on February 23, 2021). The imposition of further restrictions along with the vaccine rollout led to cases falling to less than 12,000 average daily cases in June 2021. Cases rose again during summer 2021 as most remaining restrictions were lifted, reaching over 165,000 average daily cases in August, after which cases started to decline. The Omicron variant (first detected in the United States on December 1, 2021) led to a rapid surge in new cases, peaking at over 800,000 average new daily cases in January 2022. The Omicron variant was generally considered more transmissible but less severe than previous variants. Cases declined later in the year due to vaccines and natural immunity buildup, averaging around 50,000 to 70,000 new daily cases in the latter half of 2022.

Impact of the COVID-19 Pandemic on U.S. Air Passenger Traffic

The COVID-19 pandemic resulted in the single largest downturn in global aviation activity in the history of modern aviation. Air passenger traffic across the United States, as with nearly all other nations, was most drastically and negatively impacted after the initial onset of the pandemic in 2020.

By the end of 2020, monthly national passenger volumes had recovered to only 37% of prepandemic levels, and traffic for the entire year was down 61% relative to 2019.

The pandemic-driven drop in total passenger enplanements in 2020 and 2021 is compared against the longer timeline of annual traffic since 2000 in Figure 4. (Year-over-year declines are noted in red; green figures indicate periods of significant growth.) Prior events driving annual declines in passenger volumes include the aftermath of the 9/11 terrorist attack and the U.S. recession in 2001, followed by the Great Recession of 2008–2009. The impacts of these events, which resulted in single-digit percentage declines in annual passengers are significantly smaller than that of the COVID-19 pandemic.

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Table 1. Pandemic timeline.

Date	Milestone
December 2019	First reports of a novel coronavirus in Wuhan, China.
January 9, 2020	World Health Organization (WHO) releases a statement regarding the identification of a new virus.
January 20, 2020	Centers for Disease Control and Prevention (CDC) begins screening for the virus at John F. Kennedy International Airport, San Francisco International Airport, and Los Angeles International Airport (the airports where most passengers to/from Wuhan connect).
January 21, 2020	First U.S. coronavirus case is reported in Washington State.
January 30, 2020	WHO declares a global public health emergency.
January 31, 2020	**United States bans travel from China for non-U.S. citizens.**
February 6, 2020	First COVID-19-related death in the United States.
February 11, 2020	WHO officially names the disease caused by the new coronavirus COVID-19.
March 11, 2020	WHO declares COVID-19 a pandemic.
March 13, 2020	United States declares COVID-19 a national emergency. **Travel ban on non-U.S. citizens traveling from Europe goes into effect.**
March 19, 2020	**U.S. State Department issues a Level 4 travel advisory: U.S. citizens should not travel abroad due to the global impact of COVID-19.**
March 19, 2020	California is the first state to issue a stay-at-home order requiring all but essential workers to work from home and avoid nonessential travel (Puerto Rico issued an order on March 15). A total of 40 U.S. states issued stay-at-home orders during late March and early April (the remaining 10 states issued advisories).
March 20, 2020	**U.S. closes land borders with Canada and Mexico to all nonessential travel.**
March 27, 2020	**The Coronavirus Aid, Relief, and Economic Security (CARES) Act is signed into law, providing $2.2 trillion in aid to the U.S. economy.**
April 3, 2020	CDC recommends individuals wear face masks in public spaces to reduce the spread of COVID-19.
April 26, 2020	Colorado and Montana become the first states to lift stay-at-home orders.
June 11, 2020	New Hampshire becomes the final state to remove stay-at-home orders.
July 27, 2020	Moderna and Pfizer begin Phase 3 (final stage) clinical trials of their COVID-19 vaccines.
December 11, 2020	Pfizer-BioNTech COVID-19 vaccine receives emergency use authorization from the Food and Drug Administration (FDA) for individuals 16 years and older.
December 14, 2020	First doses of the Pfizer COVID-19 vaccine begin to be distributed in the United States.
December 18, 2020	FDA authorizes the Moderna COVID-19 vaccine for emergency use in individuals 16 years and older.
December 27, 2020	**The Consolidated Appropriations Act is signed into law, providing $900 billion in stimulus relief to facilitate national economic recovery.**
April 30, 2021	By the end of April, 46% of the eligible population had received at least one dose of the COVID-19 vaccine.
March 11, 2021	**American Rescue Plan Act is signed into law, providing $1.9 trillion in stimulus and support to the U.S. economy.**
May 20, 2021	Lifting of restrictions on businesses and public spaces underway in all states that had stay-at-home or shelter-in-place orders.
August 2, 2021	United States reaches 70% of the eligible population vaccinated with at least one dose of the COVID-19 vaccine.

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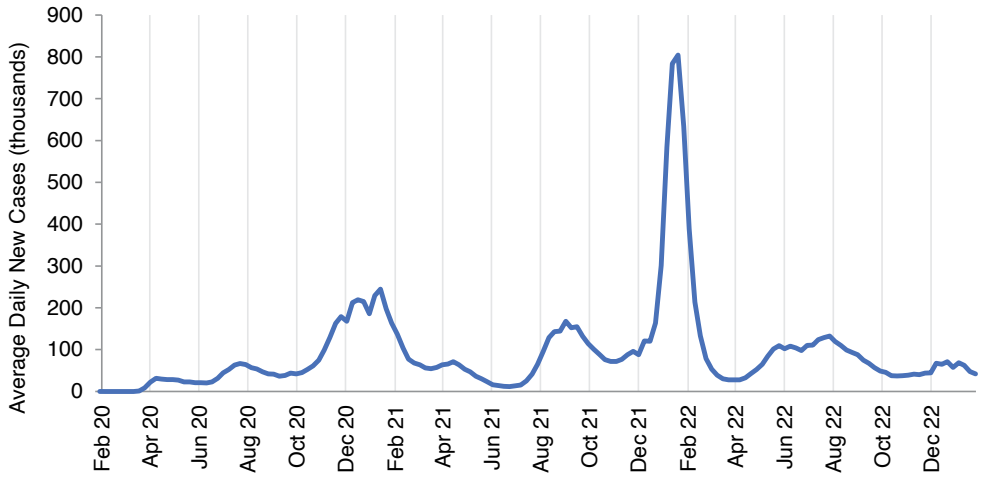
Table 1. (Continued).

Date	Milestone
August 23, 2021	Pfizer COVID-19 vaccine becomes the first COVID-19 vaccine to receive full approval from the FDA.
September 9, 2021	New executive order requires federal employees and federal contractors to be fully vaccinated with COVID-19 vaccines.
October 31, 2021	By the end of October, 78% of the eligible population had received at least one dose of the COVID-19 vaccine.
November 8, 2021	**United States reopens borders to all foreign national travelers provided they are fully vaccinated with the COVID-19 vaccine, with limited exceptions. Land borders with Canada and Mexico reopen to nonessential travel.**
November 29, 2021	**Entry of non-U.S. citizens from eight southern African countries was suspended due to concerns over the new Omicron variant originating from that region.**
January 31, 2022	FDA approves a second COVID-19 vaccine (the Moderna vaccine, marketed as Spikevax).
February 25, 2022	CDC updates its guidance regarding mask-wearing based on risk.
June 12, 2022	**The requirement for a negative COVID-19 test before boarding a flight to the United States is rescinded, regardless of vaccination status.**

SOURCE: U.S. Department of Defense 2023; AJMC Staff 2021; Wu et al. 2020; The New York Times 2021; Elassar 2020.

By the end of 2020, monthly national passenger volumes had recovered to only 37% of pre-pandemic levels, and traffic for the entire year was down 61% relative to 2019 (U.S. DOT, n.d.).

U.S. air passenger traffic partially recovered in 2021, reaching 70% of 2019 annual levels, due to the removal of some domestic travel restrictions and the deployment of the first vaccines across the United States. As shown in Figure 5, by summer 2021, monthly passenger traffic had increased to between 80 and 90% of prepandemic 2019 levels and generally maintained a similar level of recovery in most months up through the first half of 2022. By the second half of 2022 and into 2023, total system traffic was approaching 100% recovery to 2019 levels, though monthly international passenger recovery remained below domestic but was trending toward a full recovery in summer 2023.



NOTE: Calculated as the average number of cases reported each week divided by seven.

Figure 3. Average daily new COVID-19 cases per week in the United States, February 2020 through December 2022 (as sourced from the CDC).

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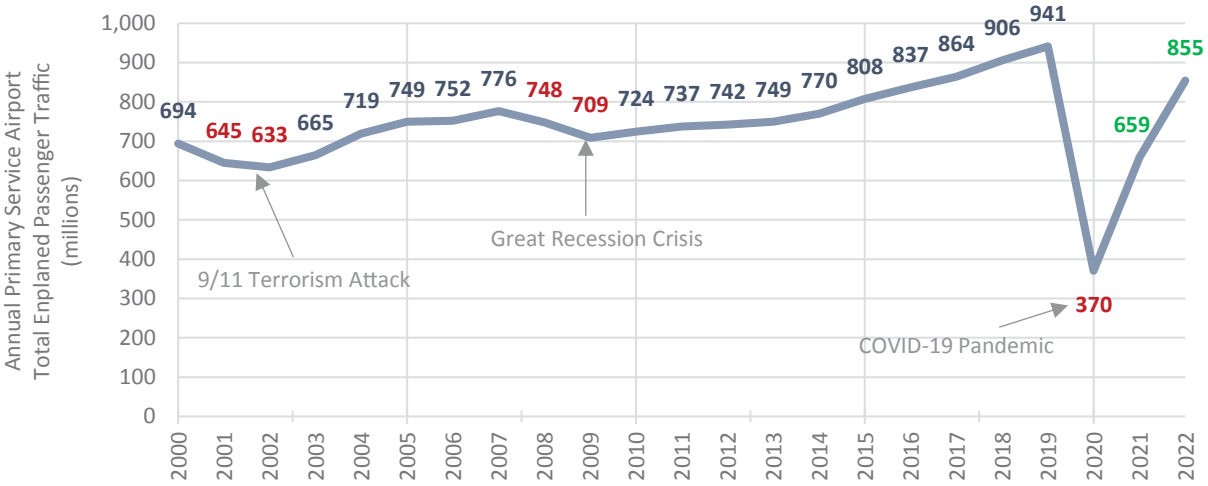


Figure 4. Annual enplaned passenger traffic at U.S. Primary Service Airports, 2000–2022 (U.S. DOT, n.d.).

As of June 2022, the recovery of international passenger traffic nationwide has lagged behind domestic traffic by roughly one year.

The initial rebound in passenger traffic throughout 2020 and 2021 was driven by domestic travel, due to relaxing many state-level lockdowns and the implementation of the Coronavirus Aid, Relief, and Economic Security (CARES) Act to support the U.S. aviation industry. Conversely, international traffic remained hampered by federal government restrictions regarding border crossings along with restrictions imposed by foreign countries. By the end of 2020, domestic traffic had recovered to 39% of 2019 levels, while international traffic was at 26%. Further relaxation of restrictions and the mass vaccination of U.S. citizens and residents beginning in early 2021 then led to a major uptick in domestic air travel starting in March 2021, with domestic traffic reaching 89% of 2019 levels by July 2021 compared with just 47% for international traffic. The gradual growth of international traffic was more heavily oriented to destinations in the Caribbean and Latin America with many pre-COVID-19 top destination countries (e.g., Canada, United Kingdom, Japan, China) seeing

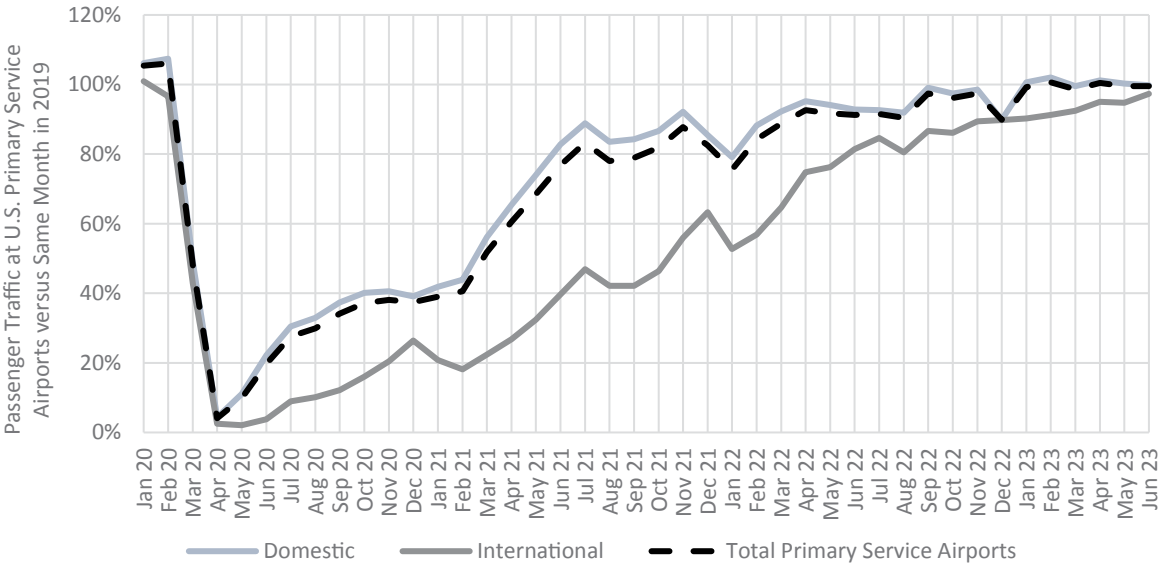


Figure 5. Monthly passenger traffic at U.S. Primary Service Airports versus the same month in 2019, January 2020 through June 2023 (U.S. DOT, n.d.).

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limited recovery due to travel restrictions in those countries on nonessential travel. Monthly international traffic experienced a stronger recovery from the start of 2022, although exceeding 80% of 2019 levels only by June 2022, roughly 1 year later than when domestic traffic had reached the same recovery point. International passenger volumes saw steady but slowing recovery rates in the latter half of 2022 and into 2023 and were approaching 100% recovery at a monthly level by the summer of 2023.

Impacts by Airport Size and Type

Across various types of airports, the recovery of traffic has remained largely consistent. Airport types are defined by the National Plan of Integrated Airport Systems (NPIAS), which includes 396 Primary Service Airports (FAA 2020). These Primary Service Airports, which process nearly all of the commercial passenger traffic in the United States, are categorized into the following types:

- **Large hubs:** Airports that receive more than 1% of annual U.S. enplanements. This category comprises major carrier hubs, international gateways, and large origin/destination (O/D) airports such as
  - Los Angeles International Airport (LAX),
  - San Francisco International Airport (SFO),
  - John F. Kennedy International Airport (JFK),
  - Hartsfield-Jackson Atlanta International Airport (ATL),
  - Denver International Airport (DEN),
  - Harry Reid International Airport (LAS), and
  - Daniel K. Inouye International Airport (HNL).
- **Medium hubs:** Airports that receive 0.25% to 1% of annual U.S. enplanements. This category includes airline regional hubs, focus cities, and larger metropolitan area airports such as
  - San Francisco Bay Oakland International Airport (OAK),
  - Raleigh-Durham International Airport (RDU),
  - Louis Armstrong New Orleans International Airport (MSY), and
  - Nashville International Airport (BNA).
- **Small hubs:** Airports that receive 0.05% to 0.25% of annual U.S. enplanements. This category accounted for 9% of passenger traffic in 2019 and comprises spoke airports to major hubs, medium-sized metropolitan areas such as
  - Memphis International Airport (MEM),
  - Will Rogers World Airport (OKC),
  - Reno-Tahoe International Airport (RNO), and
  - Boise Airport (BOI).
- **Nonhubs:** Airports that process more than 10,000 annual passengers but fewer than 0.05% of annual system traffic (approximately 500,000 passengers in 2019). This category of 266 airports accounted for less than 4% of total NPIAS passengers in 2019.

The 30 large hub airports accounted for 71% of total NPIAS traffic in 2019, while medium hub airports accounted for a further 17%. Therefore, these two airport categories represent most of the commercial passenger traffic in the United States.

Figure 6 shows the recovery of passenger traffic, carried on both U.S. and foreign carriers, at the four NPIAS airport types relative to the same month in 2019. All Primary Service Airport types experienced very similar declines and recoveries in passenger traffic. Smaller airports, such as the small hub and nonhub airports, have fared slightly better in the recovery than the bigger large and medium hubs. This can be attributed to much lower exposure to international traffic

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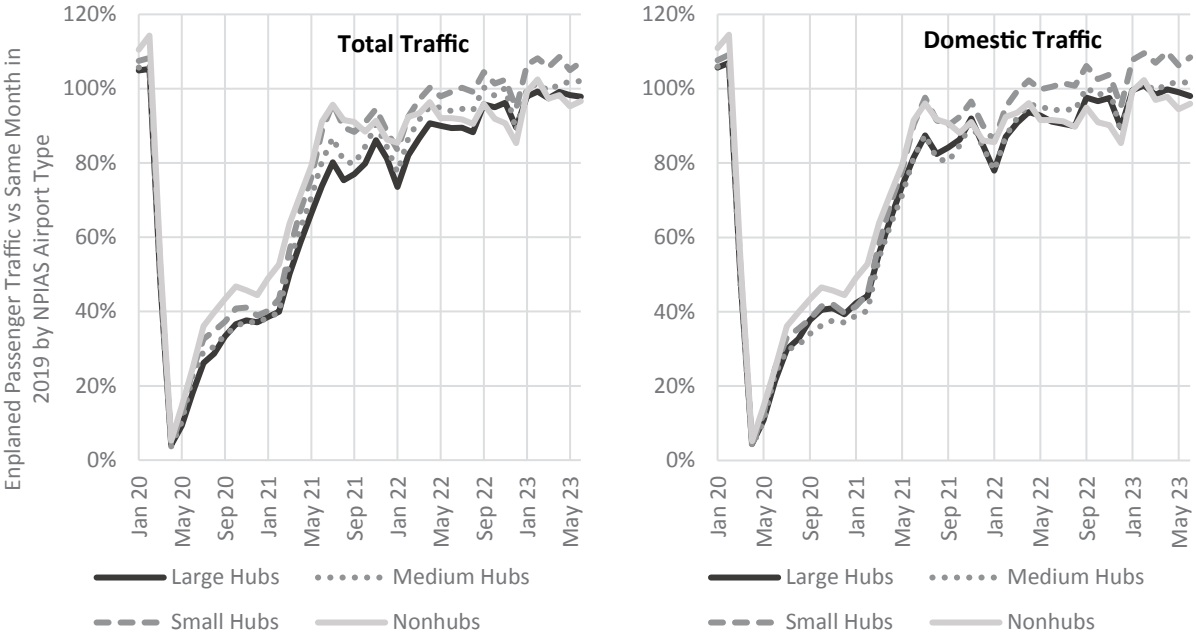


Figure 6. Enplaned passenger traffic recovery by NPIAS airport type, all carriers (including U.S. and foreign), January 2020 to June 2023 (U.S. DOT, n.d.).

at these airports, with nonhub airports recovering domestic traffic slightly better than other airport categories. This may be due to the lower level of traffic and services at these smaller airports as well as the impacts of government support mechanisms such as the Essential Air Services (EAS) program and pandemic-related relief programs (discussed later in this chapter).

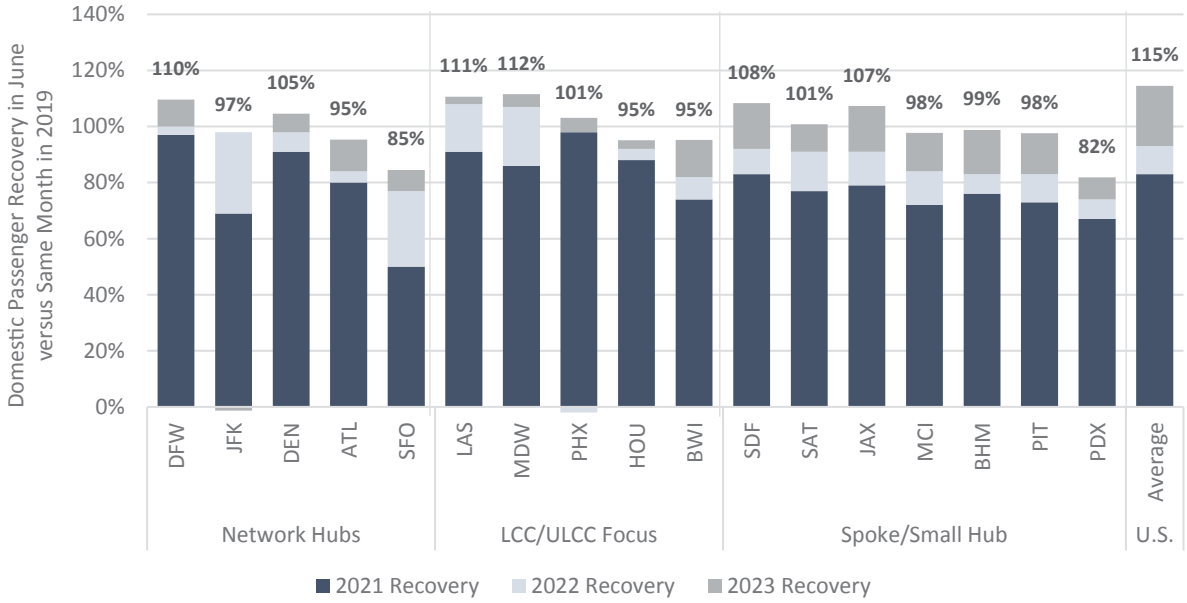
Passenger traffic recovery at individual U.S. airports has also depended on the types of airlines served and each airport's role in those airlines' networks. Figure 7 shows the extent of domestic passenger volume recoveries in June 2021, 2022, and 2023 versus June 2019 at selected U.S. Primary Service Airports with various roles and network structures. These examples are grouped into three categories: major network hubs; airports that are primarily focused on service by low-cost carrier (LCC) and ULCC airlines; and spoke airports that are primarily connected to larger network hubs or are primarily O/D airports. These categories are not always mutually exclusive (e.g., DEN is displayed as a large hub airport given its relationship with United, but it also serves as a hub/focus city for LCCs Frontier and Southwest Airlines); however, the analysis is indicative of key trends by different network types.

Airports whose capacity was more oriented to LCC and ULCC traffic initially fared the best in their recovery to prepandemic levels, due to focus on leisure, VFR, and domestic point-to-point travel.

Up until June 2021, airports whose capacities were more oriented to LCC and ULCC traffic generally fared the best in their recovery to prepandemic levels. With travel demand in 2021 being dominated by leisure and "visiting friends and relatives" (VFR) travel, ULCCs have had the opportunity to serve that pent-up demand without the challenges some network carriers may have faced due to the slower recovery of business and international passenger demand. Network hubs experienced further traffic recovery up through June 2022, but there was considerable variation due to exposure to slower-recovering international markets. LCC- and ULCC-oriented airports experienced further recovery up to June 2022, with some exceeding 2019 levels. By mid-2023, the trends seen in 2022 had largely continued with some network hub airports, particularly those with a large proportion of international service, remaining at a lower level of recovery than some smaller airports.

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NOTE: MDW = Midway International Airport; PHX = Phoenix Sky Harbor International Airport; HOU = William P. Hobby Airport; BWI = Baltimore-Washington International Airport; SDF = Louisville International Airport; SAT = San Antonio International Airport; JAX = Jacksonville International Airport; MCI = Kansas City International Airport; BHM = Birmingham-Shuttlesworth International Airport; PIT = Pittsburgh International Airport; PDX = Portland International Airport.

Figure 7. Domestic passenger recovery at selected U.S. airports by network structure, June 2021, 2022, and 2023 versus June 2019 (U.S. DOT, n.d.).

Finally, spoke (nonhub) airports have generally recovered close to but still below the national average traffic due to a variety of outcomes from the pandemic. Among these outcomes are financial strains, which have changed the relationships between legacy network carriers and their regional affiliates, with some regional airlines, such as Trans State Airlines and Compass Airlines, ceasing operations altogether. To curb costs, airlines have also accelerated the retirement of smaller 50-seat regional jets, a shift that predominantly impacts smaller communities and spoke airports for markets that may not generate sufficient demand to fill mainline narrow-body aircraft at a frequency to provide adequate connectivity throughout the network.

While airline networks and services shifted in many ways throughout the pandemic, the overall impact on O/D versus connecting (transit and transfer) traffic nationwide has followed similar trajectories. Both O/D and connecting passengers declined by 96% compared to 2019 levels in April 2020, which was the most restricted month for air travel during the pandemic, as shown in Figure 8. By June 2021 and 2022, O/D traffic at U.S. Primary Service Airports had recovered to 81% and 93%, respectively. At the same time, connecting passengers had recovered to 57% in mid-2021 and were at 86% of 2019 values in June 2022. The gap in recovery between O/D and connecting passenger volumes is attributed to two factors. First, with international traffic having only recovered to much lower levels than domestic, there is a portion of connecting traffic that is yet to recover in the domestic-to-international connections, primarily at large hub gateway airports. Second, there has been a greater recovery in LCC and ULCC traffic during the pandemic, which is primarily point-to-point versus connecting itineraries. Both of these effects point to a relatively lower level of connecting traffic among total enplanements in the United States. However, the difference between the two categories of traffic is fairly small, and there is no evidence that carriers significantly changed the mix of O/D and connecting traffic in the operations, even as traffic volumes declined.

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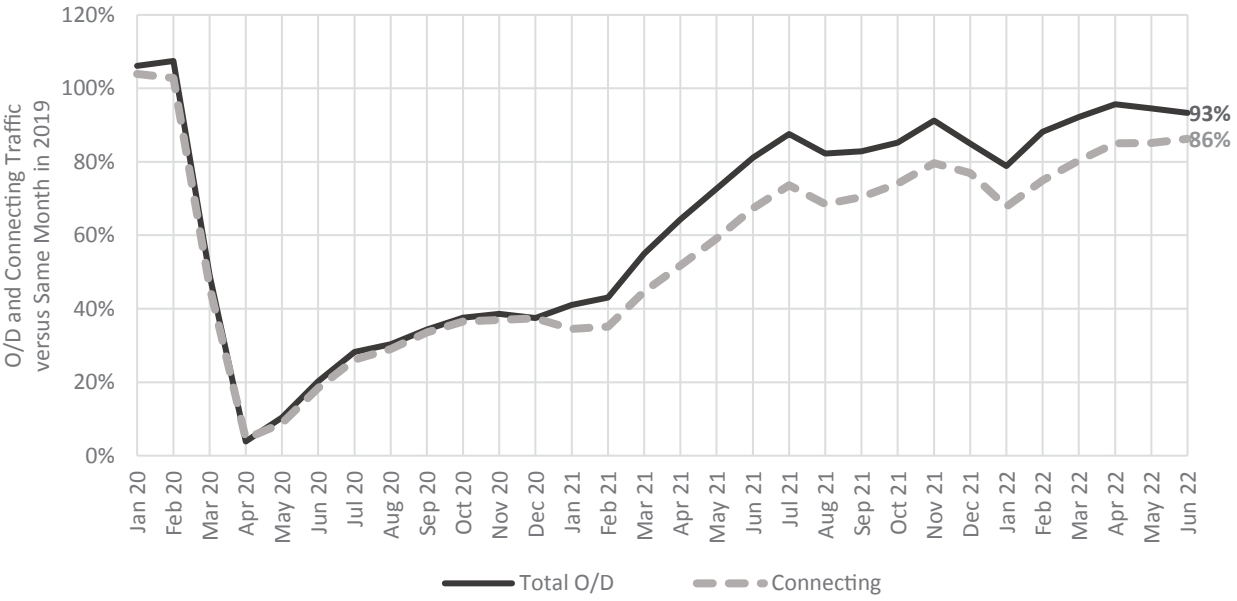


Figure 8. Monthly O/D and connecting passenger volumes from U.S. airports, January 2020 through June 2022 (Cirium 2023).

Impacts on Aircraft Operations

Commercial aircraft operations did not decline to the same extent as passengers due in part to government support to airlines.

Compared to commercial passenger traffic, commercial aircraft operations at U.S. Primary Service Airports experienced a smaller reduction in activity during the COVID-19 pandemic. As shown in Figure 9, total commercial aircraft operations dipped to a low of just 33% of 2019 monthly levels in April 2020 versus total passenger traffic dropping to a low of 4% of 2019 traffic in April 2020. The smaller dip in operations throughout 2020 and 2021 is attributable to a number of factors, but it is primarily attributed to the effect of government funding programs, which required airlines to maintain minimum levels of activity. This led to higher levels of aircraft movements relative to passenger traffic and reduced load factors throughout the pandemic. The relatively lower reduction in aircraft activity compared to passenger traffic also likely assisted airports' finances through this time by inducing a relatively smaller reduction in aeronautical revenues (i.e., landing fees) compared to passenger-related revenues (i.e., concession revenues), as discussed in a later section. However, some airports indicated that this reduced level of aircraft activity was increasingly concentrated in peak periods—particularly in 2020—placing peak demands for gates and other airport infrastructure at levels closer to prepandemic while remaining very quiet during off-peak periods. The recovery of commercial aircraft operations plateaued around summer 2021 and has remained around 90% up to and through June of 2023. International operations, after seeing a cut back in early 2022, likely associated with the Delta and Omicron waves leading to renewed health restrictions in some countries, have seen a fairly steady increase now nearly matching domestic's recovery by mid-2023.

Impacts on Air Cargo and General Aviation

While commercial passenger service experienced unprecedented declines during the pandemic, air cargo activity grew, with national tonnage increasing throughout the onset of the pandemic for several reasons including 1) higher e-commerce and integrator usage as consumers

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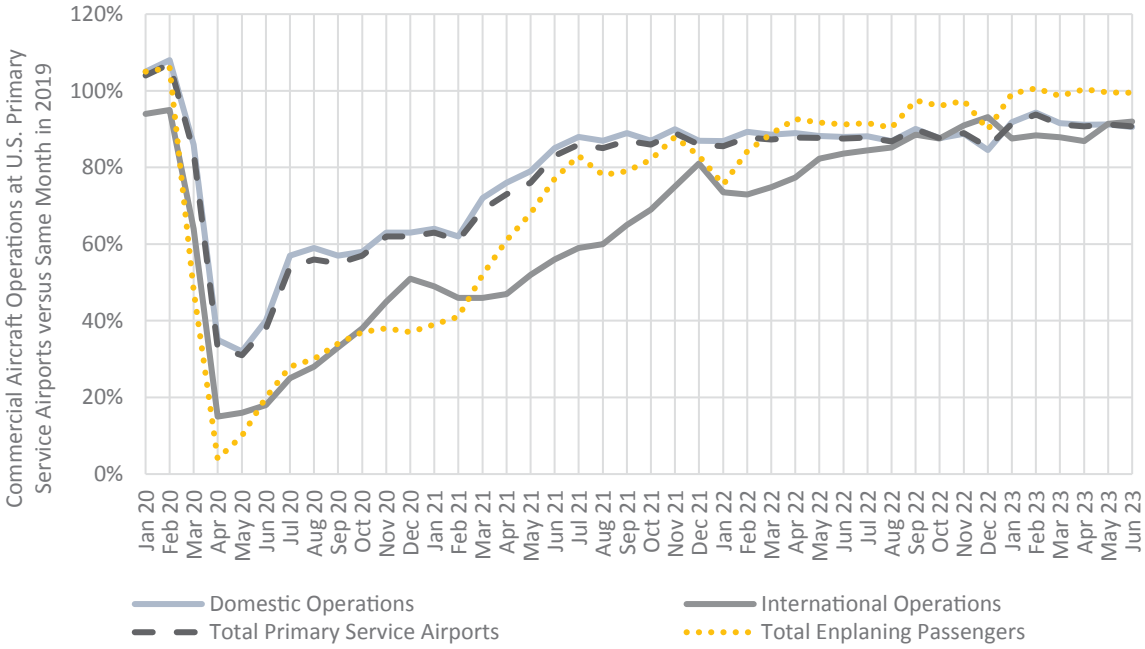
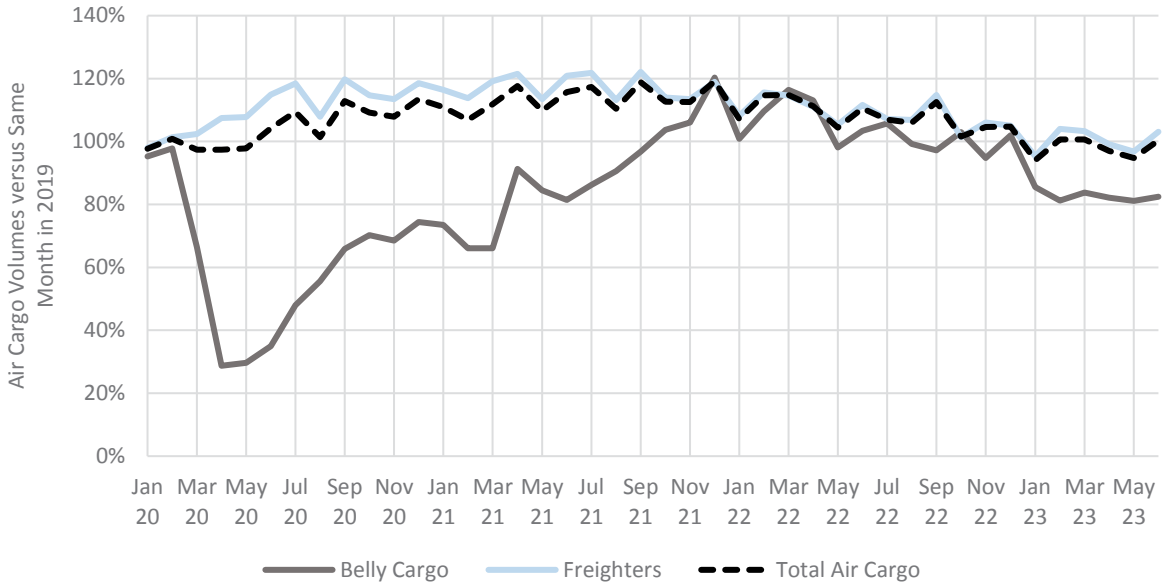


Figure 9. Commercial aircraft operations at Primary Service Airports versus the same month in 2019, January 2020 through June 2023 (U.S. DOT, n.d.).

While commercial passenger service experienced unprecedented declines during the pandemic, air cargo activity grew.

shifted to online shopping; 2) greater substitution toward air service as a more reliable transportation method for shippers in the wake of congestion and rising prices in the maritime sector; and 3) emergency use of air service as the fastest shipping method for transporting medical supplies, devices, and eventually vaccines to combat the pandemic. Figure 10 shows air cargo volumes at U.S. airports during the COVID-19 pandemic, beginning in January 2020, relative to the same month in 2019. Total tonnage—including both freight



NOTE: Air cargo includes both freight and mail.

Figure 10. Air cargo weight (pounds) at U.S. airports versus the same month in 2019, January 2020 to June 2023 (U.S. DOT, n.d.).

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and mail—saw only a very minor dip below 2019 levels during the early stages of the pandemic in the first two quarters of 2020. From mid-2020 to late 2022, total air cargo volumes remained above 2019 levels. The drawing back of cargo volumes in 2023 was likely reflective of the lessening of global supply chain issues, particularly in the maritime shipping sector, which supported some of the rapid growth in air cargo in 2020 and 2021. In many cases, air cargo movements increased during the pandemic, with a greater share of volume carried on freighter aircraft or converted passenger aircraft ("preighters") to meet the rising demand for air service and to offset the loss of considerable belly capacity due to passenger service reductions by commercial airlines.

While total air cargo volumes across U.S. airports increased in total in 2020 and 2021 compared to 2019, not all airports experienced equal growth. Cargo tonnage at large hub airports declined by 3.5% in 2020 compared to 2019 as a result of the reduction in passenger activity, particularly in international passenger services, which normally carry cargo in the belly hold of wide-body aircraft. As large passenger hub airports do not typically receive significant amounts of freighter traffic (due to capacity and/or slot restrictions), the reduction in passenger services and belly cargo led to a reduction in cargo at these airport types as a group. Medium hub airports saw the largest growth in cargo activity in 2020 at 14% due to growth in cargo activity at the nation's larger cargo airports, which are medium hubs in terms of passengers (e.g., Anchorage, Indianapolis, Cincinnati, Ontario). Small hub airports saw a 5% gain in cargo volumes, though that growth was driven almost entirely by growth in cargo at Memphis (MEM) and Louisville (SDF), which are global hubs for FedEx and United Parcel Service, respectively.

General aviation (GA) was less negatively impacted by the pandemic and recovered more quickly relative to commercial passenger operations. As depicted in Figure 11, itinerant GA and local civil operations at U.S. airports in April 2020, declined to 45% of the April 2019 level (versus a 67% decline for commercial passenger aircraft). However, by summer 2020, GA and local civil operations had recovered to 90% or higher (compared with less than 60% for commercial passengers). By summer 2021, activity levels were back at or above 2019 levels. The relatively modest decrease in GA and civil local operations can largely be attributed to private and business aviation being utilized as an alternative to commercial flying during a period when commercial air travel was reduced and restricted due to health measures. An increase in local operations versus pre-COVID levels in 2022 and 2023 may be related to increased training activity, potentially as a response to demand for commercial pilots and re-training following airline furloughs during 2020–21.

Aeronautical revenues declined less than nonaeronautical revenues (–11% versus –32% for all airports).

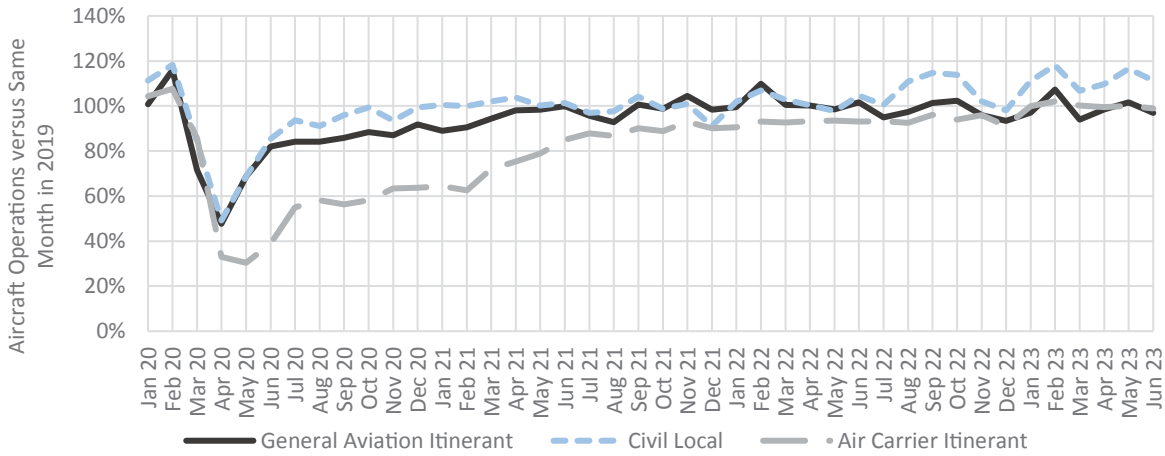


Figure 11. Aircraft operations at towered airports versus same month in 2019, January 2020 through June 2023 (FAA, n.d.-a).

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Airport Financial and Business Impacts

With sustained reductions in commercial passenger traffic, airports faced a substantial financial downturn. To provide a national assessment of the impact of the pandemic on airport financials, analysis was conducted on FAA Certification Activity Tracking System (CATS) Report 127 data, an online source of financial data for airports, both individual and aggregate (FAA, n.d.-b). There are differences in how individual airports report their financials, including which financial year they report—some use a calendar year while others use fiscal years ending April 30, June 30, or September 30. Nevertheless, FAA CATS provides a useful dataset for observing the overall trends in airport financials.

Figure 12 presents year-over-year trends for U.S. airport aeronautical revenues, nonaeronautical revenues (parking, car rental, food and beverage, retail, etc.), and total revenues for fiscal years (FY) 2019 and 2020 [operational revenues only, i.e., excluding nonoperational revenue such as Passenger Facility Charges (PFCs), interest income, or grant receipts]. FY 2019 is a prepandemic year, and depending on the fiscal year calendar used by individual airports, FY 2020 includes a portion of the year that is prepandemic and a portion of the year during the pandemic.

Across all airports, total revenues declined by 19% versus a 40% drop in passengers in the relevant fiscal year (the reported decline in passengers differs from the figure previously reported due to the mixing of fiscal years in the data). These equate to an operating revenue decline of $4.7 billion across all airports (from $24.7 billion in FY 2019 to $20.0 billion in FY 2020). Large hub airports experienced the greatest decline in revenues due in part to their greater exposure to international traffic, which has been slower to recover. Large hub airports accounted for 76% of total airport revenue decline (large hubs accounted for 71% of passenger traffic prepandemic). In all cases, aeronautical revenues declined less than nonaeronautical revenues (–11% versus –32% for all airports). Aeronautical revenues are more linked to aircraft operations (and landed weight) which, as previously discussed, did not decline to the same extent as passengers. Nonaeronautical revenues are more directly related to passenger volumes. Airports either experienced a decline in customers at their nonaeronautical businesses or received reduced payments from concessionaires that depend on passenger volumes (in many cases, the airports provided relief to these concessionaires). The airport financials for FY 2021 in the FAA CATS Report 127

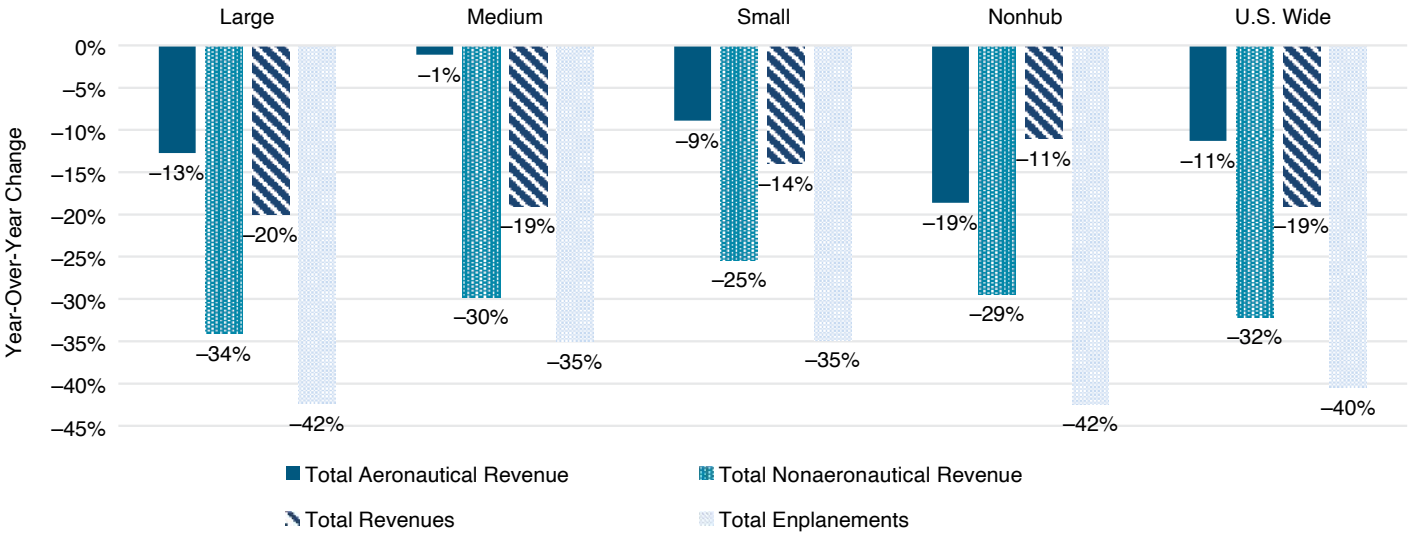


Figure 12. Airport operational revenue year-over-year trend by hub size, FY2019 and FY2020 [InterVISTAS's analysis of FAA CATS Report 127 (FAA, n.d.-b)].

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showed a similar decline in revenues, with total U.S. aeronautical revenues down by 9% relative to FY 2019 and nonaeronautical revenues down by 35% (FAA, n.d.-b). The similarity in results is due in part to the mixing of fiscal years (e.g., an airport using a fiscal year starting from April experienced the initial declines in FY 2021, while an airport using a fiscal year starting from January or September experienced the initial declines in FY 2020).

Government Support

In response to the financial and economic risks posed by the pandemic, the federal government introduced several support programs providing funding for the aviation industry, including airports. These programs were the CARES Act in March 2020; the Coronavirus Response and Relief Supplemental Appropriations (CRRSA) Act in December 2020; and the American Rescue Plan (ARP) Act in March 2021. The CARES, CRRSA, and ARP Acts collectively allocated approximately $20 billion in support for airports and their concessionaires over the course of several years, as summarized in Figure 13.

Aside from direct grants, airports also benefited from the support provided to U.S. airlines. For example, the CARES Act provided support totaling $62 billion for airlines and related contractors. The aid to airlines was split with approximately 50% going to direct aid to pay salaries and benefits and 50% going into loans and loan guarantees for general business purposes. There were various conditions and restrictions for accepting this aid, which did influence the degree to which individual airlines participated in the aid program. The conditions most important for operations at airports included no involuntary staff furloughs, maintenance of employment at 90% levels, and maintenance of minimum air service levels. As noted previously, aircraft operations did not decline to the same extent as passenger numbers, which was due in part to these conditions. The airline support in the CARES Act and subsequent bills also avoided the bankruptcy of any major U.S. airlines, which could have impacted airport traffic in the short and long term.

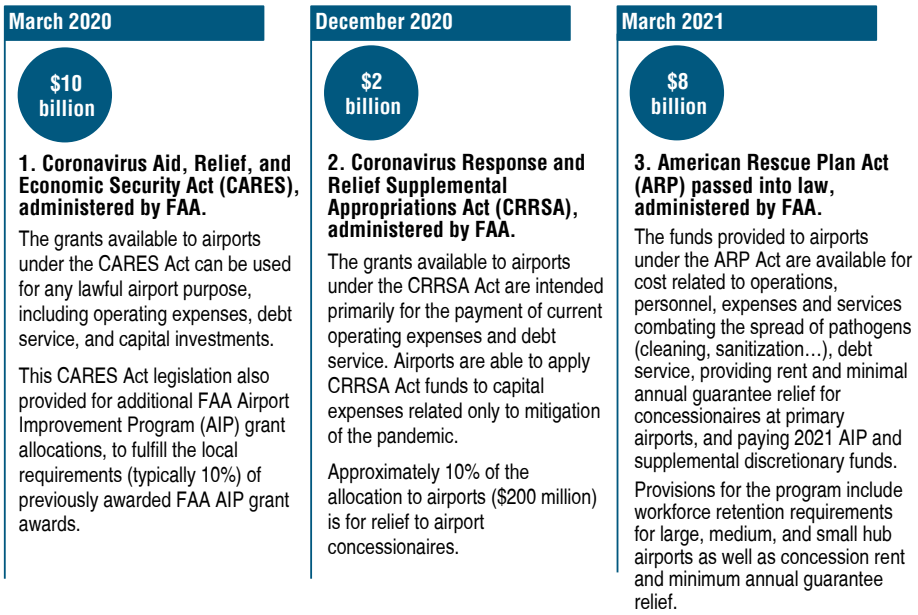


Figure 13. Summary of COVID-19-related federal funding programs to U.S. airports.

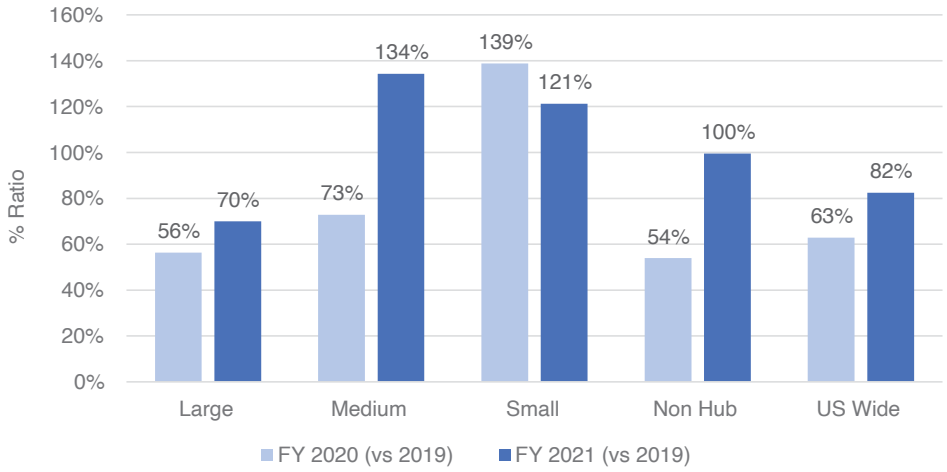
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Analysis suggests that federal support to airports offset approximately 63% and 82% of the operating revenue losses experienced by airports in FY 2020 and FY 2021, respectively.

In many cases, U.S. airports allocated CARES Act funds toward their airline cost centers to reduce or maintain airline rates and charges. Measures targeting concessionaires, terminal service providers, and ground transportation operators included the waiver of minimum annual guaranteed payments, payment deferral, and rent relief. The different federal aid packages have been used to cover operating expenses, debt service requirements through the expected recovery period, and to maintain or reduce airline rates and charges. Many airports did not fully utilize funds in fiscal years 2020 and 2021, with funding available for future years. Airports nationwide drew on governmental support, in part to help offset these revenue losses. The CATS Report 127 data show that airports received grants totaling $5.5 billion in FY 2020 and $6.8 billion in FY 2021, an increase of 135% and 193%, respectively, relative to FY 2019 ($2.3 billion) (FAA, n.d.-b). Much of this increase can be attributed to support provided by the CARES Act, although it likely also includes other grants unrelated to the pandemic. Figure 14 compares the level of support provided to airports relative to revenue losses, calculated as the increase in grant receipts in FY 2020 and FY 2021 relative to FY 2019, divided by the decline in operating revenues between the same years. The analysis suggests that federal support to airports offset approximately 63% and 82% of the operating revenue losses experienced by airports in FY 2020 and FY 2021, respectively. This masks how the government support was used by airports (such as debt serving) and may include other non-COVID support (which may explain why support to smaller airports was over 100% of revenue losses). Nevertheless, it does illustrate the scale of support provided by the federal government.

Operating Expenses, Debt, and Capital Planning

In response to diminishing revenues, airports looked to manage other major line items on their financial statement. Starting in FY 2020, airports attempted to realize savings in operating expenses through hiring freezes or slowdowns, reduced reliance on contractual services, and suspension of nonessential travel and expenses for staff. In many cases, airport tenants were allowed to limit their operating hours. For example, most airport parking operators consolidated their operations by closing some of their parking facilities (often the remote lot), reducing their



NOTE: Calculated as an increase in grant receipts in FY 2020 and 2021 versus FY 2019 divided by the decline in operating revenues in those years versus FY 2019.

Figure 14. Increase in grant receipts as a percentage of operating revenue losses [analysis of CATS Report, January 2023 (FAA, n.d.-b)].

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busing operations, and temporarily relocating their employees to close-in facilities. Additionally, larger airports reduced their debt service for FY 2021 and FY 2022 through bond refunding and restructuring, made possible in part with the use of CARES Act funds. In some cases, airports issued debt to reimburse cash expenditures for airport-funded capital projects. Finally, many airports reviewed their capital improvement plans. Some projects, like airside rehabilitation, were sped up to take advantage of quieter operations; other projects were deferred or rescoped to limit capital spending.

The response of airports to the pandemic is explored in more detail in the next chapter, using case studies at 11 airports.

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CHAPTER 4

Further Examination of the Impacts of COVID-19: Airport Case Studies

This chapter explores the impacts of the COVID-19 pandemic at 11 airports, captured in key themes told through 10 case studies (two San Francisco Bay Area airports were paired into one study). While all airports and the aviation industry in general were impacted by the COVID-19 pandemic, not every airport was impacted in the same way. The mix of operations and air service markets were major factors in determining how traffic at individual airports was affected by the pandemic. With differences in airport operations mix—passenger, cargo, and GA—airport revenues were impacted in different ways as well.

The impacts of COVID-19 on these 11 airports were analyzed through a combination of published data sources, contemporary news stories, and remote interviews involving personnel employed in planning, forecasting, air service development, and finance. Six key themes were identified; they are summarized in the following sections and illustrated with information for selected case study airports.

Table 2 provides an overview of the 11 case study airports. The studies included large, medium, and small hub airports, analyzing the differential impacts on passenger traffic at a variety of airport types. In addition, airports were selected to examine the impact on cargo traffic and GA. The table provides information on the maximum monthly recovery of the relevant activity level—passengers, cargo, or operations—compared to the same month in 2019. While some case study airports have seen their activity grow beyond pre-COVID levels, a number of passenger airports have yet to see recovery to pre-COVID levels as of June 2022.

Case Study: San Francisco Bay Area Airports Divergent Recovery Trends

The two largest airports serving the San Francisco Bay area are San Francisco International (SFO) and Oakland International (OAK). SFO is the largest airport in the region, the primary international gateway airport for the region, and a hub for United Airlines and Alaska Airlines. In 2019, 27% of SFO's passengers were international with a significant air service focus on Asia and trans-Pacific traffic. OAK is the second largest airport in the region, served primarily by low-cost carriers, and a major base for Southwest Airlines.

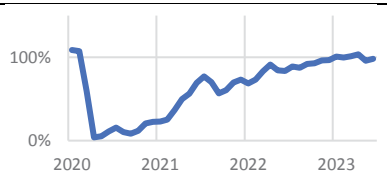
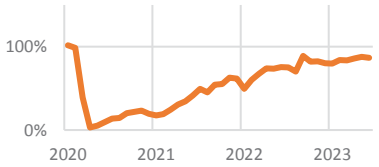
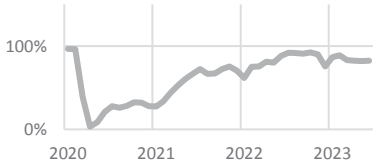
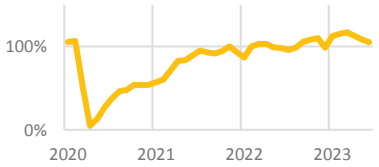
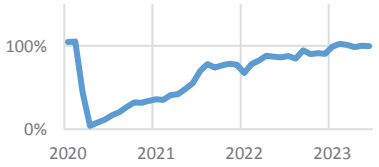
Among the two airports, OAK's traffic recovery has outpaced that experienced by SFO throughout the entire pandemic. While OAK does have some international services, those are oriented largely at leisure destinations in Mexico and the Caribbean, which were aligned more closely with the surge in leisure-travel demand in 2021 than the long-haul international services and trans-Pacific markets best served by SFO.

Key Theme #1: Traffic Reductions and Recovery Rates Depended on Destination Mix and Airline Mix

Throughout the pandemic there was a clear distinction between the impacts and recovery of domestic versus international travel demand and capacity. Across the U.S. aviation industry, domestic passenger traffic recovered faster than international due to the varying range of travel restrictions based on health policies on international travel—from outright travel bans to policies disincentivizing international travel, such as quarantine requirements and testing. Because of these effects, airports that had a high proportion of domestic passengers and service prepandemic (e.g., DEN, OAK) tended to recover faster than airports with a larger international air service market (e.g., SFO, IAD).

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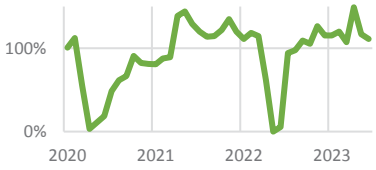
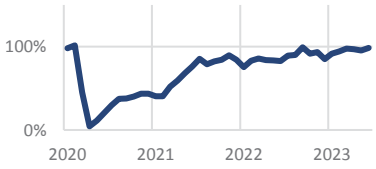
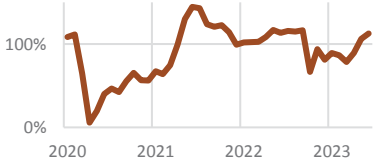
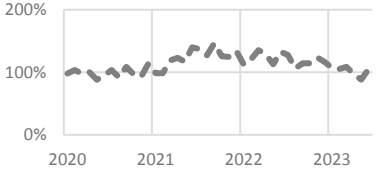
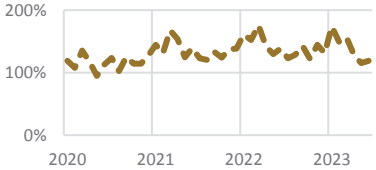
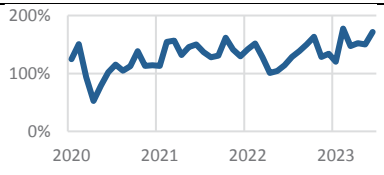
Table 2. Comparison of key impact and recovery metrics for case study airports.

Airport	Airport Type	2019 Traffic	COVID-19 Impact and Recovery *(% Relative to 2019)*	Maximum Recovery by June 2023	Unique Impacts
Honolulu, HI HNL	Large hub, primary gateway to Hawaii	10.3 million enplaned passengers		103%	Impacted by state policy to limit visitations in 2021 and slower recovery of international travel demand.
San Francisco, CA SFO	Large hub, major transpacific gateway, United Airlines hub	27.9 million enplaned passengers		89%	Deeply impacted by restrictions on international travel and ongoing slow recovery of Asia market.
Oakland, CA OAK	Medium hub, 2nd largest in Bay Area, Southwest Airlines hub	6.8 million enplaned passengers		92%	Regional LCC focus saw expansion of international services to sunspot leisure destinations.
Denver, CO DEN	Large hub, primarily domestic traffic, hub for United Airlines and Frontier	33.9 million enplaned passengers		117%	High proportion of domestic and LCC traffic led to rapid recovery compared to other large hubs.
Washington, DC IAD	Large hub, primary international gateway to DC, 35% of passenger traffic international	12.0 million enplaned passengers		102%	Large hub with more rapid transcontinental recovery from focus on European market.

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Table 2. (Continued).

Airport	Airport Type	2019 Traffic	COVID-19 Impact and Recovery *(% Relative to 2019)*	Maximum Recovery by June 2023	Unique Impacts
Jackson Hole, WY JAC	Nonhub airport, outdoor leisure-focused destination	0.5 million enplaned passengers		>140%	Outdoor leisure-focused destination in high demand during early pandemic. Temporary traffic shut down in first half of 2022 due to runway rehabilitation project.
Birmingham, AL BHM	Small hub, spoke airport	1.6 million enplaned passengers		99%	Spoke airport with average recovery due to network reorientation by legacy carriers.
Fort Myers, FL RSW	Medium hub, leisure destination with international services	5.1 million enplaned passengers		>140%	Injection of LCC traffic to domestic leisure-focused market produced a rapid recovery of passengers but saw a reduction in traffic in 2022/23 as carriers reduced injected capacity as demand diversified from just leisure travel.
Pittsburgh, PA PIT	Large hub, cargo	83 million pounds enplaned air cargo		>145%	Expansion of cargo services and major cargo tenants during pandemic helped offset financial impacts of reduced passenger services.
Rockford, IL RFD	Nonhub cargo airport, regional integrator/courier hub	387 million pounds enplaned air cargo		>175%	Surge in integrator and e-commerce demand supported strong cargo growth.
Van Nuys, CA VNY	Nonhub GA airport, busiest GA airport in the United States	212,000 aircraft operations		>160%	Growth in flight training activity spurred significant increase in operations.

NOTE: IAD = Washington Dulles International Airport; JAC = Jackson Hole Airport; RSW = Southwest Florida International Airport; RFD = Chicago Rockford International Airport; VNY = Van Nuys Airport.

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As the COVID-19 pandemic was a global phenomenon, each country and regional bloc adopted different policies and stances that restricted or enabled travel, either domestically or internationally. This patchwork of policies and regulations led to greater uncertainty for both travelers and airlines in reestablishing international travel. Some nations retained travel restrictions and sanitary policies longer than others, which allowed certain markets to open up faster, while other markets remained slow to recover or effectively closed even into 2022. China, the world's second-largest aviation market, put in place a "zero COVID" policy that involved effectively closing the nation's borders to foreign travel for an extended period. By contrast, the United States and Europe saw an earlier opening of borders through a managed set of health policies and risk management but only to certain country and region markets. Therefore, airports that had greater non-stop connectivity or O/D demand to regions or countries that experienced greater travel restrictions and longer returns to allowing discretionary travel, generally experienced greater negative impacts compared to airports with air services and demand oriented toward more rapidly reopening markets.

Key Theme #2: Traffic Reductions and Recovery Rates Were Influenced by Passenger Market Makeup

Travel patterns and market demand between business and leisure travel were also significantly impacted during the pandemic. As travel restrictions and health policies began to be relaxed starting in the latter half of 2020 and into 2021, leisure travel (including VFR) was one of the first travel segments to begin recovering. Partly as a function of pent-up demand as well as shifts toward domestic leisure travel while foreign markets continued to see travel or health restrictions, the U.S. domestic leisure air travel market saw a relatively quick recovery from the initial impacts of the pandemic.

There was also an observed trend of preference, particularly in 2020 and 2021, for leisure destinations that featured primarily outdoor recreation and activities. With health risks still present and mass vaccination not started or only partially underway, the early phases of the pandemic saw a greater demand for travel to leisure locations with lower population densities than major cities and/or outdoor leisure opportunities. The case studies of Jackson Hole and Southwest Florida are two examples of these types of leisure-focused destinations, with Denver being another case study that capitalized on the outdoor recreation draw of the region. In contrast, Las Vegas (LAS), one of the nation's top tourist destinations and a major conference and exhibition destination, saw a relatively slow recovery of its traffic demand in 2020 and into 2021. This was likely a combination of the relatively low desirability of Las Vegas' high-density indoor attractions (e.g., casinos and shows) as well as the restrictions and lack of demand for large-scale conferences and exhibitions during the earlier phases of the pandemic.

In 2020, 2021, and through much of 2022 business and conference travel remained largely below prepandemic levels in the United States and globally. Business travel was impacted by two major factors. First, changes in the ways of working during the pandemic with the rise of work-from-home in the

Case Study: Southwest Florida International (RSW) and Jackson Hole Airport (JAC) Leisure-Travel Demand Recovery

Both RSW and JAC are leisure-focused airports, the former located on Florida's southwestern coast and the latter serving a number of ski resorts and national parks. Leisure demand was one of the first segments of travel demand to return as travel and health restrictions began to be relaxed starting in the summer of 2020. Both case study airports are also gateway destinations for primarily outdoor recreational activities such as beaches, mountains, and national parks rather than dense urban environments.

Both of these airports saw an increase in annual traffic in 2021 compared to 2019 as travelers shifted demand toward domestic destinations amid a period of pent-up demand for leisure travel in 2021 after limitations on travel throughout 2020. Both airports continued to see above-2019 levels of traffic in the first half of 2022 despite dips during the winter season due to the Delta and Omicron waves of COVID-19.

RSW is also an international leisure destination with non-stop air services to Canada and Europe, which grew in 2022 above 2019 levels. As these are leisure-focused services primarily operated by vacation scheduled charter airlines, the airport has been able to grow its international market amid an industry-wide surge in leisure demand in 2021–2022.

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Case Study: Washington Dulles International Airport (IAD) Business Travel Market Impacts

IAD is one of three airports serving the Baltimore-Washington Metropolitan area and the region's primary international gateway airport. The airport is largely served by network carriers. IAD is also a major connecting hub, especially for international connections as approximately one-third of its pre-pandemic international passenger volumes were connections. Based on these factors the airport is estimated to have a fairly high proportion of business travelers, also due to its catchment area for the nation's capital.

Throughout the pandemic, IAD's traffic recovery lagged behind national averages. The major decline in international travel demand and capacity slowed the airport's recovery in 2020 and 2021, coupled with an only modest recovery in domestic traffic volumes. A part of this relatively slower recovery than national averages is attributed to the slower progress of recovery in business travel, including government and conference markets.

However, as of 2022, the airport's international sector experienced a strong uptick in recovery likely related to the ongoing recovery of the international business travel market as well as the airport's focus on European connectivity, which has recovered faster than Asian air travel markets, especially China.

services sector and the use of video conferencing technologies replaced in-person meetings and visits. Initially, this was due to the necessity of health and travel policies, but as the pandemic progressed, it became more of a norm to conduct business—both internally and externally—virtually rather than in person. The second key factor was company policies that forbade or limited business travel, not only for health safety reasons but also to reduce costs during the economic downturn in the early stages of the pandemic.

While the business travel market (in-person conferences, trade shows, and other large meetings) did continue to recover largely in 2022, there remained components of the business travel market—particularly short-haul travel and internal meetings—that were slow to recover.

Key Theme #3: Not All Areas of the Aviation Industry Were Similarly Affected

While global passenger traffic was deeply affected by the impacts of the COVID-19 pandemic, the aviation industry also includes cargo and GA activities that experienced their own unique effects from the pandemic. Chapter 3 provides an overview of the aggregate impacts on cargo tonnage and aircraft operations on the U.S. aviation system. Two case studies—PIT/RFD and VNY—discuss the local impacts of the pandemic on the cargo and GA industries.

Key Theme #4: Localized Pandemic Management Strategies Influenced Impact and Recovery of Aviation Activity

Health policy and regulatory responses to the COVID-19 pandemic saw a significant variation among countries and even within countries themselves. The United States was no exception, with individual states setting their own policies regarding stay-at-home orders, vaccination rollout plans, and travel advisories as public health regulations are generally within the purview of state governments. Across the nation there was a wide range of approaches, with some states adopting stricter measures (e.g., longer stay-at-home orders, limitations on public gatherings, proof-of-vaccination to access businesses or services) while others relaxed measures earlier in the pandemic or never implemented them.

To evaluate the differential impacts of health policy, two sets of states were compared: one group that implemented more stringent restrictions in response to the COVID-19 pandemic and a second group of states with less stringent responses (Figure 15). The "more stringent" group is made up of California, Hawaii, Illinois, New Mexico, New York, Vermont, and Washington, while the "less stringent" group includes Alabama, Florida, Iowa, North Dakota, South Dakota, and Texas (data from Hallas et al. 2021). Analysis of domestic passenger volumes for these two groups reveals that states with fewer restrictions generally recovered traffic faster in 2021 and the first half of 2022 than those states with more stringent public health policies. It should be noted though that the grouping of states is not equally representative in the types of airports and destination markets. States in the more stringent policy response group tended to

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Case Study: Cargo Activity at Pittsburgh International Airport (PIT) and Chicago Rockford Airport (RFD)

PIT is a medium hub airport serving the Greater Pittsburgh area with both commercial passenger and cargo services. The airport has developed a cargo market particularly focusing on integrator traffic with UPS and FedEx operations starting in the early 2000s. Like most airports, the COVID-19 pandemic led to a major downturn in passenger activity at the airport, but cargo volumes were maintained throughout 2020 due to growing e-commerce and business-to-business activity. In 2021, PIT annual cargo volumes increased due to expanded UPS operations as well as the planned introduction of Amazon Air. Cargo freighter activity and associated land leasing helped financially offset reductions in passenger-based revenues during the pandemic. The airport evaluated infrastructure plans regarding cargo operations at the airport during the pandemic and designated air cargo as a priority for PIT in planning, focusing on leveraging the airport's abundant available land for future cargo growth.

Chicago Rockford Airport (RFD) is a nonhub airport located approximately 85 miles northwest of Chicago. The airport has limited passenger services but is UPS' second largest hub in the United States and is a regional hub for Amazon Air. Both UPS and Amazon Air increased their flight operations and cargo throughput as the demand for air cargo increased during the pandemic, particularly spurred on by the increase in e-commerce demand during 2020. Even though stay-at-home orders were rescinded largely by mid-2020, demand for integrator cargo activity continued to grow at RFD. As of June 2022, cargo volumes remain up more than 25% above prepandemic levels and have consistently been above the national average for cargo volume recovery.

Case Study: General Aviation (GA) at Van Nuys Airport (VNY)

VNY is the nation's busiest GA airport, located in the San Fernando Valley just north of Los Angeles. In 2019 the airport handled 219,000 aircraft operations and was home to more than 700 based aircraft. Like many GA airports, VNY's operations are a mix of private operations, noncarrier commercial operations, and business aviation activities.

After initial stay-at-home orders were relaxed in the summer of 2020, VNY experienced a rapid recovery in its aircraft operations—both itinerant and local. Flight schools in the region took advantage of the reduced activity of passenger carriers and the resulting less-active airspace in the busy southern California region, to ramp up training activities. Local aircraft operations at VNY were at times double or triple prepandemic levels in 2020 at a time when passenger carrier activity nationwide was significantly depressed due to the pandemic. There was also a rapid recovery and even growth in business aviation traffic, which was understood to be driven by challenges in the commercial aviation sector regarding capacity and health safety perceptions.

While VNY does not collect landing fees, the rapid recovery in aircraft operations helped produce a quicker recovery in airport revenues through fuel sales as well as supporting local businesses such as fixed-based operators (FBOs) and maintenance businesses. As of 2022 the airport had experienced a 30% increase in annual aircraft operations compared to 2019.

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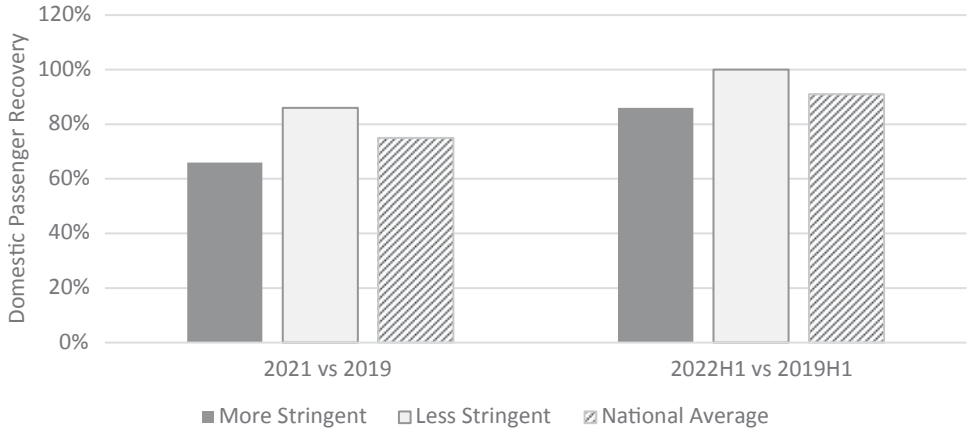


Figure 15. Recovery of domestic passenger volumes by select state groups (U.S. DOT, n.d.).

have large hub airports and be more densely populated (e.g., California, New York, and Illinois), whereas the less stringent policy grouping states did not feature large hub airports (e.g., North Dakota, Iowa) or were high leisure destinations (e.g., Florida). While there are a multitude of factors contributing to the recovery of aviation activity at the state level, government policy with respect to health and travel policies was often a contributor to how quickly air travel recovered within a state.

Case Study: State Health Policies on Travel in Hawaii

Daniel K. Inouye International Airport (HNL) serving Honolulu is the largest airport in Hawaii as well as the largest international gateway airport in the state. In addition to domestic and Canadian destinations, HNL had non-stop services to 12 countries in Asia and the Pacific, with Japan being its largest international market. Like all airports, the onset of the pandemic saw a major downturn in demand, but the airport's international demand recovery was strongly influenced by government policies at home and abroad.

Early in the pandemic, the Government of Hawaii imposed a 14-day quarantine period for visitors (including U.S. residents) and returning residents upon arrival to the state. This policy heavily impacted the tourism industry early in the pandemic, but the introduction of a testing policy—allowing travelers to skip quarantine with a negative test result—starting in October 2020 allowed tourism and travel to resume. All testing and quarantine restrictions for arrivals to the state were not lifted until January 2022.

These health policies have had a notable impact on the recovery of international travel demand and capacity at the airport. As of mid-2022, HNL had recovered less than 30% of its prepandemic international travel demand. This may be attributed both to the travel policies and outbound travel demand and travel policies in HNL's largest foreign market of Japan. However, with the start of mass vaccinations in 2021 and increased testing capacity, HNL experienced a rapid recovery of its domestic travel market (above 100% of 2019 levels in the winter of 2021/2022) likely spurred by pent-up demand as well as the accessibility of Hawaii as an outdoors domestic destination for U.S. travelers.

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Key Theme #5: Financial Recovery Was Aided by Government Support

As the COVID-19 pandemic negatively impacted travel demand and capacity, airports along with the rest of the aviation industry and associated businesses, experienced major downturns in revenues. As previously documented, U.S. airports experienced a significant decline in operating revenues as traffic volumes dropped during the pandemic.

The U.S. federal government provided a series of financial support packages to the wider economy and specifically to the aviation industry to help provide much-needed financial aid at a time when passenger aviation activity was reduced in part to meet health policy goals. The financial support impacted airports in several ways, including the following.

- Financial aid provided much-needed cash flow to ensure that airports could maintain operations, required maintenance, and debt servicing at a time when revenues were severely depressed due to the downturn in aviation sector activity. This reduced the need to take on other debt through bond issues or other instruments to merely maintain operations. However, many airports did engage in bond refinancing (e.g., OAK, DEN, BHM) as part of a strategy to reduce debt servicing obligations.
- Stipulations on financial aid packages largely required that airport operators could not reduce headcounts to reduce costs. This ensured that airport operators had the necessary staffing levels to provide safe operation of the airport and to provide ongoing financial support to households during a period of industry downturn. Hiring freezes were generally instituted and positions that were voluntarily vacated (e.g., retirement, early retirements, and employees voluntarily changing jobs) were generally not filled back.
- Airline-specific subsidies also required maintaining minimum levels of service early in the pandemic, which provided a base revenue stream for airports. For smaller spoke airports (e.g., BHM or JAC), this was likely more valuable as it ensured levels of service to drive revenues and connectivity for their communities that may have otherwise been threatened if airlines looked to rationalize their networks during the downturn in air travel demand.
- Federal aid programs included requirements targeted to concessionaires, terminal service providers, and ground transportation operators, which helped support associated airport businesses and industries beyond the airport operators themselves. These targeted supports were viewed as important for keeping the airport ecosystem functioning amid a period of unprecedented low demand. Some airports (e.g., DEN and HNL) were able to proactively provide these supports based on strong financial positions entering the pandemic that could be further backstopped through the targeted federal support programs.

Interviews with airports of various sizes and hub types all indicated that the financial support provided by the government was a significant help in maintaining operations and ensuring financial stability during the pandemic. The provision of significant funds for airports, airlines, and associated businesses created stability in the industry in a time of major disruption and was a key source in airports being able to meet the return in demand for air travel as travel picked up in 2021 and 2022. Clauses to ensure that airports could

Case Study: Denver International Airport

Denver International Airport (DEN) is a large hub airport serving the metropolitan Denver area and the primary air gateway to the state of Colorado. The airport is a major hub for United Airlines, Southwest Airlines, and ultra low-cost carrier Frontier Airlines.

The airport entered the pandemic in a strong cash position with a number of major infrastructure projects planned and underway that were to revitalize and expand DEN's terminal facilities. Because of this strong cash position and the presence of federal supports targeting airports and the aviation industry, the airport was able to proactively offer financial support and relief to its airline stakeholders and concessionaires. These airport-level supports and initiatives were part of a strategy to ensure strong stakeholder relations during the challenging times of the pandemic and help position stakeholders to be able to react to the anticipated traffic recovery.

To accomplish this, the airport placed its relief grants into a targeted fund solely for the purposes of debt servicing. This decision allowed the airport to focus its financial activities on supporting local stakeholders (in addition to federal targeted supports such as those directed at airport concessions) and prioritize key capital improvements, such as a major concourse and gate expansion that would be needed as traffic recovered from the impacts of the pandemic.

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not reduce headcounts to cut costs were also seen as valuable in ensuring that there were staff available to meet the demands of the traveling public as recovery began.

More cargo-focused airports, such as those in the case studies (PIT and RFD, previously in Key Theme 3) experienced growth in cargo operations, particularly during the early phases of the pandemic. This included increased freighter activity, as cargo capacity previously provided by passenger services was significantly reduced and more business-to-consumer and business-to-business activity moved online. As e-commerce in the United States had been on a growing trend before the pandemic, airports that had an existing base of integrator and e-commerce activity saw increases in activity as the pandemic changed consumption patterns. While lockdown and mobility restrictions were largely lifted in the United States by the end of 2020, e-commerce and integrator air cargo traffic remained strong, enabling growth at airports where they are a focus market for both activity and land use.

Key Theme #6: Airports Implemented Differing Operational and Capital Financial Management Strategies

As the COVID-19 pandemic severely limited air travel demand and capacity in 2020, airports across the United States were forced to develop strategies to adapt to a situation of low revenues while having to remain open and maintain safe operations. While the federal government

Case Study: Birmingham-Shuttlesworth International Airport (BHM)

BHM is a small hub airport and the largest in Alabama, serving the Birmingham Metropolitan Area. The airport is a 'spoke' airport with non-stop service primarily to major airline hubs, but also a variety of destinations through low-cost carriers.

In response to the downturn in traffic and revenues from the pandemic, airport staff began an extensive review of all airport contracts and functions resulting in a much leaner and financially streamlined operation. Any capital projects not being funded with federal funds were halted, as was a land acquisition project. Unlike some airports, BHM did not accelerate any planned capital projects as primary planned projects were relating to the terminal, including a passenger lounge project, which were put on hold until traffic levels increased again. Operationally, the airport renegotiated many of its service provider contracts or acquired new service providers and shuttered some of the least profitable concessions at the airport.

Financially, the airport was in a process of restructuring its debt. The airport's governing authority was able to secure a restructuring program in 2020 which would save $33 million over the lifetime of the bonds. The airport saw operating revenues in FY2020 drop by –15%, with non-aeronautical revenues falling by –20%. This was less than the national average, and generally less severe an impact than was observed at large and medium hub airports. By FY2022, total operating revenues still remained down by –7% compared to FY2019. However, the airport reduced its operating expenses by –24% and significantly reduced its total indebtedness highlighting the success of cost-cutting and efficiency measures undertaken during the pandemic.

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provided substantial funding in the form of grants to airports and airlines, in the earliest phase of the pandemic airports were faced with developing strategies to mitigate the major reduction in revenues.

Across the nation, several strategies evolved toward reducing operating and capital costs borne by airports at a time of significantly reduced revenues. Based on interviews with airports and observations of industry data, the following major strategies were observed:

- Closures of concourses and terminals to reduce operating costs. With declines in passenger traffic, particularly international traffic, many airports would close portions of passenger terminals, ranging from a few gates to entire terminals or concourses. Temporary closure of facilities reduced operating and maintenance costs ranging from utilities to janitorial services. When demand began to rise, airports were able to recommission these facilities, often in a staged fashion to meet demand. OAK and HNL both closed entire concourses/terminals. While the former did so to reduce operating costs, the latter closed terminals that were no longer in use due to restrictions on international travel.
- Many airports deferred or delayed capital improvements to trim costs. Airports interviewed for this research indicated that they conducted reviews of existing capital plans to determine which projects could be deferred to a later date, which were necessary (e.g., safety improvements, utility upgrades), and which should be continued as the cost of deferral was deemed to be too high.
- Some airports accelerated infrastructure development plans, particularly those focusing on airside (runway, taxiway, apron) and terminal facilities to take advantage of reduced operations and passenger activity. With fewer operations, airside and terminal construction was less disruptive to airport activities, and schedules for projects were brought up to reduce the cost of disruptions if projects were carried out during periods of normal activity levels. DEN maintained pace on a major concourse and gate expansion project during 2020 as airline stakeholders encouraged the airport to continue with construction to prevent capacity constraints from hindering the recovery of traffic once demand regrew to near prepandemic levels.
- Airport concessionaires asked for, and in many cases received, waivers and relief for contracted Minimum Annual Guarantees. With passenger-based revenue sources being so majorly impacted, concessionaires were often more negatively impacted than the airport operators themselves. Airports that had entered the pandemic in good financial positions with large cash reserves were mostly able to provide support to concessionaires and other service providers before federal grant programs targeting those sectors (CRRSA) were implemented.
- Parking and ground transportation services, such as shuttle buses and valet parking, were reduced or suspended. Many airports closed their long-term parking lots and kept open only the lots or parking garages closest to the terminal during times of major traffic reductions. OAK, DEN, and BHM all engaged in these closures based on the case study research. This was often accompanied by strategies to flatten parking rate structures (e.g., high-value parking garages adjacent to terminals could be used at low-value, long-term parking lot rates as seen at DEN) to improve passenger experience despite missing out on potential revenues. Some airports repurposed temporarily closed parking lots for functions such as COVID-19 testing and vaccination sites or community amenities like temporary drive-in movie theaters.

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CHAPTER 5

Other Shock Events and Their Impacts on Airports and the Aviation Sector

This chapter highlights other examples of events causing shocks in the aviation industry. A major part of being able to better plan for and mitigate future shock events is understanding historical shocks and how they impacted the aviation industry and airports. Some shocks may be unique and true one-off events, while others are variants of a common theme. Typical examples of the latter are natural disasters or the entry of a new carrier leading to rapid growth. Although considered unanticipated by the airports experiencing them, shock events have precedence in history.

Infectious Disease Event: SARS (2002–2003)

The SARS outbreak brought to light the modern reality that aviation can simultaneously be a potential transmission medium and a victim of pandemics.

The COVID-19 pandemic has been the largest infectious disease event to impact the aviation sector, but it was certainly not the first. The 2002–2003 outbreak of SARS marked the first major outbreak of a new infectious virus in the 21st century and was followed by other disease events, including MERS of 2012–onwards, multiple Ebola outbreaks in Africa starting in 2014, and various zoonotic influenza viruses such as H5N1 (Avian Flu) of 2005, H1N1 (Swine Flu) of 2009–2010, and H7N9 (Avian Flu) of 2013 (these outbreaks generally had less impact on aviation than SARS or COVID-19). Several researchers have argued that the likelihood of pandemics has increased over time, driven by increased mobility, growing urbanization and population density, industrialized food production, globalized supply chains, and the expansion of global transportation networks including air travel (Hall, Scott, and Gössling 2021).

The 2002–2003 global outbreak of SARS lasted approximately 9 months from November 2002 to July 2003, marked by a precipitous rise and then fall of cases throughout affected regions, primarily in Asia. In total, there were nearly 8,100 cases and nearly 800 deaths from SARS worldwide throughout this period, with 96% of cases occurring in Asia. Air travelers served as vectors for the disease in the early stages of the global outbreak, flying it to several countries where onward transmission would then occur primarily with infected patients' family members and within hospitals. At the peak of the global SARS outbreak, more than 30 international airports in 20 countries applied measures to detect infected travelers. Protocols that directly impacted airports and airlines included travel alerts and advisories issued by governments and WHO, passenger screening questionnaires, temperature screening of passengers, contact tracing, media and information campaigns in terminals, quarantine measures, and in-flight protocols such as recommendations to wear masks. In the case of SARS, international coordination and the rate of initial emergency response (or rather lack thereof) proved to be critical in determining the timeline and severity of the pandemic as well as the ability for the disease to spread across borders. Once an outbreak occurred, physical containment efforts were effective in mitigating

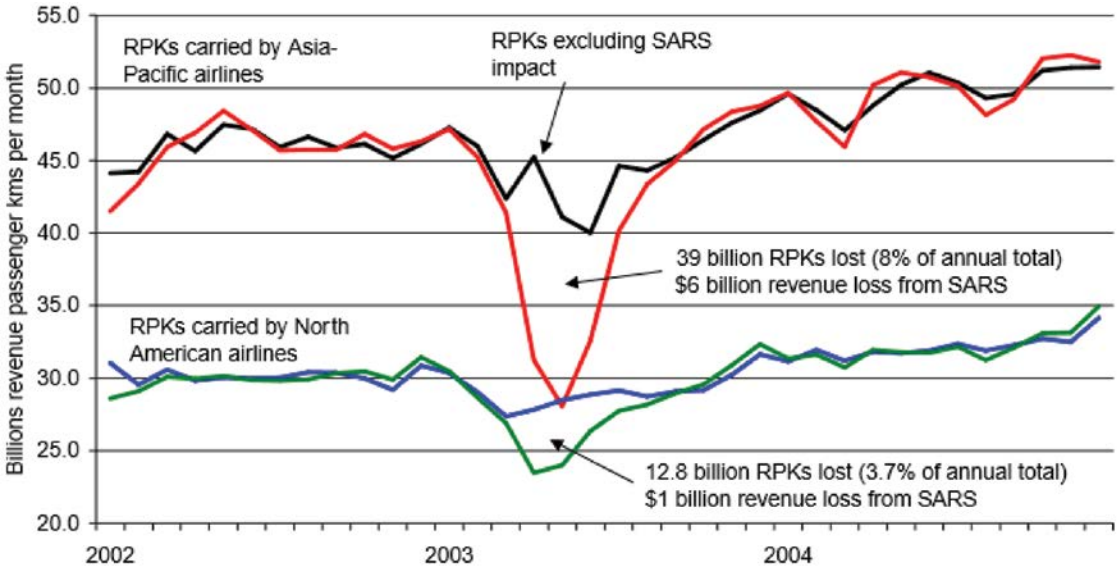
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further spread, although certain measures taken at airports, particularly temperature screenings, were not necessarily effective given the difficulty in identifying infected travelers.

The impact of SARS on the aviation industry was characterized by a dramatic yet transitory event that primarily impacted international air passenger traffic, most notably marked by a V-shaped recovery, which took less than a year (Figure 16). According to the International Air Transport Association (IATA), it is estimated that Asia-Pacific airlines lost approximately 8% of their annual international traffic (in terms of revenue passenger kilometers or RPKs), amounting to $6 billion in lost revenue. North American airlines were far less involved directly with the outbreak but still lost 3.7% of their annual international traffic amounting to an estimated $1 billion in lost revenue.

Overall, it took 9 months before global international passenger traffic returned to precrisis levels, though the recovery timeframe varied widely across affected airports. Monthly traffic levels declined by as much as 40% to 80% (relative to 2002 levels) at major Asian airports in Beijing, Hong Kong, Taiwan, and Singapore. In Toronto, the largest hotspot for SARS cases outside of Asia, quarterly total air traffic at Toronto Pearson Airport declined by as much as 14% relative to the prior year. Passenger declines lasted anywhere between 6 and 10 months before returning to precrisis (2002) levels. Conversely, in the United States, which was not significantly affected by SARS directly, declines in air traffic were limited to passengers coming from hotspot regions. During the months of the global SARS outbreak (March to July 2003), international traffic to the United States declined by 4% (U.S. DOT, n.d.). LAX and SFO, which had the most passengers from China (including Hong Kong and Taiwan) and Singapore, saw traffic from these Asian hotspots fall by nearly 50% during the global outbreak (relative to 2002 levels). However, passengers from these places accounted for less than 5% of total traffic at those airports, making the overall impact relatively small.

Airports, government authorities, and other stakeholders took several actions in the aftermath of the global SARS outbreak to aid traffic recovery and plan for future health crises. Various forms of financial relief were administered to help shoulder the costs associated with lost air traffic during the outbreak as well as to help stimulate the return of travelers. Beyond immediate



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Figure 16. Impact of SARS on Asia-Pacific and North American airlines' international traffic (Pearce 2006).

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recovery efforts, governments and airports contended with the need to improve emergency response protocols for future crises. Some governments restructured or centralized their public health systems to help establish clearer protocols and hierarchies across governing bodies, including transportation authorities. For instance, the SARS outbreak was a pivotal event for Taiwan; the slow initial response to combat and contain SARS incited the government to establish a centralized public health system responsible for crisis management over all major authorities, including those that govern or operate transportation, customs, and immigration. This centralized agency enabled processes to rapidly implement passenger screening, case monitoring, and contact tracing for travelers and the country as a whole. These measures have been credited with helping Taiwan manage subsequent emergencies, particularly COVID-19 (Hall, Scott, and Gössling 2021).

For airport infrastructure, research across the industry emphasizes the benefits of airport planning and facility designs that incorporate open terminal layouts and flexible or movable infrastructure to help quickly implement emergency processes like fever screening, infection control, and physical distancing (HOK Architects 2020). Flexibility, in particular, is key given that not all emergencies are the same; even minor differences between SARS and COVID-19 resulted in far different pandemics, and not all of the physical containment efforts that worked for SARS were effective for COVID-19 (Wilder-Smith, Chiew, and Lee 2020).

The SARS outbreak brought to light the modern reality that aviation can simultaneously be a potential transmission medium and a victim of pandemics. The international connectivity and speed afforded by air travel when compared to the incubation period of many infectious diseases, means that air travel can be a significant factor in turning a local epidemic into a global pandemic under the right conditions. Airports, particularly major international or gateway airports, are often identified (fairly or unfairly) as the key transit points through which diseases may be imported and therefore will directly bear the impact of restrictions on movement and travel.

9/11 Terrorist Attacks

The terrorist attacks of 9/11 were one of the most significant shock events affecting the United States and global aviation industries in the last 30 years. The attack was carried out by the extremist group Al-Qaeda, involving the hijacking of four commercial U.S. airplanes with the intent of crashing them into major U.S. commercial and political institutions. Two of the planes were flown into the twin towers at the World Trade Center complex in New York City, a third plane was flown into the Pentagon in Arlington, Virginia, and the last plane was destined for Washington, DC but instead crashed into an empty field in Pennsylvania following a revolt against the hijackers by the passengers onboard. Altogether, the attacks claimed the lives of nearly 3,000 people, including passengers onboard the hijacked flights as well as civilians and first responders caught in the targeted areas on the ground.

As the events of the morning of September 11th unfolded, the FAA suspended all civilian flights and shut down U.S. airspace, ordering all flying aircraft to land at the nearest available airport as well as all inbound international flights to divert elsewhere. Due to the shutdown of U.S. airspace, inbound international flights to the U.S. were either returned to their origin airport or redirected to land at airports in Canada or Mexico. All civilian flights in the United States remained grounded for 3 days. Certain airports were shut down for longer: Ronald Reagan Washington National Airport in Washington, DC, was closed for 23 days and opened gradually over a span of 6 months, with restrictions such as bans on large aircraft, GA, nighttime operations, and certain flight paths.

When commercial air travel resumed, airlines and airports had to contend with a drop in demand amid dampened economic conditions and a lower willingness among the public to fly,

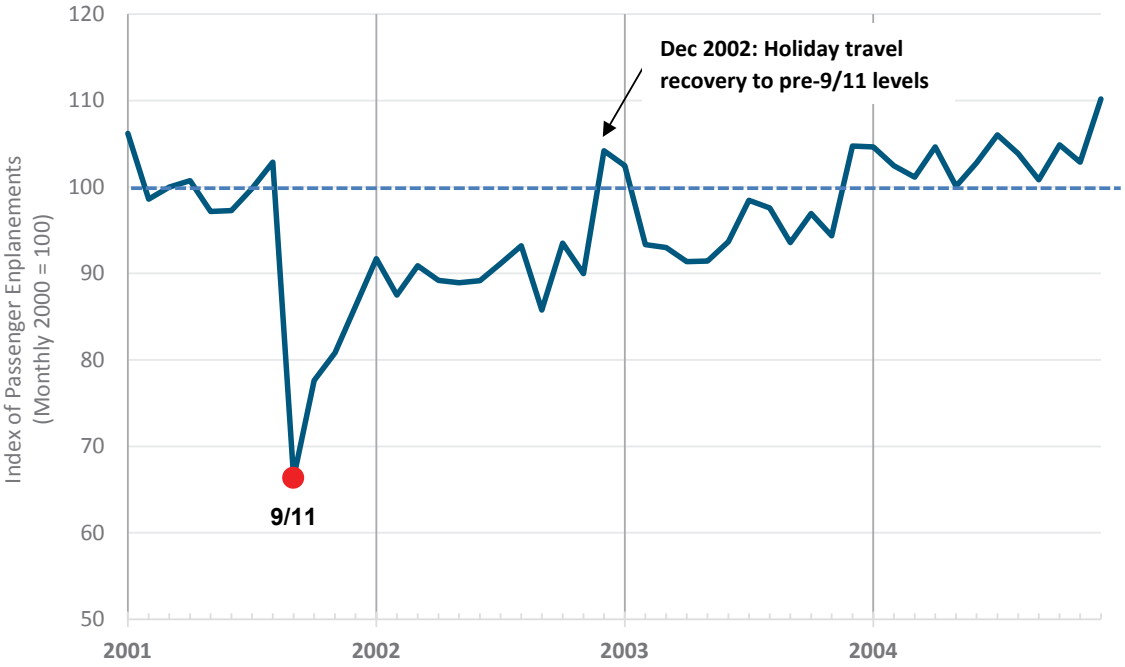
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along with immediate changes to enhance airport security in the interim. For instance, in the months following 9/11, airport security screening was conducted with the assistance of armed National Guard soldiers and local and state police in some places. Security lines grew as officers introduced additional screening measures like searching carry-on bags and patting down passengers even before more formal regulations were introduced.

Figure 17 shows monthly passenger traffic at commercial airports within the United States in the year after 9/11. September 2001 traffic dropped by one-third relative to September 2000, with subsequent months in 2001 remaining 15% to 20% below the prior year's levels. Nationwide, monthly air traffic would not return to pre-9/11 levels until the holidays in late 2002, while nonholiday seasonal traffic took until 2004 to recover and begin growing again. More broadly, 9/11 diminished consumer confidence and slowed economic growth, which had already been dampened by the dot-com bust in preceding years, while fears over the safety of air travel persisted. The downturn in air traffic added pressure on U.S. airlines, driving cost-cutting measures and reductions in capacity and network realignments across the country. To support U.S. airlines for lost revenues associated with the mandatory groundings in September 2001 as well as additional anticipated losses in air traffic as a result of 9/11, the federal government introduced the Air Transportation Safety and System Stabilization Act (2001). This provided $5 billion in compensation and $10 billion in loan guarantees for approved U.S. airlines.

The 9/11 terrorist attacks led to a transformation of aviation security. The Aviation and Transportation Security Act (ATSA) in November 2001 federalized the security screening of passengers at airports and established the Transportation Security Administration (TSA) to manage it. In addition, the ATSA mandated the screening of all checked baggage. Prior to the ATSA, aviation security was privatized and fell under the responsibility of airlines, which often hired screening companies to perform actual duties at airport security checkpoints. By the end of 2002, the TSA began overseeing

The 9/11 terrorist attacks led to transformation of aviation security impacting airport operations and planning.



NOTE: Each month is indexed against the same month in 2000 (e.g., Sept 2001 versus Sept 2000, Oct 2003 versus Oct 2000).

Figure 17. U.S. passenger traffic index, 2001–2004 (monthly 2000 = 100) (U.S. DOT, n.d.).

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security operations and federal screening mandates at all commercial airports in the United States, thus ushering in many of the security screening measures and luggage limitations that are now commonplace for commercial air travelers. By the end of 2002, the newly established TSA had hired approximately 56,000 agents for both passenger and baggage screening at airports, nearly 3.5 times the estimated 16,200 private security screeners that were employed at U.S. airports before 9/11 (Blalock, Kadiyali, and Simon 2007). This headcount was later scaled back to just over 45,000 screeners by 2004.

These new regulations substantially changed airport operations and had a particular space-constraining effect on preboard security facilities that were originally designed for a pre-9/11 operating environment. In particular, the added passenger and baggage screening protocols created the need for expanded inspection areas to accommodate more security agents and other personnel, larger screening equipment, and longer lines and processing times of passengers and baggage. Due to more intensive security screening requirements, the processing rate at passenger screening checkpoints dropped from 500 to 600 passengers per hour per lane before September 2001, to 100 to 150 passengers after 9/11 (de Barros and Tomber 2007). This translated into much longer queues and wait times for passengers and additional space requirements to accommodate longer queues.

Beyond the resource-intensive nature of the screening procedures themselves, ATSA impacted other airport planning considerations, such as the clearer delineation between pre- and post-security terminal areas and the prohibition of nonpassengers from the post-security side. Before 9/11, nontraveling family and friends could accompany passengers to their gates, adding to the number of customers potentially spending money at terminal concessions, which was a source of revenue for airports. The new regulations not only reduced this source of concession demand but also forced airports to reconsider the layout and accessibility of concessions (e.g., food and beverage, retail) between pre- and post-security locations.

The reality remains that much of the airport infrastructure in the United States was not originally designed for a post-9/11 operating environment. According to a 2021 infrastructure assessment by Airports Council International-North America, the average U.S. airport terminal is over 40 years old, which means that many facilities were built before 9/11 and have undergone multiple rounds of structural and operational changes to accommodate the changing security landscape (ACI-NA 2021).

Extreme Weather Event: Hurricane Katrina

On August 29, 2005, Hurricane Katrina came ashore about 60 miles southeast of New Orleans, with wind speeds of up to 127 miles per hour and a storm-driven wave surge of up to 30 feet. The size and strength of the storm and subsequent flooding resulted in one of the largest natural disasters in U.S. history. Storm waters flowed over floodwalls and breached levees in Louisiana's Orleans and neighboring parishes, causing widespread flooding, billions of dollars of property damage, and, according to the Federal Emergency Management Agency (FEMA), more than 1,300 deaths in the United States. Although the airfield of Louis Armstrong New Orleans International Airport (MSY) did not experience flooding during Hurricane Katrina, the airport's facilities suffered extensive wind and water damage (the airport is just 3.7 ft above sea level and has a perimeter dike to limit damage from storm surge). During the immediate period of the storm and recovery, MSY was closed to commercial air traffic but supported humanitarian and rescue operations.

Figure 18 shows enplaned passenger traffic at MSY in the years before and after Hurricane Katrina. The hurricane led to a 39% decline in enplanements between 2004 and 2006 (from

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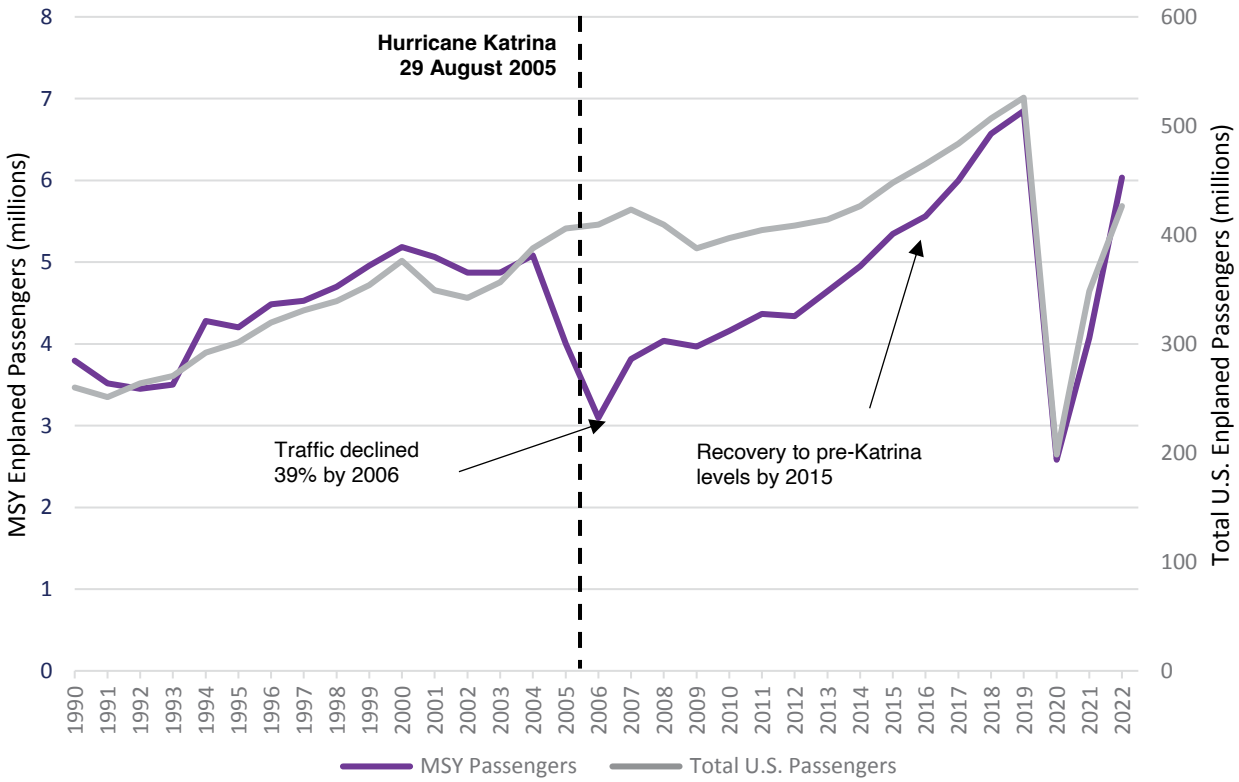


Figure 18. Enplaned passengers at MSY and nationally (U.S. DOT, n.d.).

5.1 million passengers to 3.1 million). Before the hurricane, traffic development at the airport was broadly following the U.S. average (the trend in total U.S. traffic), but after Katrina, traffic at the airport greatly diverged from this trend, with the result that passenger enplanements did not return to prehurricane levels for 10 years. In more recent years, traffic growth was one of the highest among U.S. airports, due to the entry and build up of services by ULCCs (Spirit and Frontier Airlines) along with growth by established carriers. With the COVID-19 pandemic in 2020, traffic at the airport declined by 62% that year, in line with the national average.

The protracted recovery after Hurricane Katrina was due in large part to two key factors outside the control of the airport—population decline and tourism recovery. The U.S. Census Bureau estimates that the population of the City of New Orleans, which was already slowly declining before Hurricane Katrina, dropped by 54% in 2006 due to the loss of housing and economic activity caused by the hurricane, which prevented those who had been evacuated from returning to the city. About half of the initially lost population was recovered by 2010, and by 2019, the city's population was 14% less than the 2005 level. Similarly, the population of the wider metropolitan statistical area was 9% lower in 2010 (than in 2005) and 3% lower in 2020. Tourism, in addition to convention tourism, is a major sector in New Orleans, attracting both domestic and international leisure travelers. The City is also a home port for cruise ships where passengers start or end their cruise holiday, thus generating traffic for MSY in the form of passengers. The University of New Orleans Hospitality Research Center reported that visitors to New Orleans declined by 63% between 2004 and 2006 (from 10.1 million to 3.7 million) and did not recover to pre-Katrina levels until 2016.

Following the hurricane, the airport put airline incentives in place to stimulate the return of air services, offering reduced charges for carriers reintroducing capacity or adding new routes. More recently and before the pandemic, the airport had attracted additional international air

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service to the UK and Germany. In May 2021, Breeze Airways (a domestic LCC formerly known as Moxy Airways, founded in 2018), announced the development of an operating base at MSY (Louis Armstrong New Orleans International Airport 2021).

In 2013, the airport undertook a Long-Term Strategic Development Plan that included construction of a new North Terminal (opened in 2019). The terminal was built with storm resilience in mind, with the use of extensive wind-tunnel modeling and on-site testing to ensure that the terminal could withstand hurricane-force winds. On August 29, 2021, New Orleans was hit by Hurricane Ida, a Category 4 hurricane, which was more powerful but more compact than Hurricane Katrina when it made landfall. The hurricane did not bring about the same scale of devastation to New Orleans as Hurricane Katrina, owing to the levees that had been upgraded after Katrina and did not fail this time. The airport remained technically open throughout the hurricane although all flights were cancelled for several days. The airport reported that no significant damage was caused to the new terminal.

Great Recession of 2008–2009

Economic recessions are not surprise events in the sense that they happen relatively frequently and with considerable historical precedent. According to the National Bureau of Economic Research (NBER), there were 10 U.S. recessions between the end of the Second World War up to 2007, roughly once every six years. Nevertheless, the timing and extent of the recession, and their implications for aviation, are impossible to accurately predict.

The Great Recession (also known as the Great Financial Crisis) of 2008–2009 was the largest recession experienced in the United States since the Great Depression (1929–1933). NBER records that the United States was in recession for 1.5 years (December 2007 through June 2009) and experienced a 5.1% decline in gross domestic product (GDP) (peak to trough). Previous recessions since the Second World War generally lasted 6 to 12 months and caused GDP declines of 0.3% to 3.7%. The economic recovery was also slow in comparison with past recessions—GDP did not recover to prerecession levels until 2013, and even then, it was considerably below the prerecession trend.

Figure 19 shows that, at a national level, passenger enplanements declined 8.7% between 2007 and 2009. Passenger traffic did not recover to prerecession levels until after 2014, 5 years after the decline and 7 years after the start of the recession. In comparison, traffic took 2 years to recover after 9/11 (the traffic decline following 9/11 was also 8.7%). Having recovered to prerecession levels, there was robust growth in passenger traffic averaging 4.1% per annum, between 2014 and 2019.

The impact of the Great Recession differed by airport size with medium and small hub airports experiencing the largest declines and slowest recovery.

The recovery from the Great Recession differed considerably by airport size. Large hub airports (30 airports, which accounted for approximately 71% of passenger traffic in 2019) experienced the smallest declines on average and had the fastest recovery (by 2012). On average, medium and small hubs (17% and 9% of total traffic in 2019, respectively) had steeper declines in traffic and a recovery that was 3 to 4 years longer than for large hubs. In contrast, nonhub airports (4% of passenger traffic) generally experienced a smaller decline and a faster recovery.

The uneven nature of the recovery is further illustrated by analyzing the proportion of airports that had lower traffic levels in 2014 and 2019 than they did in 2007, shown in Table 3. Approximately 55% of airports had not fully recovered traffic lost during the Great Recession by 2014, and 32% had not recovered traffic by 2019. Only two out of 30 (7%) large hub airports had not fully recovered traffic by 2019 (IAD and HNL), whereas approximately a third of

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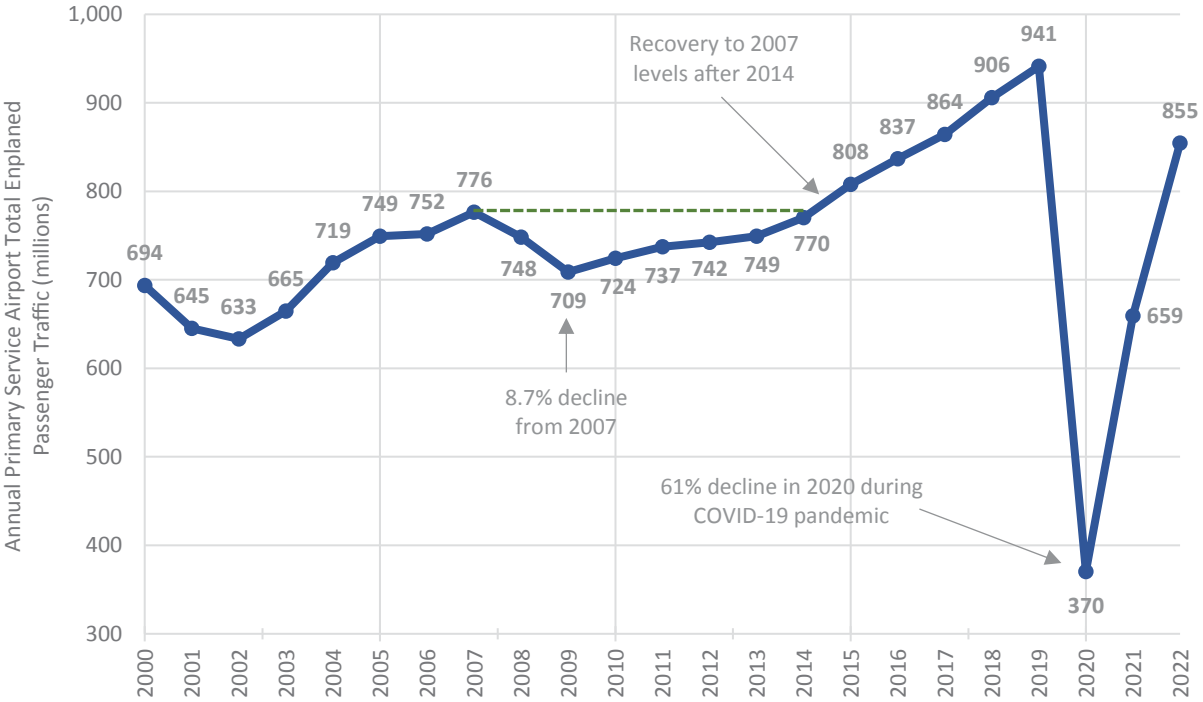


Figure 19. U.S. passenger traffic, all carriers, 2000–2022 (U.S. DOT, n.d.).

medium, small, and nonhubs had not recovered to 2007 traffic levels (13 medium hub airports, 29 small hub airports, and 138 nonhub airports). By 2014, 20% of these airports were down by 25% or more of their 2007 traffic levels, and 13% were still in this position in 2019.

There are multiple reasons for the lack of recovery at one-third of airports, many specific to the individual airports. However, significant factors were airline service reductions, dehubbing, and station closures (particularly at medium hubs) that took place in response to the recession, although airline consolidation played a role in some cases. In 2008, seven major airlines accounted for 80% of seat capacity in the United States: Southwest Airlines, American Airlines, Delta Air Lines, US Airways, United Airlines, Continental Airlines, and Northwest Airlines. With a number of mergers between 2008 and 2013, these seven carriers were reduced to four: Southwest Airlines, American Airlines, Delta Air Lines, and United Airlines. While the Great Recession was not the sole reason for these mergers, the financial difficulties that some airlines experienced over this period likely contributed to the rationale for the mergers. As a result of the Great Recession and airline mergers, several airports experienced a significant loss of capacity or airline dehubbing.

Table 3. Percentage of airports by hub type with lower traffic levels in 2014 and 2019 than in 2007.

Airport Type	Airports with Lower Traffic in 2014 than in 2007	Airports with Lower Traffic in 2019 than in 2007
All Airports	**55%**	**32%**
Large Hubs	43%	7%
Medium Hubs	81%	35%
Small Hubs	61%	31%
Nonhub	51%	35%

SOURCE: U.S. DOT, n.d.

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The period after the Great Recession also saw the development of ULCCs such as Spirit Airlines, Allegiant Air, Sun Country Airlines, and Frontier Airlines. ULCCs are characterized by very low base fares with ancillary charges for additional services and lower unit costs than traditional LCCs achieved through reduced labor costs and higher aircraft utilization. According to U.S. DOT Form 41 T100 Segment data (U.S. DOT, n.d.), ULCCs accounted for 30% of seat growth between the trough of the traffic decline in 2009 and 2019. Furthermore, ULCCs have tended to deploy more of their capacity at smaller airports. Between 2009 and 2019, 41% of ULCC capacity was deployed at medium hubs, small hubs, and nonhubs, compared with 19% for all other carriers. ULCCs added four times as much seat capacity at nonhub airports as all other types of carriers. The impact of carrier exit and entry on individual airports is examined in the next sections.

Loss of a Carrier or Dehubbing

The loss of a major carrier (or carrier dehubbing) can lead to a substantial loss of air service and with it, passenger (or cargo) volume. It can also lead to a fundamental change in the long-term course of traffic development. This section examines three examples of carrier loss or dehubbing:

- St. Louis Lambert International Airport, Missouri (STL) – loss of TWA hub (Exhibit 1).
- Cleveland International Airport, Ohio (CLE) – United Airlines dehubbing (Exhibit 2).
- Wilmington Air Park, Ohio (ILN) – loss of cargo carrier services (Exhibit 3).

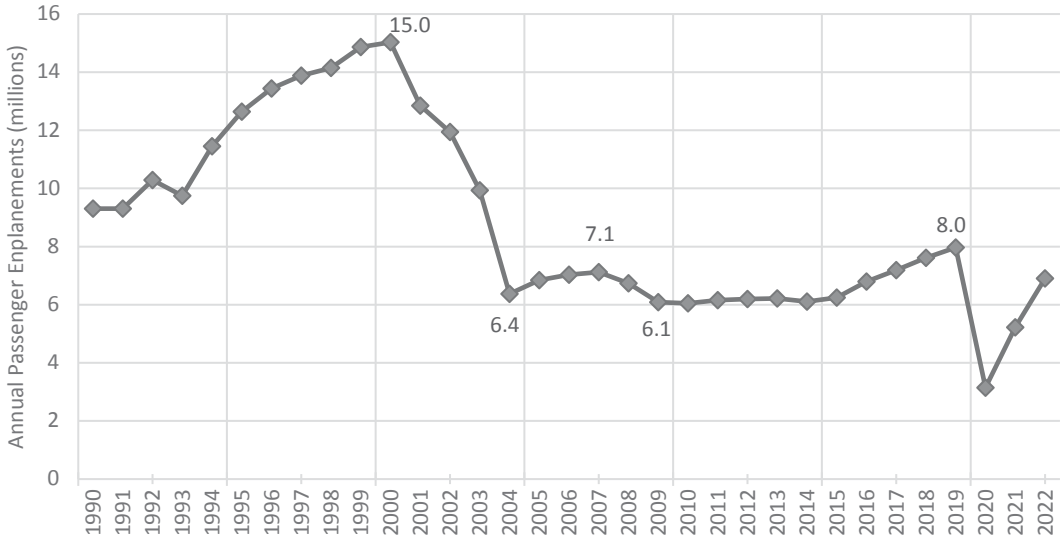


Exhibit 1. Passenger enplanements at STL, all carriers, 1990–2022 (U.S. DOT, n.d.).

In 1982, TWA named STL as its principal domestic hub resulting in rapid passenger growth at the airport during the 1980s. Despite entering bankruptcy in 1992 and 1995, TWA continued to grow traffic at STL, with the airport reaching 15 million enplaned passengers in 2000 (TWA accounted for 71% of passengers in that year). However, in 2001 TWA declared bankruptcy for the third time, which resulted in American Airlines acquiring TWA's assets. American Airlines initially indicated that it planned to keep STL as a hub, partially due to congestion at Chicago's O'Hare International Airport (ORD). However,

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with the severe downturn in traffic that followed 9/11, American Airlines closed its STL operations, focusing more on its main hub operations at ORD and DFW. As a result of cutbacks by American, passenger volumes declined to 6.4 million passengers by 2004 representing a 57% decrease in airport traffic since its peak in 2000. There was a modest recovery in traffic to 7.1 million in 2007 before the Great Recession led to further declines.

The loss of traffic following the TWA/AA dehubbing was largely through connecting passengers, as O/D traffic was far less affected. O/D traffic was 5.7 million enplanements in 2000, which declined by 12.1% in 2004 but recovered to 5.9 million in 2019 (slightly higher than in 2000). Much of this traffic growth since 2010 was driven by LCCs such as Southwest Airlines and Frontier Airlines. However, STL's traffic has fundamentally changed since the dehubbing with the majority of current traffic comprising originating passengers, whereas during TWA hub operations pre-2001, more than half of all enplanements were attributed to connecting passengers.

Cleveland International Airport (CLE)

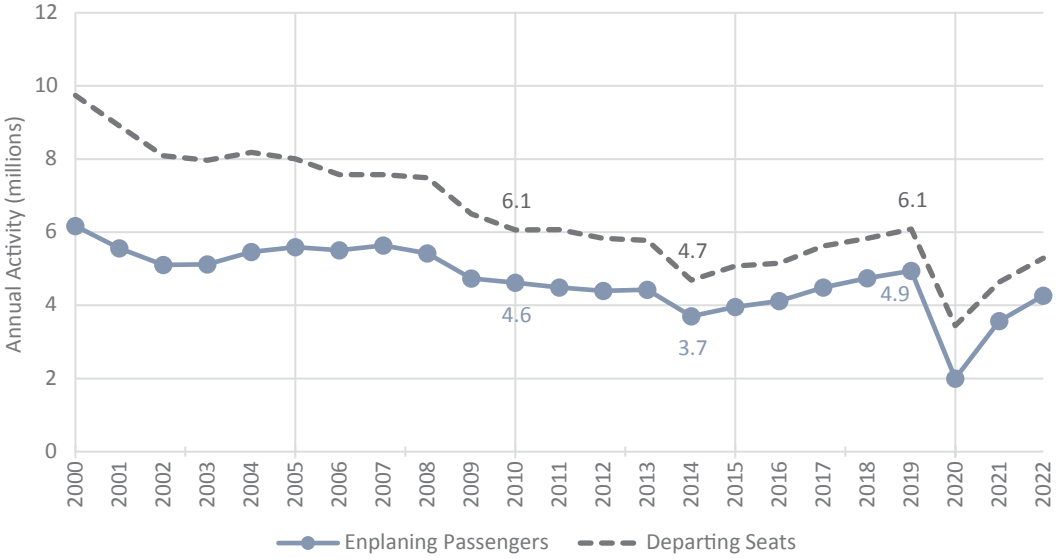


Exhibit 2. CLE onboard passengers and total seats, all carriers 2000–2022 (U.S. DOT, n.d.).

CLE had been a hub for Continental Airlines since the 1990s, with the airline accounting for 70% of passengers by 2010. When Continental merged with United Airlines in 2010, the merger agreement (with the Ohio Attorney General) required the airline to maintain 90% of its premerger capacity for 2 years. In 2014, the merged United Airlines announced the termination of CLE as a hub and a 60% service reduction (relative to 2010), citing long-term losses at the airport. Despite United's dehubbing of CLE in 2014, seat capacity actually increased every year until the COVID-19 pandemic. LCCs and ULCCs, such as Southwest Airlines, JetBlue Airways, Frontier Airlines, and Spirit Airlines, grew capacity at CLE as did Delta Air Lines and American Airlines. While O/D passengers grew at an average of 7.9% per annum between 2013 and 2019, or in aggregate 57%, connecting passengers declined by 89% over the same period.

After dehubbing CLE, the airport's Concourse D was closed, and United consolidated its operations into Concourse C. However, the carrier was still required to pay the airport $1.1 million per month in rent for the facility until 2027. Concourse D remained vacant

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as of 2022. The airport completed a master plan update in 2021, which recognized the need for the facility to focus on O/D traffic rather than connecting flows as it had done during the era of Continental and United's hub operations. For example, the potential for increased demands on parking facilities, roadways, and ticketing areas as a higher proportion of future traffic demand is expected to be from originating passengers, rather than connecting.

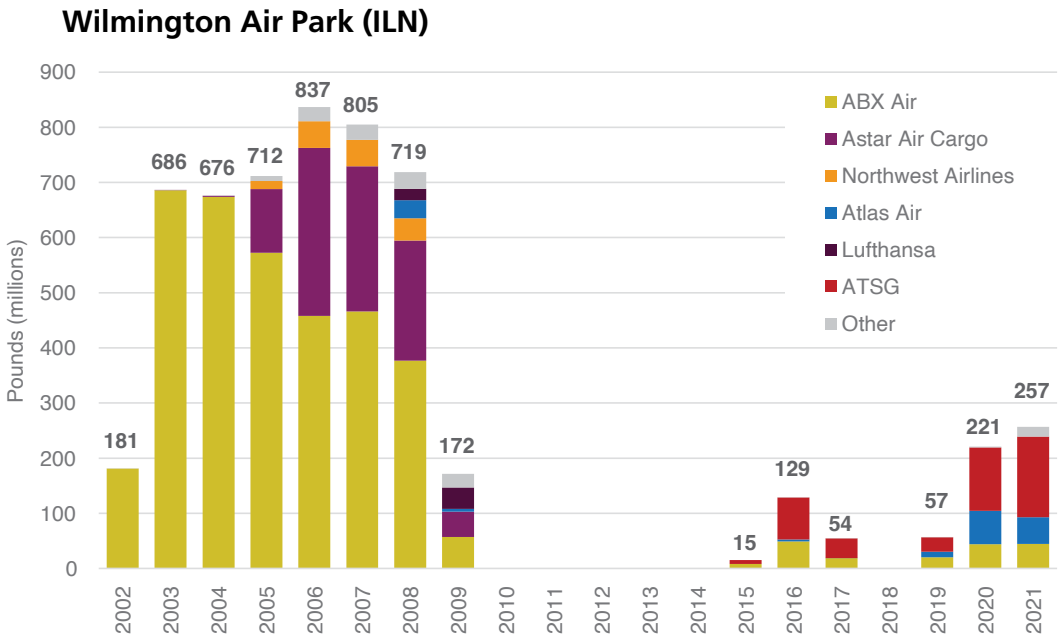


Exhibit 3. Cargo volumes by carrier, 2002–2021 (U.S. DOT, n.d.).

Previously an air force base, ILN, located in Southwest Ohio, was purchased by cargo airline Airborne Express in 1980, making it the first airline to purchase its own airport in the United States. In 2003, Airborne Express was acquired by German logistics and courier company DHL. DHL took ownership of the airport's ground operations, while the air business was spun off as a separate company called ABX Air, which operated services under contract to DHL. As a result, cargo volumes, aircraft activity, and employment at the airport expanded rapidly. In November 2008, DHL announced that it would shut down its express delivery service within the United States, including its ILN operations, due to severe financial losses in the U.S. domestic market. The withdrawal of DHL resulted in the loss of almost 8,000 jobs in Wilmington and the cessation of almost all cargo operations by carriers contracted to DHL such as ABX Air, Atlas Air, and Astar Air Cargo. DHL agreed to transfer ownership of the airport and facilities to Clinton County.

Air cargo at the airport was largely dormant until 2015 when a number of carriers (ATSG, ABX, and others) began operating flights for Amazon's e-commerce deliveries using Boeing 767s, which was later formalized under the Amazon Air brand. In early 2017, Amazon decided to move flight operations from ILN to Cincinnati/Northern Kentucky International Airport (CVG) where it was building its main hub, resulting in the loss of all operations in 2018. However, in 2019, Amazon Air returned to ILN to meet growing e-commerce demand as its new CVG hub facility was under construction, with a rapid expansion in services driven by the growth of e-commerce activity during the pandemic.

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Conclusion

In all three cases, the shock event was caused by the business decisions of the airport's largest customer. In 2019, 57% of commercial airports had over half their passenger seat capacity operated by a single carrier (including regional subsidiaries), including 43% of large hub airports. This creates a risk that the airport will be subject to the actions of that carrier creating a shock event not anticipated in air traffic forecasts. In the case of STL and CLE, airline dehubbing led to a decline predominately in connecting traffic, while O/D traffic was less affected. This does not mean that the local community is not impacted—connecting traffic allows carriers to support a higher level of service (frequencies and routes) than can be achieved from O/D traffic alone. All three airports sought to diversify their carrier mix and revenues, leading to a partial recovery in traffic, but undoubtably behind the trend anticipated before the shock event.

Carrier Entry or Rapid Traffic Growth

Rapid carrier development at an airport, especially smaller airports, can also be a shock event that while being positive for business development can present challenges in facility management and design. This section examines the impacts of rapid traffic development at three airports:

- Bellingham International Airport, Washington (BLI)—Allegiant and Southwest Airlines (Exhibit 4).
- Orlando Sanford International Airport, Florida (SFB)—Allegiant Airlines (Exhibit 5).
- Williston Basin International Airport, North Dakota (XWA) (Exhibit 6).

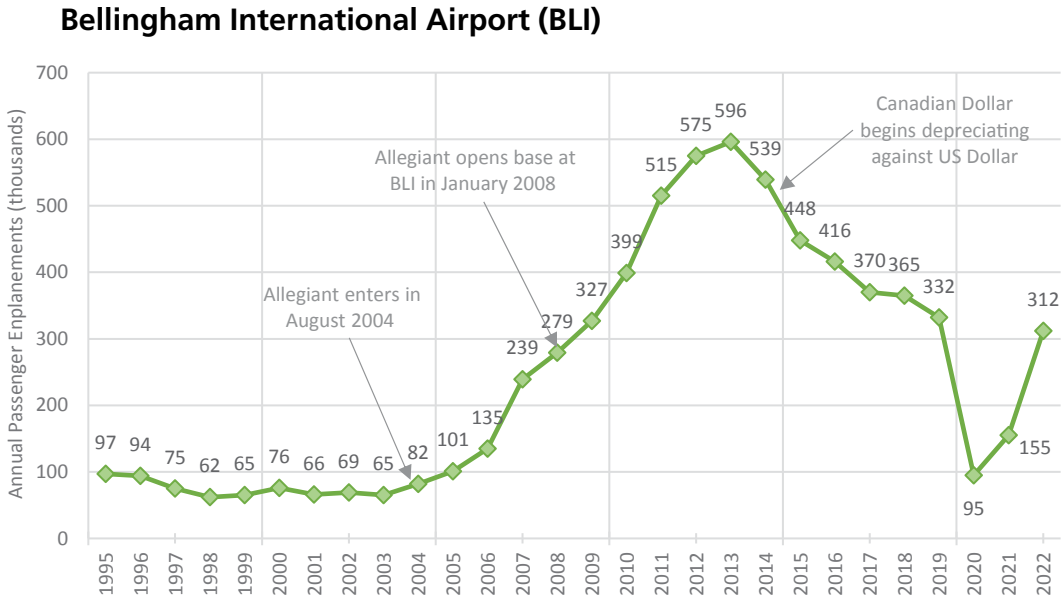


Exhibit 4. Passenger enplanements at BLI, all carriers, 1995–2022 (U.S. DOT, n.d.).

Before 2004, passenger enplanements at BLI averaged around 75,000 per annum, with the majority of capacity being operated by Horizon Air (the regional subsidiary of Alaska Airlines) with service to Seattle-Tacoma International Airport. In August 2004, ULCC Allegiant Air entered the market at BLI, with service to Las Vegas. As Allegiant grew its operations at BLI it opened its sixth base at the airport in 2008. Passenger enplanements grew from 64,975 in 2003 to 596,383 in 2013, a ninefold increase and an average growth rate of 24.8% per

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annum, with service to 10 destinations. A large proportion of BLI passengers originated from Canada, as the airport was located 21 miles south of the border with Canada and offered considerable fare and tax savings relative to Canadian airports (Vancouver and Abbotsford in particular), especially with a historically strong Canadian dollar over this period (versus the U.S. dollar). As the Canadian dollar started to weaken in 2014, passenger volumes at BLI started to decline. Allegiant reduced capacity in response, and by 2019 passenger volumes at BLI had declined 44% relative to the peak in 2013.

To address rapid traffic growth when Allegiant Air entered the market, BLI accelerated its expansion program, completing a $29 million resurfacing project for the runway, followed by a terminal expansion project completed in 2014, which tripled the size of the old terminal building. The airport sought to remain competitive and attractive to both new and existing carriers by maintaining low-cost levels and offering incentive programs to carriers. In November 2021, Southwest Airlines began daily direct service from BLI to Oakland, California, and Las Vegas, and by the end of 2022 was offering service to eight destinations.

Orlando Sanford International Airport (SFB)

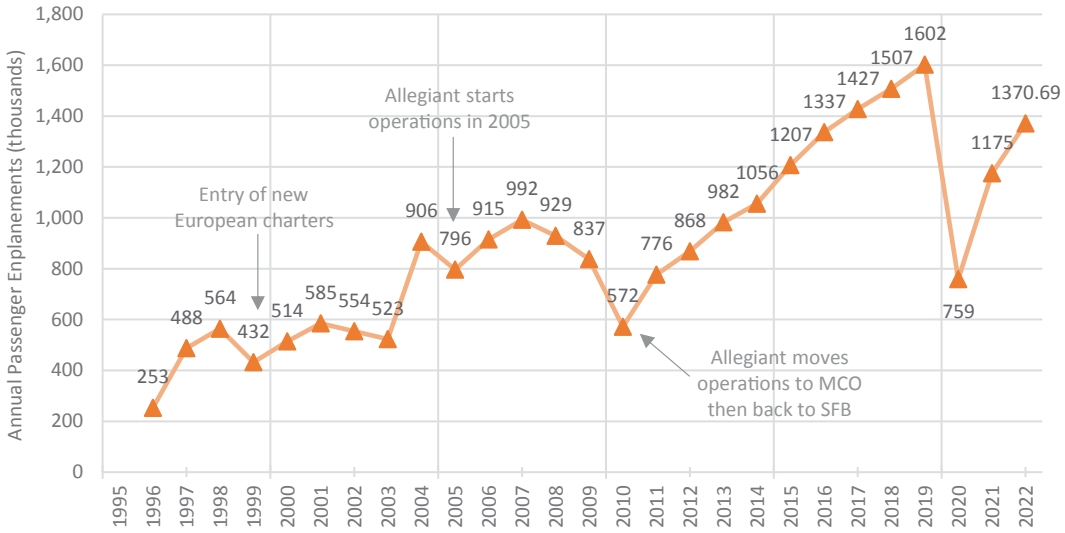


Exhibit 5. Passenger enplanements at SFB, all carriers, 1995–2022 (U.S. DOT, n.d.).

Previously a military and then GA facility, SFB started to receive commercial air service in the mid-1990s following the building of a new passenger terminal. In the late 1990s and early 2000s, commercial operations largely comprised charter services from Europe (particularly the UK). Allegiant Airlines started operations in 2005 and rapidly grew operations (offsetting the loss of some international charter services) and by 2009 accounted for 70% of passengers at the airport. In 2010, Allegiant announced it was moving all of its operations to Orlando International Airport (MCO) in part to compete with AirTran Airways at the larger airport. However, the airline moved all operations back to SFB by February 2011 due to customer feedback. Since returning in 2011, the airline almost tripled its seat capacity at SFB by 2019, operating over 70 routes and accounting for 93% of seat capacity at the airport. Allegiant has been almost the sole driver of growth, with capacity by other (largely international) carriers declining by 61% between 2011 and 2019 due to carriers failing or moving to MCO. Passenger volumes declined 53% in 2020 due to the pandemic, less than the national average of 61%.

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Allegiant further expanded its presence at the airport in 2016 by building its East Coast Training Center at Sanford. The Center can train 150 pilots, 500 flight attendants, and 100 mechanics annually on a full range of flight simulators, cabin trainers, and classroom resources for A320 and MD80 aircraft. Sanford Airport Authority has sought to diversify its carrier mix and in late 2021 attracted two Canadian LCCs, Swoop Airlines and Flair Airlines, to begin service between Sanford and Canadian cities in late 2021. Given the depth of the Orlando area destination market, the airport has sought to respond to opportunities as more international cities became available prospects for air service development.

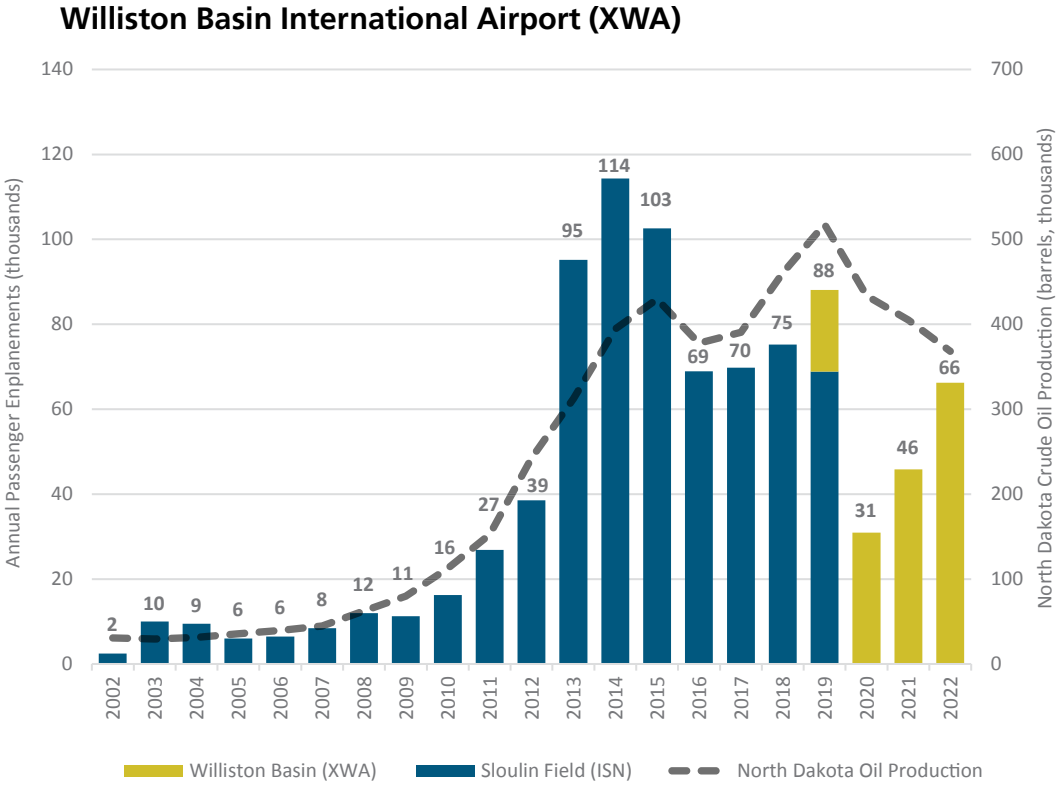


Exhibit 6. Passenger enplanements at Sloulin Field International Airport (ISN) and XWA, 2002–2022 (U.S. DOT, n.d. and the U.S. Energy Information Administration).

XWA is a new airport that opened in October 2019 to replace Sloulin Field International Airport (ISN) as the latter was unable to handle the increased traffic in the region.

Williston, North Dakota, became a center for the shale oil industry following the discovery of the Parshall Oil Field in 2006, part of the Bakken Formation. As oil industry activity increased between 2006 and 2014, passenger traffic at ISN increased rapidly, peaking at 114,325 in 2014, a tenfold increase over 7 years. The initial growth was supported by regional carrier Great Lakes Airlines, operating six routes using 30-seat Embraer EMB 120s. In 2012, United Airlines entered with service to Denver and later Houston (operated by a regional subsidiary). Delta Air Lines also started service in 2012 to Minneapolis—St. Paul, operated by a regional partner. Due to the significant increase in demand, ISN was not able to cope with the additional activity and capacity, and plans were implemented to construct a new airport to meet the demand of the growing region and energy industry.

XWA construction started in 2016 as a replacement for ISN due to the latter's capacity and operational constraints (building a new airport was determined to be cheaper than

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upgrading ISN). Traffic volumes at the airport declined from 2014 due to reduced oil industry activity (the chart above shows a strong correlation between oil production and air traffic). Great Lakes Airlines pulled out from the airport in 2014 citing a pilot shortage and both United Airlines and Delta Air Lines reduced frequencies at the airport, with United cancelling the Houston service in 2016. The COVID-19 pandemic resulted in a 65% decline in passenger traffic in 2020, with United and Delta operating reduced service levels in line with passenger traffic but have continued services into 2021. ULCC Sun Country Airlines started twice weekly service to Las Vegas in September 2021.

Conclusion

All three airports faced the challenge of growing facilities as a consequence of rapidly growing demand while also being cognizant of the risk from competing airports, economic conditions, and carrier decisions that could result in fluctuating traffic volumes. Small airports are particularly challenged by this form of shock event as introductions of new carriers or substantial increases in service can move airport activity levels well above previously forecasted long-term trend levels. While traffic growth is generally welcomed, these airports do not necessarily have the facilities to accommodate rising traffic volumes, leading to the need for considerable capital development.

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PART II

Approaches to Addressing Shock Events

The research team conducted an extensive literature review on identifying, modeling, and managing shock events. The literature review examined a range of industries and academic disciplines, including the aviation and transportation industry, risk and decision analysis, management theory, the financial and insurance industries, as well as other network industries such as telecommunications and energy generation and distribution. The research team also consulted previous work commissioned by ACRP on risk, forecasting, and planning for the aviation industry to build upon the base of knowledge already developed in areas relevant to this research project.

The findings are based primarily on a review of industry publications and scholarly journal articles. It also leverages the collective knowledge and experience of the research team in the treatment of shock events. In some instances, the methodologies examined are related to uncertainty more generally rather than shock events specifically; these have been included as they can apply to shock events as well.

Details of the documents reviewed, including a summary of the contents, are provided in Appendix B.

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CHAPTER 6

Defining and Identifying Shock Events

Defining Shock Events

Much of the literature in this area uses a variety of terminology and definitions regarding shock events. However, in general, there is broad agreement that shock events are those whose likelihood, timing, and magnitude are hard to predict or even anticipate but which when they occur, impact business in significant or profound ways (positive or negative). These have also been referred to as "high-impact, low-probability events" (Lee, Preston, and Green 2012), "unknown unknowns" (a term popularized by former Secretary of Defense Donald Rumsfeld during a press conference in 2002), and "black swans" (Taleb 2007). Taleb codified this conceptualization of a black swan as an event or world state that cannot be (or is not) imagined and therefore is not considered even as a possibility. Feduzi and Runde (2014) contrast this with known unknowns, which is an uncertainty that has some real possibility of occurring. They provide an important caveat that "an event experienced as a black swan by one person may not come even as a mild surprise to the next person" based largely on the information available to an individual (Feduzi and Runde 2014, 274).

There is broad agreement that shock events are those whose likelihood, timing, and magnitude are hard to predict or even anticipate but which, when they occur, impact business in significant or profound ways (positive or negative).

In fact, Taleb (2007) argues that often we are dealing with grey swans—events for which there is little information, but at some level, we can anticipate or be aware of them. Taleb has argued that COVID-19 was not a black swan event and was entirely predictable (Avishai 2020).

As an example of the difference in information, and by some measure imagination, in whether an event that unfolds is a black or a grey swan (a known unknown), Feduzi and Runde (2014) discuss the 9/11 terrorism event. They argue that for a great many people, this was a black swan surprise event, but to North American Aerospace Defense Command (NORAD) this was a known unknown as they had simulated the potential impacts of an attack like 9/11 on targets including the World Trade Center and the Pentagon. This is an important distinction between what may be considered a shock or surprise event, particularly deeming something an unknown unknown, as the ability to contemplate the possibility of uncertain events can transform it into a known unknown. The director of the National Commission on the Terrorist Attacks upon the United States (Kean and Hamilton 2004) called the misreading of known terrorist group activities (previous hijackings, flying lessons, etc.), "a failure of imagination."

Other authors have argued that true black swans may not exist in the sense that the market or scenario in which a surprise event occurs may be created by the setting in which it occurred. Walter (2020) posits that financial black swans, such as market crashes or recessions, are "illusory" as they are a construction of the regulations of the market themselves, among other factors. Walter (2020) goes on to suggest that how the financial sector measures and quantifies risk, with regulations intended to minimize that risk, creates opportunities for shock events but that

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those should not be considered as black swan events as they arise from the design of the market itself. This distinction is useful as it reminds the reader that shock events can be from known or imagined sources, and it is through the deeper examination of risk and risk events can the unknown unknowns be revealed as potential events.

Wucker (2016) takes this one step further, referring to "grey rhinos"—high-impact but high-probability events that we are blinded to or actively choose to ignore. Wucker suggests that the 2007–2008 Global Financial Crisis was one such example, where there were plenty of warnings (and plenty of people warning) that the financial system was at high risk, but policymakers chose not to address them until the crisis hit (the rhino charged, in Wucker's analogy). Wucker argues that some shock events are missed or downplayed not because of a lack of information but due to an unwillingness to consider uncomfortable outcomes.

Shock events may be becoming more frequent due to such factors as the internet and information technology, globalization, and climate change.

Aggarwal and Bohinc (2012) suggest that the underlying frequency and impacts of shock events are increasing over time due to such factors as the internet and information technology, globalization, and climate. The scale and scope of shock events can be linked to how increasingly interconnected the global society and the global economy have become. With less interconnection, the spread or speed of the consequences of a shock event may be less and more local. Aggarwal and Bohinc propose looking at indirect as well as direct impacts from black swan events and as an example, put forth terrorism-related supply chain disruptions. Therefore, individuals and organizations dealing with risk must project trends beyond simply looking at historical data and how the future developments of major trends will affect the probability and magnitude of future potential shock events. A document by Munich Re (2018), a reinsurance company, discusses trends in shock events, particularly how the frequency and impact of natural catastrophes appear to be on the rise due to the impacts of climate change. Through analysis of loss statistics, this insurer has "demonstrated the plausibility of climate change already influencing some types of events in certain regions" (Munich Re 2018, 23) in its work to reduce the potential risk loss from catastrophic events by better informing their actuarial models about future risk potential. In its continued research, Munich Re has looked at how the probability of catastrophic natural events, as well as the number of claims and value of damage, has grown as climate change progresses and included trend projections in its risk analysis and management.

Risk researchers and actuaries are always trying to move events that initially seemed like shocks or even black swan events into standard risk calculations. Once a shock or surprise event occurs, ex post evaluation of its impacts and causes can allow what may have been previously considered an unknown unknown to become a known unknown. It is argued in the literature that greater awareness of risks and uncertainties could improve an individual's or organization's ability to postulate new risks and shocks and potentially convert what would otherwise be considered a black swan surprise event into a known uncertainty, even where information is limited.

Identifying Shock Events

Individuals and organizations tend to err on the side of overconfidence, particularly when projecting future activities.

There is an abundance of empirical research that indicates humans are too often surprised. Studies have shown that when asked to estimate the 10th and 90th percentile ranges (or some other extreme) of some distribution, the true range turns out to be considerably outside of the estimated range (Morgan, Henrion, and Small 1990). Individuals and organizations tend to err on the side of overconfidence, particularly when projecting future activities. This has been explored in the realm of airport traffic forecasting

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in the United States (Suh and Ryerson 2019) from a variety of causes, including the underrepresentation or appreciation for the potential of significant contractions in aviation traffic. An optimism bias, or overconfidence in the ability to estimate probabilities, will tend to lead to more positive outlooks and generally downplay the probability or impacts of negative events.

The gambler's fallacy, also known as the Monte Carlo fallacy, is another psychological phenomenon that creates practical challenges when attempting to model and predict shock events. In this psychological phenomenon, people may be led to believe that if a certain event or outcome occurs more frequently than in the past, then that outcome is less likely to occur in the future, despite that the outcome of the event is not conditional on previous outcomes (i.e., that the probability of any given outcome of the event is statistically independent). Psychologists have analyzed this phenomenon, which frequently uses gambling as an example, using games involving independent probabilities like dice rolls or spins of a roulette wheel, and have extended the examination of the phenomenon to behavior such as observing an athlete's performance and predicting future outcomes based on whether they have happened recently or not. As a form of cognitive bias, the gambler's fallacy can be difficult to overcome, especially when evaluating the likelihood of a future event or outcome when it is not fully understood whether or not the event's probability is independent or conditional on past outcomes or some other force.

Methods for eliciting or uncovering the potential risks that seek to avoid or minimize human biases are raised in much of the literature on risk analysis and strategic planning under uncertainty. *ACRP Report 76* summarizes several approaches—such as Delphi, statistical groups, nominal groups, and unstructured interacting groups—for eliciting such information from experts and stakeholders (Kincaid et al. 2012). The Delphi method, also known as estimate-talk-estimate technique, is one of the more commonly used methods for forecasting future outcomes and identifying emerging trends. Developed by the RAND Corporation in the 1950s, the Delphi method is a systematic and qualitative method of forecasting by collecting opinions from a group of experts through several rounds of questions. It consists of providing a questionnaire to relevant selected experts to give their opinion on the topic at hand (in this case, potential shock events that could impact an airport and the likely results of those shock events). After the experts answer the questionnaires, the facilitator collects all the answers and hands out a summary report of the answers to each expert. The experts then review the summary report and either agree or disagree with the other experts' answers. Another questionnaire is completed that allows the experts to update their opinions and challenge the opinions of others. The Delphi method becomes complete when a consensus of forecasts is achieved via this iterative process.

Goodwin and Wright (2010) argue that Delphi and similar methods are less effective in identifying shock or rare events, as they remain heavily influenced by past experience and cognitive biases (availability and linearity biases). These methods rely on consensus views, while the authors suggest in the field of rare events, more weight should be given to minority opinions that challenge the consensus. They argue for the use of scenario planning that is less focused on predicting the future and allows for less mainstream but still uses plausible outlooks for the future to be considered and played out. The authors acknowledge that even these methods are problematic for aiding the anticipation of rare events but may aid in the development of protective strategies. The authors propose the use of "devil's advocacy and dialectical inquiry," whereby the Delphi approach is modified to include groups who are challenged with offering contradictory views, and there is less focus on consensus.

In a similar vein, Bonabeau (2008) proposes the use of "augmented paranoia." Writing in the context of predicting terrorism and related activities, he argues for a mix of methodologies: outreach using nontraditional experts from other fields, additive aggregation; exploiting the "wisdom of crowds" and self-organization; and working outside of the organizational hierarchy

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Identifying shock events requires being open minded and a creative viewpoint to considering the risks (and shock events) facing an organization, including uncomfortable and extreme scenarios.

to avoid groupthink and thought hijacking. ["Wisdom of crowds" is the idea that large groups of people are collectively smarter and better informed than individual experts (Surowiecki 2004).] Walker et al. (2019) propose a number of methodologies for decision-making under deep uncertainty that do not focus on predicting the future but rather on considering the range of possible futures and ensuring that decision-making is robust enough to support that range of possible futures.

The goal of all these approaches is to ensure an open-minded and creative viewpoint when considering risks (and shock events) facing an organization, including uncomfortable and extreme scenarios. Paté-Cornell (2012) argues that many previous supposed black swans were no such thing but rather failures of analysis and imagination, citing examples such as the Fukushima tsunami and nuclear accident, the Columbia space shuttle explosion in 2003, and the 9/11 terrorist attack.

Risk Identification in Aviation

For most airports, air traffic forecasts play a key role in airport planning and development. The standard master plan approach can be characterized as follows:

- Determination of the forecast, and
- Selection of a plan that best suits this forecast.

Even broader airport strategic planning is guided by air traffic forecasts, although in this case, it can be argued that the relationship is more two-way as the airport strategic plan can impact future traffic through marketing and business planning.

Therefore, air traffic forecasts can play a key role in the identification of risk factors. As documented in *ACRP Synthesis 2: Airport Aviation Activity Forecasting* (Spitz and Golaszewski 2007) and master planning guides such as the *Airport Development Reference Manual*, edition 12 (IATA 2022), forecasting generally involves the identification of key drivers or factors affecting traffic development, such as economic growth, fuel prices, airline behavior, and government policy. Uncertainty about the future development of these factors can be considered by changing their assumed values in the forecasting process to produce low- and high-scenario forecasts, as documented later. Arguably, this remains the most common approach to addressing uncertainty in air traffic forecasts. However, this approach does not formalize the identification of risk and may miss out-of-model factors such as shock events.

ACRP Report 76 (Kincaid et al. 2012) proposed the use of a risk register, whereby information is elicited from airport management and other stakeholders on the risks and uncertainty facing the airport. Using a combination of judgment-based and data-based methodologies, risks (including shock events) are identified and evaluated in terms of their likelihood (or probability) of occurring and the anticipated magnitude of impacts (positive or negative). This information is cataloged in a risk register that can feed into subsequent tasks (such as forecasting) and ideally be updated regularly. These risks can be represented in a graphical form, as shown in Figure 20. Note that the register includes a shock events category with "Pandemic" indicated as one of the risks. The probability and scale of impact are relatively low, likely reflecting experience with past pandemics such as SARS in 2013, which had minimal and short-lived impacts on U.S. airports.

Examples of this approach to strategic risks and shock events in the aviation industry appear fairly uncommon, certainly in the public domain. One example is the "Opportunities and Risks Report" produced by Munich Airport as part of its *Integrated Annual Reports* (covering finance, environment, social, employees, and infrastructure) (Munich Airport 2018, 2019). The risk

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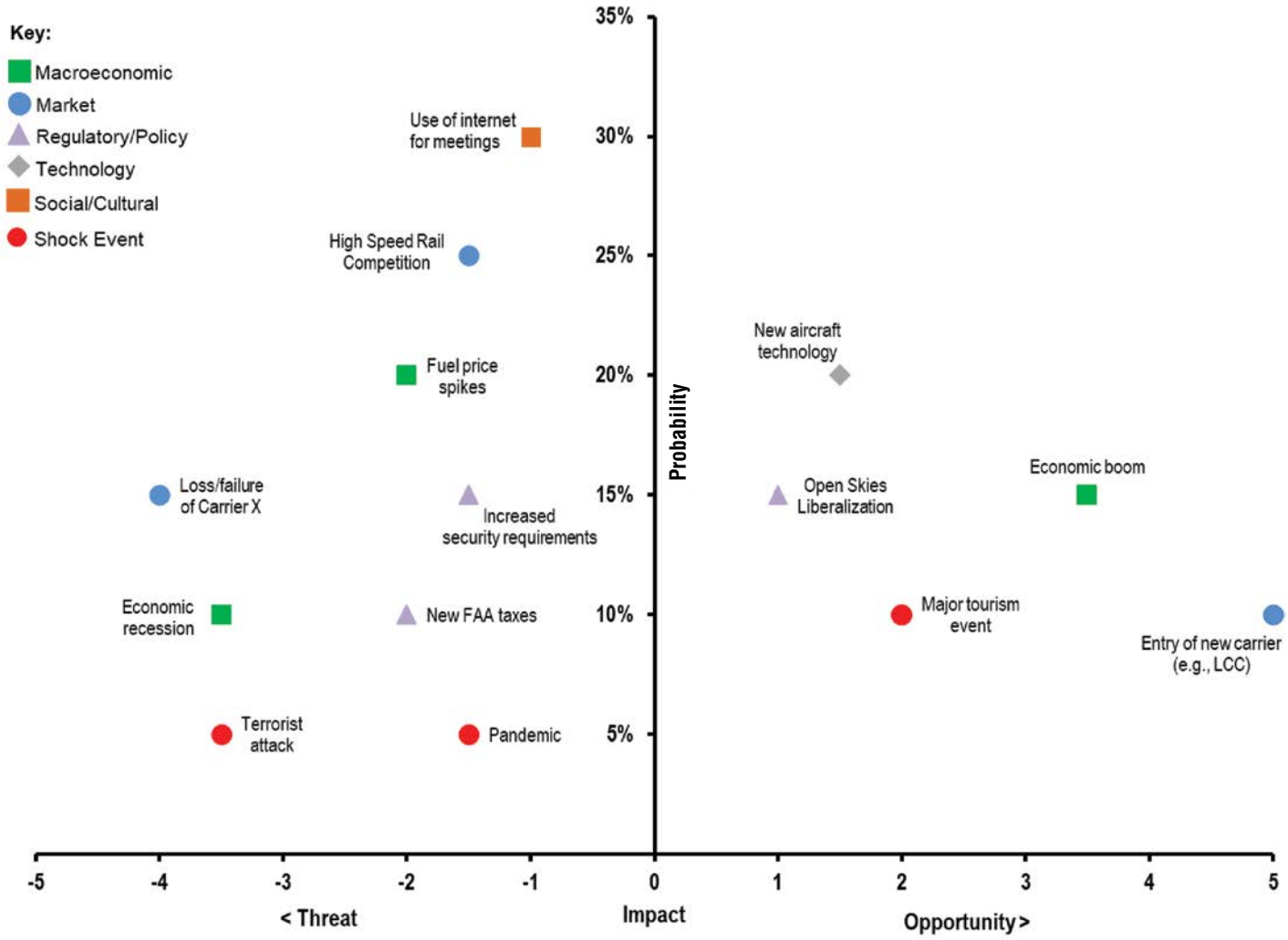


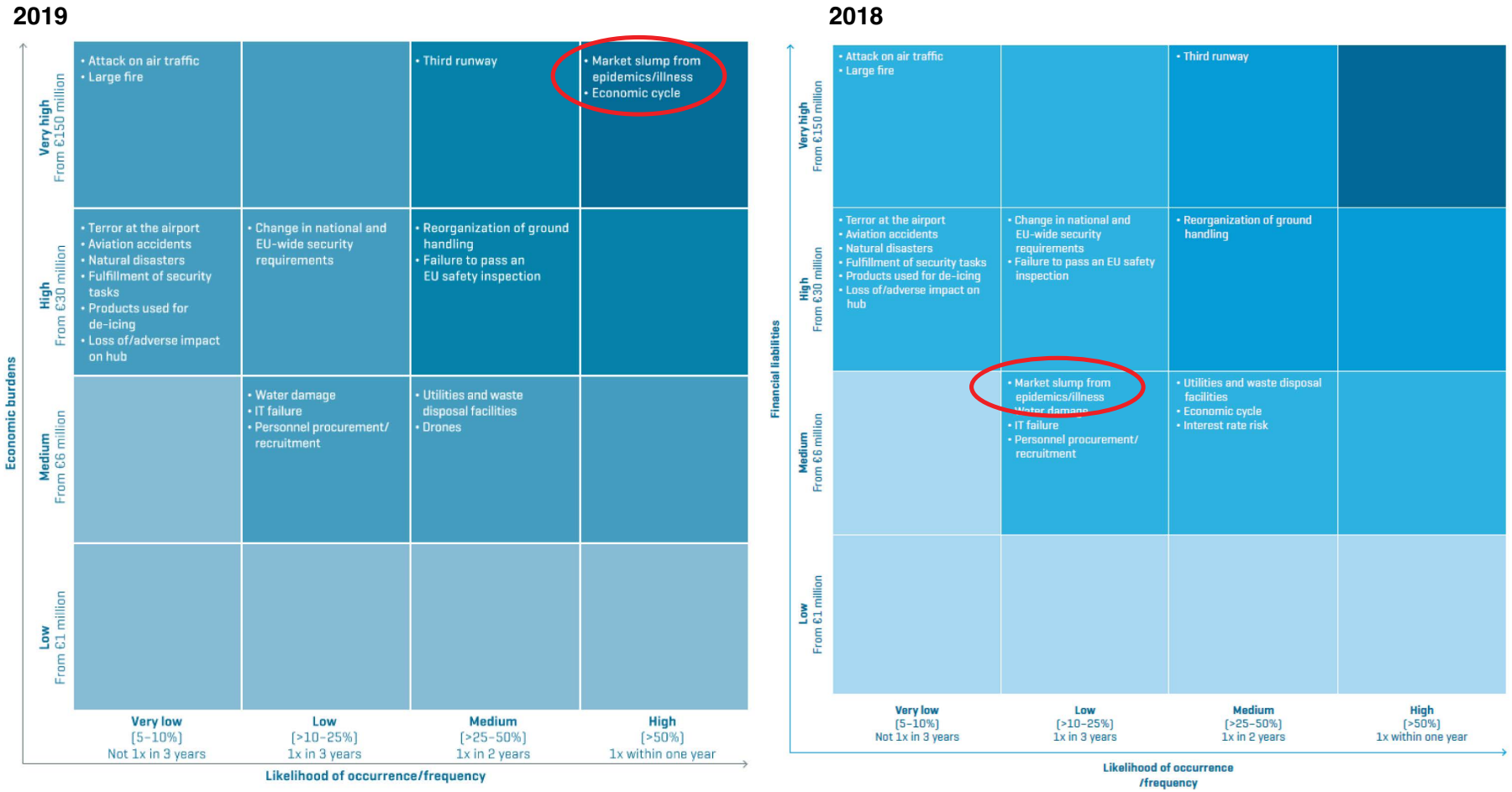
Figure 20. Illustrative example of a graphical representation of the risk register (Kincaid et al. 2012).

assessment covers various categories of risk: force majeure (shock events such as terrorist attacks, pandemics, and natural disasters), market risk (carrier loss and economic cycle), operating risks, legal risks, and financial risks. Figure 21 shows the risk assessments done in 2018 and 2019. The 2019 assessment was started in late 2019 and completed by April 2020 and so it incorporates the emerging COVID-19 pandemic. As a result, "market slump from epidemics/illness" changes from a medium likelihood/medium risk factor in 2018 to a very high-impact/high-likelihood in 2019.

For each risk factor, the report indicates the countermeasures the airport has in place and evaluates how these countermeasures change the risk outlook facing the airport (the "net risk"), with some risk factors being removed from the matrix. In 2018, the countermeasure for "market slump from epidemic/illness" indicated that "Due to a relatively high fixed cost ratio, Munich Airport's ability to react to market downturns is limited."

The focus of this research has been on strategic risks and shock events rather than those related to safety, security, or operations. Airports and the aviation sector more generally have sophisticated and well-established approaches to safety and operations, as documented in *ACRP Report 131: A Guidebook for Safety Risk Management for Airports* (Neubauer, Fleet, and Ayres 2015) and *ACRP Synthesis 71: Airport Safety Risk Management Panel Activities and Outcomes* (Ayres and Parra 2016).

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Figure 21. Overview of gross risks at Munich Airport, 2018 and 2019 (Munich Airport 2018, 2019).

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Similarly, there are risk management processes for capital and maintenance projects as documented in *ACRP Report 116: Guidebook for Successfully Assessing and Managing Risks for Airport Capital and Maintenance Projects* (Price 2014).

Many U.S. airports, particularly large airports, do have dedicated risk managers and departments and have adapted enterprise risk management (ERM) practices. Airports Council International-North America has a Risk Management Committee with representatives from a large number of airports. The focus of this work is fairly broad, ranging from loss control and prevention, airport insurance agreements, regulatory requirements, and various safety issues to more strategic aspects related to the air market and the economy. For example, *ACRP Synthesis 115* (Murphy et al. 2021) documents emergency plan practices at airports, including a case study of Charlotte Douglas International Airport (CLT) that developed a Communicable Disease Response Plan in 2018 through planning meetings with various stakeholders, including the Centers for Disease Control and Prevention (CDC). The result was a plan that could be activated or triggered by sick passengers at CLT who meet specific characteristics. The aim of the plan was containment of a communicable disease rather than the wider implications of a pandemic outbreak. Some of the airports interviewed for Part I indicated that they had developed business contingency plans or emergency plans to address how the airport would continue to operate given potential disruptive situations. While these plans did not consider a pandemic specifically, some did consider declines in traffic volume. However, in the context of long-term forecasting and planning, the evaluation and incorporation of shock events are generally not addressed in a fairly formalized manner.

Airports typically have well-established procedures to identify and devise mitigation strategies for operational risks. Airport operations manuals will typically include response strategies and mitigation activities for operational risks such as fires, aircraft accidents, security incidents, power losses, and so forth. ACRP has published two guidebooks on this topic including *ACRP Report 131* (Neubauer, Fleet, and Ayres 2015) on risk management for airports' safety management systems, and *ACRP Report 74* (Marsh Risk Consulting 2012) on the topic of applying ERM at airports. Similarly, there are risk management processes for capital and maintenance projects as documented in *ACRP Report 116* (Price 2014). These guidebooks provide valuable information to airports on how to identify and manage their risks, though they do not directly address the issue of accounting for risks associated with shock events. While *ACRP Report 74* (Marsh Risk Consulting 2012) does provide a framework for identifying and managing both operational (e.g., fires, security, accident response) and strategic risks (e.g., loss of a carrier, recession, challenges to workforce) its focus is on how to start and integrate an ERM process into an airport's business activities but not, unsurprisingly, how to specifically integrate shock events into processes like forecasting or planning. It has also been argued that ERM generally focuses on compliance and known or familiar risks and tends to ignore or give little weight to rare or shock events (Le Merle 2011).

Airports typically have well-established procedures to identify and devise mitigation strategy for *operational* risks.

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CHAPTER 7

Understanding and Modeling Shock Events

One of the primary challenges with predicting or modeling shock events is that there is often little historical data from which to draw predictive estimates.

Shock events are difficult to incorporate into forecasting and the planning, financial, and operational processes that rely on those forecasts. Many forecasts, including those produced within the aviation sector, are largely based on a continuation of business as usual with gradual, incremental change over the long term.

One of the primary challenges with predicting or modeling shock events is that there is often little historical data from which to draw predictive estimates. Before the start of 2020, many would be able to see the potential for another pandemic to impact the U.S. or global aviation industry using past pandemics, such as SARS, as examples of impacts and effects; however, the magnitude of the COVID-19 pandemic proved challenging to predict. When looking back through modern history for a pandemic of similar magnitude impacting the United States and the world, the 1918 influenza pandemic is the most similar event. However, historical analysis is constrained by the rapid evolution of technology, both medical and aviation. In 1918, commercial aviation was in its infancy in the United States. Medical knowledge in 1918 had not yet reached a point to recognize viruses, so the 1918 pandemic's relevance for aviation was easy to dismiss before 2020. Thus, there were few direct impacts to draw on from the influenza pandemic to compare it with the COVID-19 pandemic (the global spread of the 1918 flu was facilitated by the maritime movement of troops toward the end of the First World War). While this is an extreme example, it does highlight the difficulty that historical events cannot always be used to predict or model future potential shock events.

This challenge of the lack of historical data or inability to apply historical analysis has been noted by various authors. Gilbert, Habibi, and Nadim (2016) assert that classical statistical approaches to managing black swan events have limited value due to there being little or no historical data with which to analyze these events, by definition. These authors instead propose a new "Decision Entropy Theory" framework specifically to work in low-information environments and the use of Bayesian analysis to quantify previously unknown risks. Arney et al. (2013) highlight this challenge in the lack of historical information where surprise and the unknown are common in the paper's military strategy setting; they propose an evolution of scenario planning and simulation analysis combining rare event and tipping point analysis as ways to evaluate and explore surprise events when there is little historical information. From these examples, the literature points to analysis and, fundamentally, the use of creativity and imagination as the means to counter the lack of historical information on unknown shock events to bring them into the realm of the known, though still highly uncertain.

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Evaluating Shock Events

Lack of historical data and inherent human biases make predicting shock events difficult and attempts to do so are rarely accurate. Attempts to predict the occurrence and timing of shock events, such as earthquakes or terrorist events, require an impractical level of information and computation (McMorrow 2009), are largely unproven, and have little applicability to airport forecasting.

While predicting shock events may not be practical, there are methods to better understand how shock events may arise, develop, and impact a business.

While predicting shock events may not be practical, there are methods to better understand how shock events may arise, develop, and impact a business. There are established methods for risk assessment that can be applied to shock events impacting airports. For example, bow tie analysis is a structured approach to risk identification and management that gives a summary of all plausible scenarios that could exist around a certain shock event or risk and also captures what an organization does to control or mitigate those scenarios. The diagram is shaped like a bow tie as shown in Figure 22, with factors contributing to the shock event on the left side and consequences and reactions of the event on the right side. The advantage of this approach is that it provides an overview of complex scenarios through a simple visual explanation of a shock event. The UK's Civil Aviation Authority has used this approach in understanding and managing safety and human factors risks (UK Civil Aviation Authority, n.d.). Another visual approach is influence diagrams that can be used to illustrate and understand risk and shock factors, depicting key elements including decisions, uncertainties, and objectives (Kincaid et al. 2012).

Feduzi and Runde (2014) propose a process of "eliminative induction" to elicit and evaluate potential shock events and decision-making. The authors lay out a method for collecting evidence and generating hypotheses about future states of the world, which is valuable for risk management but is posed as a method to potentially reduce exposure to shock or black swan events by "bringing into light states of the world that might not have been uncovered otherwise" (Feduzi and Runde 2014, 20). These authors also suggest that their method helps counteract confirmation bias in the search and ideation of future states of the world, particularly with respect to looking beyond future states or events, which might be a positive for an organization. In the context of shock events for the aviation industry, the conceptualization and discussion are more often negative than positive from a forecasting perspective; still, the methodology proposed

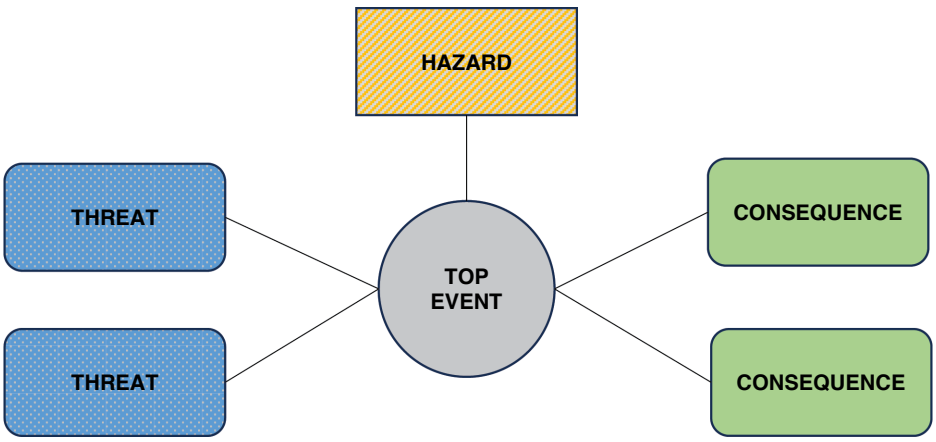


Figure 22. Visual diagram of a bow tie analysis structure.

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does provide at a conceptual level, a means to look beyond expected outcomes to attempt to see the unexpected.

Gilbert, Habibi, and Nadim (2016) propose a new framework for risk management, called Decision Entropy Theory, to help create better information for decision-making when there is limited or no information. The framework applies to the question of shock events as it is specifically designed as a starting point to assess probabilities and manage risks when there is little or zero prior information. While the mathematical approach applied is relatively advanced, the basic methodology involves exploring alternatives to existing risk-mitigation strategies or risk events—even if there is no meaningful knowledge of how likely a risk event may occur—and considering the impact of the existing or preferred methods of managing risks. So even if one cannot put an exact probability on a shock event or a known unknown happening, being aware of a potential event is still helpful in developing risk-mitigation strategies and plans. The authors highlight the importance of exploring how the possibilities for shock events and risks can help inform preferred alternatives and strategies for managing risk.

Kim (2012) develops a methodical process of evaluating potential shock events through a classification matrix. The process is designed for a project management audience, but the conceptual process is similar to other risk identification strategies discussed in other literature [e.g., *ACRP Report 76* (Kincaid et al. 2012)]. The classification approach proposes to extend the analysis of known unknowns to consider what unknown unknowns there may be throughout a project's lifetime. The author provides sample metrics for how unknown shock events may be categorized providing a useful guide to conceptualizing shock events and considering what kind of shocks could befall a project. The author provides brief case studies applied to two environmental disasters to major infrastructure/energy installations that are relatable to the capital-heavy and network-focused aviation sector.

Modeling Shock Events in Aviation Forecasting

This section examines methods used in air traffic forecasting that can be used to incorporate shock events. These include methods that are used to consider risk and uncertainty more generally.

High- and Low-Scenario Forecasts

High and low scenarios do not tend to incorporate the potential impact of shock events or large-magnitude risks in their scenarios.

One of the most common methods to model alternative forecasting scenarios is the generation of high and low scenarios developed by modifying forecasting assumptions to produce a more optimistic or pessimistic outlook, respectively. This procedure can easily be incorporated into standard forecasting techniques, and its ubiquity has led to the concept of a triad of scenarios with a central base case (also referred to as the medium or most likely forecast) and two alternatives of higher and lower growth or traffic levels. This is widely understood and intuitive for forecasters, planners, and other stakeholders.

High and low scenarios are developed by varying assumptions or inputs to the forecast, such as higher or lower economic growth driving higher traffic growth or more optimistic or pessimistic outlooks on future air service development supporting passenger growth in the forecast. However, the development of these high and low scenarios provides no information on how likely a given scenario is to occur nor do these scenarios typically have much input to further planning activities (Kincaid et al. 2012). While they may be used in some aspects of future strategic planning or financial analysis, low and high scenarios are not typically used to

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plan airport facilities. Furthermore, high and low scenarios tend to (but not always) produce a fairly narrow band of forecast outcomes, particularly if the high and low scenarios are primarily driven by varying assumptions in macroeconomic drivers of an econometric forecast model (e.g., GDP, population growth).

In addition to the above-mentioned limitations, high and low scenarios do not tend to incorporate the potential impact of shock events or large-magnitude risks in their scenarios. If the objective of the high- and low-scenario process is to create a "reasonable" forecast range, then high-impact, low-probability events are unlikely to be represented in either of the alternative scenarios.

Scenario Forecasting and What-If Analysis

Scenario forecasts, what-if or impact analysis, examine how a single event or potentially multiple events are projected to impact a forecast relative to its baseline value. In contrast to high and low scenarios, specific what-if scenario forecasting can look more deeply at how many factors in the forecast may be linked—for example, how a major external shock event, like a pandemic or military action, may have knock-on effects on the economy or the loss or exit of a major carrier.

The use of scenario or what-if forecasts are one of the more common approaches to examining the impact of shock events in air traffic forecasting.

An example of scenario forecasting before the COVID-19 pandemic can be found in *UK Aviation Forecasts* (UK Department for Transport 2017). This national-level air passenger demand forecast was evaluated over scenarios looking at three options for new runway capacity around London, England. For each of the core scenarios modeled, forecasts were developed estimating the traffic outcomes for each of the 'what-if' scenarios of capacity constraints and new runway infrastructure. While this forecast does not include a substantial assessment of major shock events, it does provide a significant example of the way scenario forecasting can be used to evaluate how different policy and planning actions can affect projected traffic volumes.

The use of scenario or what-if forecasts is one of the more common approaches to examining the impact of shock events in air traffic forecasting. The role these forecasts play in airport planning is less clear. As discussed previously, in airport master planning, the focus tends to be on the medium or most likely forecast.

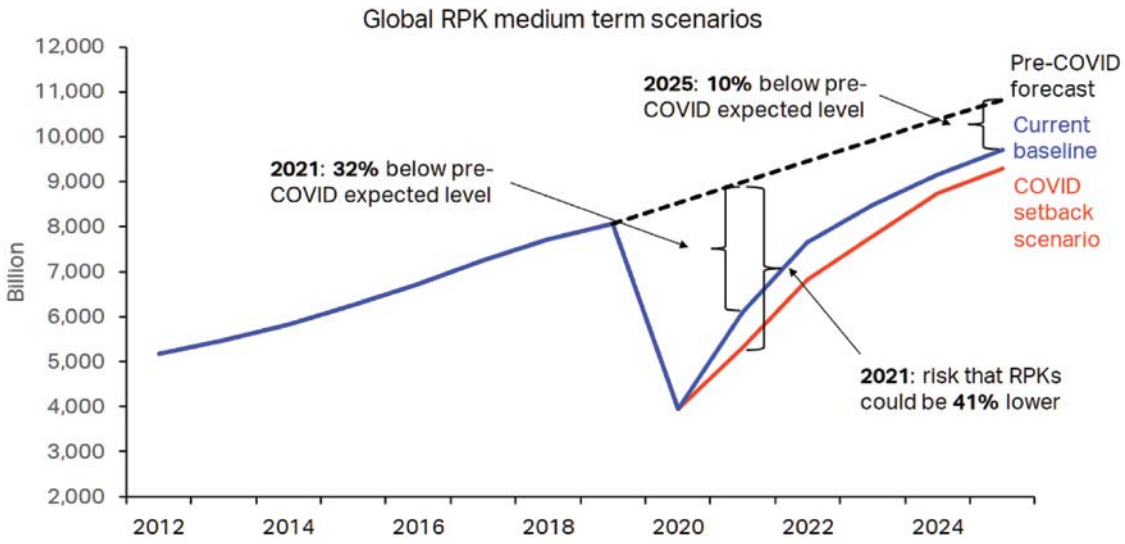
Examples of Scenario Forecasts from the COVID-19 Pandemic

With the onset of the COVID-19 pandemic in the first quarter of 2020, organizations within the aviation industry were faced with the challenge of forecasting how air traffic would recover from the impacts of the pandemic. Throughout the pandemic, these forecasts evolved as the assumptions regarding the development of the pandemic and government measures to control the outbreak were updated. Key factors, such as economic activity and recovery, the impact of government policies and regulations on travel demand and supply, traveler behavior changes, changes to airline markets and networks, and experiences of past pandemics and shock events all contributed to the formulation of scenarios for the recovery. During such an uncertain period, it was not surprising that the forecasts changed and evolved as new information became available and conditions changed.

Figure 23 and Figure 24 show the evolution of IATA's COVID-19 recovery forecasts over a 1-year period, from forecasts of April 2020 to April 2021. These forecasts reflect the information and knowledge available at the time as well as the inherent assumptions about the industry's recovery from the pandemic based on past experience and the (then) current experience of the

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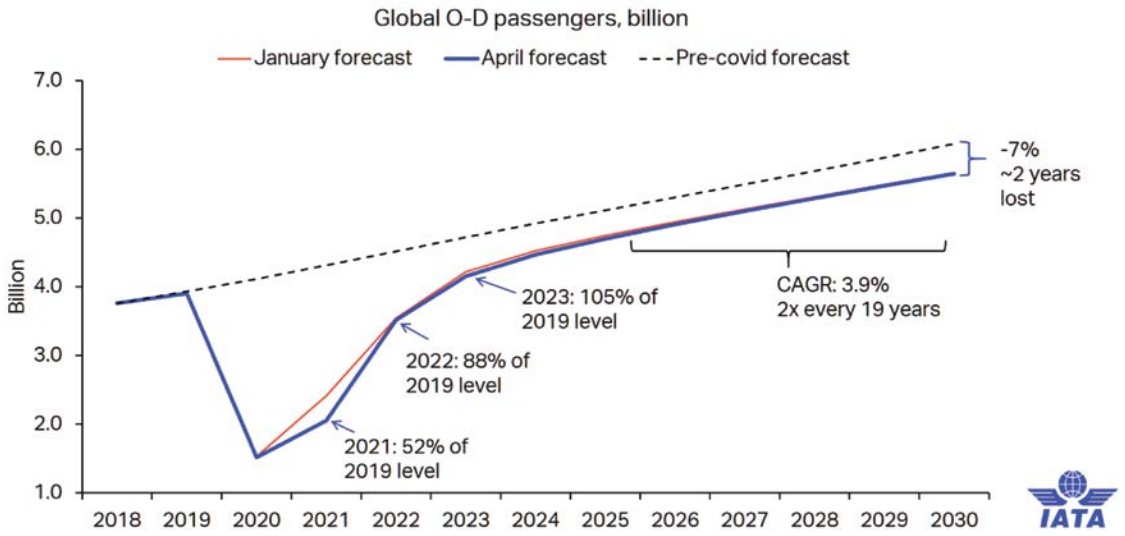


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Figure 23. IATA global air traffic recovery forecast, May 2020 (Pearce 2020).

COVID-19 pandemic. While IATA's 2021 forecast has a more pessimistic outlook on 2021 than its projections a year prior, their forecasts maintained a widely held assumption that global air traffic would return to and likely surpass 2019 levels by 2023.

As a further example, the International Civil Aviation Organization (ICAO) also developed a series of potential recovery pathways for the aviation industry from the impacts of the COVID-19 pandemic (ICAO 2020). By this time in the pandemic, it was clear that the duration of the COVID-19 pandemic was not going to be a short; transitory impact, like SARS in 2003, and the scenarios developed by ICAO in mid-2023 reflected a longer recovery pathway. ICAO, like many organizations and observers, identified that there was significant uncertainty in the future recovery path of the aviation industry and developed a series of scenarios based on varying assumptions of factors influencing air traffic recovery, demand, and supply given the information available to the forecasters at that snapshot in time.



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Figure 24. IATA global air traffic recovery forecast, May 2021 (Pearce 2021).

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Sensitivity Testing

Sensitivity testing, or sensitivity analysis, is a process in which forecasting assumptions are varied one at a time, with the resulting changes in forecast output recorded and compared as assumptions change versus the base case. A sensitivity analysis requires an existing baseline forecast with a set of assumptions ranging from input variables to an econometric model to market share projections or air service development growth outlooks.

Sensitivity testing can be helpful in the development of a forecast for a variety of reasons. Testing individual assumptions on how the forecast will react to a given change in a single variable provides forecasters with information to determine how robust the forecast is. Sensitivity testing can also be useful to understand the role of critical assumptions driving the forecast and can be explored to see how large of a departure from the baseline assumption would lead to some other decision or action resulting from the forecast. Forecast reports may include sensitivity tests as a way to demonstrate how the forecast reacts to specific changes in inputs or to allow forecast users to better understand how changes in specific assumptions (e.g., macroeconomic conditions or airline networks) could impact future traffic volumes beyond the presented scenarios.

Sensitivity testing could be used for assessing shock events as these events may result in changes to other assumptions and inputs to the model. For example, the departure of a major carrier from an airport may be due to an economic recession or major regulatory changes that could impact multiple assumptions in a model. However, as shocks often have interconnected effects across the aviation industry and economy, changing only a single variable or assumption may underrepresent the total potential impact by leaving linked or correlated assumptions unchanged.

Monte Carlo Simulation (Stochastic Forecasting)

Monte Carlo simulation (or the Monte Carlo method) is a form of simulation analysis for forecasting designed to incorporate uncertainties into the forecasting process. It makes use of randomization and probability statistics to generate a wider range of possible traffic outcomes than conventional base-case master plan forecasts and provides estimates of the probabilities of such outcomes. By modeling various uncertainties and risks to future traffic levels—such as if and when a recession might occur, what the impact of a pandemic is if one occurred, or how annual jet fuel prices may differ from the baseline forecast prices—Monte Carlo simulation allows for quantification of the uncertainty of project traffic volumes and to capture "out-of-model" factors that traditional air traffic forecasts cannot. The application of Monte Carlo to air traffic forecasts is well covered in *ACRP Report 76* (Kincaid et al. 2012).

While arguably the inclusion of shock events in Monte Carlo is beneficial in showing the wider range of outcomes, it does not highlight the significant and potentially profound effects of certain shock events on airport traffic.

Monte Carlo involves running the forecast model multiple times (e.g., 10,000 times) with each run generating different randomized values for the input variables. For example, one run might have normal economic performance but with high fuel costs and a terrorism event. Another might have weak economic performance and high fuel costs but with no terrorism event, and so on. The output from each of the runs (or iterations) is collected and can be shown as a distribution of outcomes with associated probabilities.

An illustrative example of the forecast probability risk range is presented in Figure 25, showing the central "most likely" (50th percentile) outcome as well as probability bands ranging from the 5th to 95th percentiles. For example, while the most likely forecast was 149,000 aircraft operations in Year 20, the 5th percentile was 59,000 and the 95th percentile was 218,000 (i.e., 90% of the simulations fell within that range).

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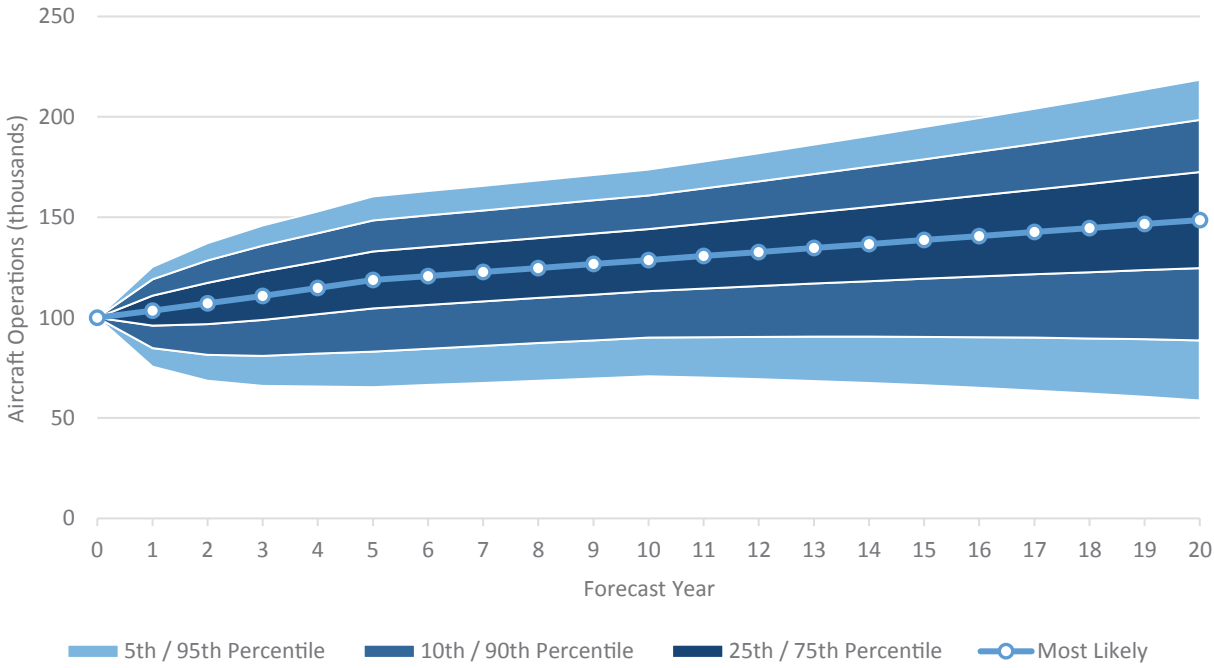


Figure 25. Illustrative forecast probability risk range of a Monte Carlo simulation (InterVISTAS analysis).

The probability distribution of forecast airport activity levels—be it passengers, aircraft operations, cargo tonnage, etc.—provides a quantitative outlook at not just the forecast range but also the likelihood of a given level of future activity. Whereas a traditional low/medium/high or any other scenario forecast cannot provide quantitative information on how likely that scenario is to occur, the risk-based forecasting method explicitly allows forecast users to understand the forecast probability of any future activity level is within the forecast time horizon.

Shock events can be and are included in a Monte Carlo simulation by specifying a probability and impact associated with the event. For example, the impact of a terrorist attack can be included by adding a variable with the following characteristics:

- Probability of occurrence (e.g., a 5% probability in any given year).
- Impact of the event (e.g., a 10% to 30% reduction in traffic, randomly determined).
- Extent of recovery and the recovery period (e.g., recovery of all lost traffic after 2–5 years, randomly determined).

Even with this approach, there are some types (or scales) of shock events that cannot be anticipated, and by itself, Monte Carlo may not be sufficient for management planning for shock events, as it does not focus on one specific stressor. It can still be valuable in evaluating the overall risk profile of the airport and the implications of that risk, but to address shock events, they may need to be combined with scenario forecasting.

However, the perceived low-probability nature of shock events means that their impact is not prominent in the aggregate averaging of 10,000 or more runs and can get missed in the mix of other factors influencing the forecasts. This is illustrated in Figure 26, which shows the distribution of forecast traffic volumes in a specified year (Year 10) at a fictional airport. The impact of randomly including shock events is to somewhat lower the median forecast and widen the "tails" of the probability distribution for traffic in Year 10. While arguably the inclusion of shock events in Monte Carlo is beneficial in showing the wider range of outcomes, it does not highlight the significant and potentially profound effects of certain shock events on airport traffic.

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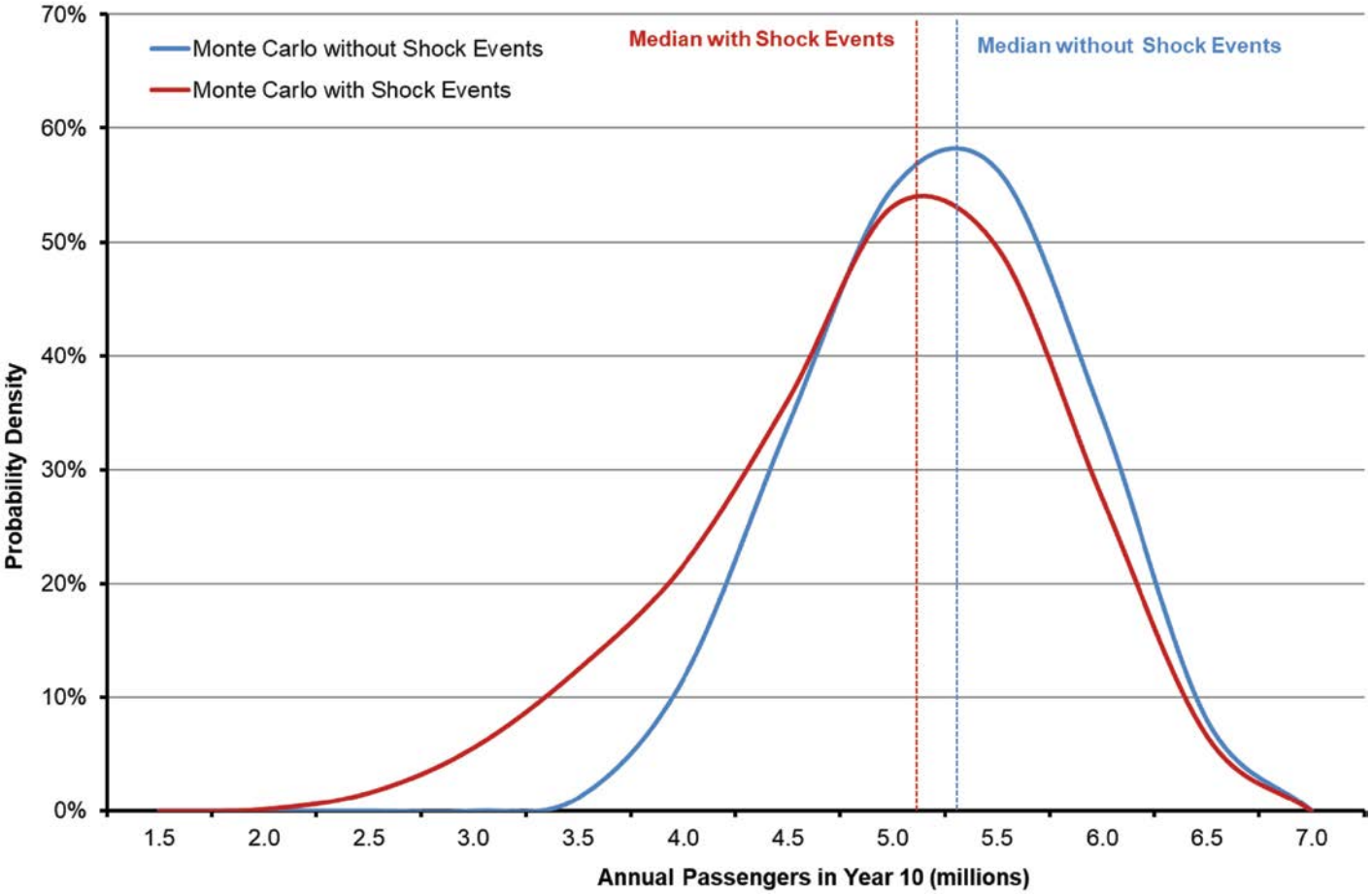


Figure 26. Illustrative impact of including shock events in Monte Carlo analysis (InterVISTAS analysis).

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CHAPTER 8

Addressing and Mitigating Shock Events

Many organizations cannot reliably predict the future and so need to deploy methods for addressing uncertainty, such as scenario planning that considers radical alternative futures and helps develop a portfolio of actions that provide flexibility to changing outcomes.

The issue of how organizations can better prepare themselves for and respond to uncertainty (including shock events) is covered in a broad range of literature on strategic planning under uncertainty (Courtney, Kirkland, and Viguerie 1997; Amram and Kulatilaka 1999; Thoren and Vendel 2019). The basic premise is that many organizations cannot reliably predict the future and so need to deploy methods for addressing uncertainty, such as scenario planning that considers radical alternative futures and helps develop a portfolio of actions that provide flexibility to changing outcomes. Linden (2021) argues that the COVID-19 pandemic and past shock events demonstrate that aviation is subject to considerable uncertainty and that approaches deployed for strategic planning under uncertainty could benefit the industry. This is illustrated in Figure 27, where traditionally, airport planning has generally followed the classical strategic management approach illustrated on the left, whereas, in fact, airports may face a more ambiguous and unpredictable future as shown on the right.

The following sections discuss methods for addressing and managing uncertainty and shock events in the general literature and within the aviation sector.

Methods for Supporting Strategic Planning Under Uncertainty

The literature surveyed generally offers the approach that risk identification is one of the primary ways to mitigate shock or surprise events. A number of papers advocate for implementing risk identification and management strategies to guard against future shock events. This "early warning" and exploration of potential threats are seen particularly in project management literature such as Hajikazemi et al. (2016), Ramasesh and Browning (2014), and Kim (2012). These authors suggest that if there can be greater attention paid during risk management analysis to imagining potential threats and their consequences, then organizations can be better prepared to avoid surprise shock events that have not been imagined. This sentiment appears to be broadly shared across industries and literature, and this approach of explicating the identification of risk factors (including shock events) and developing response strategies is the primary method put forward to deal with shock events.

Another mitigation strategy discusses the value of organizations being flexible in operations and planning in order to adapt to shock events or uncertain outcomes. Duarte Pardo (2016) discusses the benefits of flexibility in the setting of an infrastructure public-private partnership contract as well as infrastructure design that can accommodate a range of potential future outcomes.

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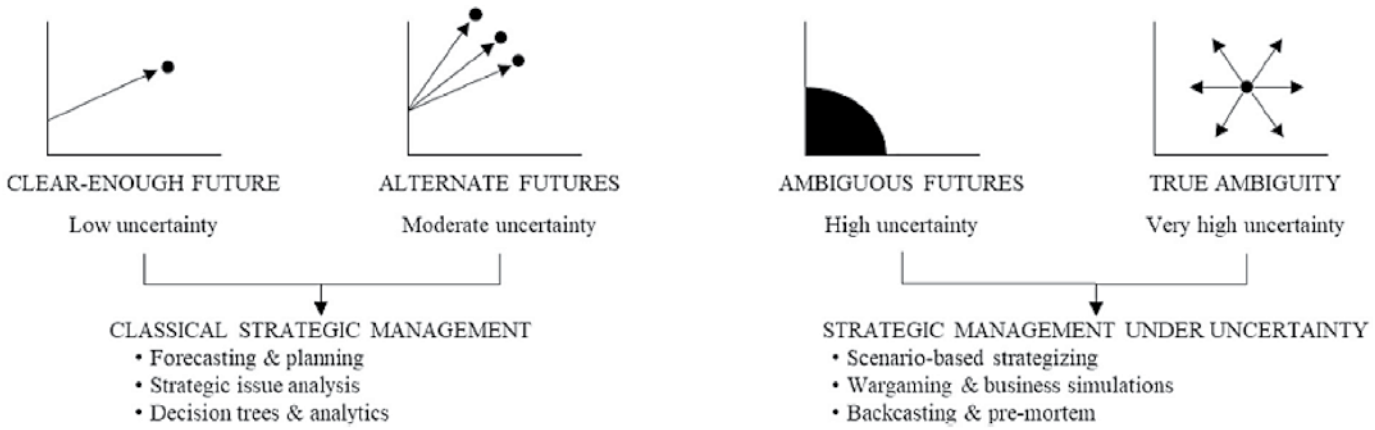


Figure 27. Classical strategic management and strategic management under uncertainty (Linden 2021).

The author points to forecasting methods that can provide quantification of risk and a range of scenarios, including consideration of potential shock events, as one strategy along with real-options analysis to value risk as part of contract development.

More broadly, this is captured in the concept of "real options" (Trigeorgis 1996). Like financial options, a real option is the right, but not the obligation, to take a certain course of action. Real options apply this approach in the real, "physical" world rather than the financial world (although real options still have financial implications). The concept started to develop in the 1970s and 1980s as a means to improve the valuation of capital investment programs and offered greater managerial flexibility to organizations. Real options and real-options analysis are used in many industries, particularly those undertaking large capital investments (e.g., oil extraction and pharmaceutical). The greater flexibility that real options provide can have significant value to the decision-maker but can often (but not always) impose a cost. The trade-off between the value of a real option and its costs will need to be evaluated by the decision-maker.

Scenario Planning

In examining the literature on strategic planning under uncertainty, the use of scenario planning (or variants thereof) was frequently discussed. Scenario planning is generally credited to Herman Kahn of the RAND Corporation who developed the methodology to aid the U.S. military, with much of the research related to shock events in support of the military and more recent antiterrorism (Chermack 2011). This approach is related to the scenario forecasts previously described but involves a more-developed approach to incorporating these scenarios into organizational management and decision-making.

An often-cited example of the successful use of scenario planning is at the Shell oil company (Wack 1985; Wilkinson and Kupers 2013). In the 1960s, Shell (then called Royal Dutch Shell) invested in emerging computer technology to forecast its business outlook. However, the forecast model could only forecast out 6 years due to computational limitations, which was too short when considering investment in oil projects, and the results were frequently wrong. Instead, Shell moved to focus on another initiative that started around the same time exploring the use of scenario planning, drawing on the work by Kahn and others. A small team within the company was tasked with developing future scenarios of the oil market and working with management to determine what actions and strategies could be taken to best manage the scenarios. This process was not

> The use of scenario planning helped break the habit, common to most corporate planning, of assuming that the future will look much like the present.

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about making better predictions of the future but rather about informing organizational processes, such as strategy making, innovation, risk management, public affairs, and leadership development. As noted by Wilkinson and Kupers (2013), the use of scenario planning helped break the habit, common to most corporate planning, of assuming that the future will look much like the present. As unthreatening stories, scenarios enabled Shell executives to open their minds to previously inconceivable or imperceptible developments. The early work included a scenario looking at the oil market switching from a buyers' to a sellers' market, resulting in oil price increases of the sort that eventually did occur in the 1970s (the 1973 oil crisis and 1979 energy crisis), albeit for slightly different reasons. This is credited with making Shell management more prepared for the eventuality, if not the timing or the cause, of this oil price shock.

There is no exact structure to the scenario planning process, and a number of variants and formats have been developed. However, the process can be generalized in the following five steps.

1. **Identify key drivers or risk factors.**

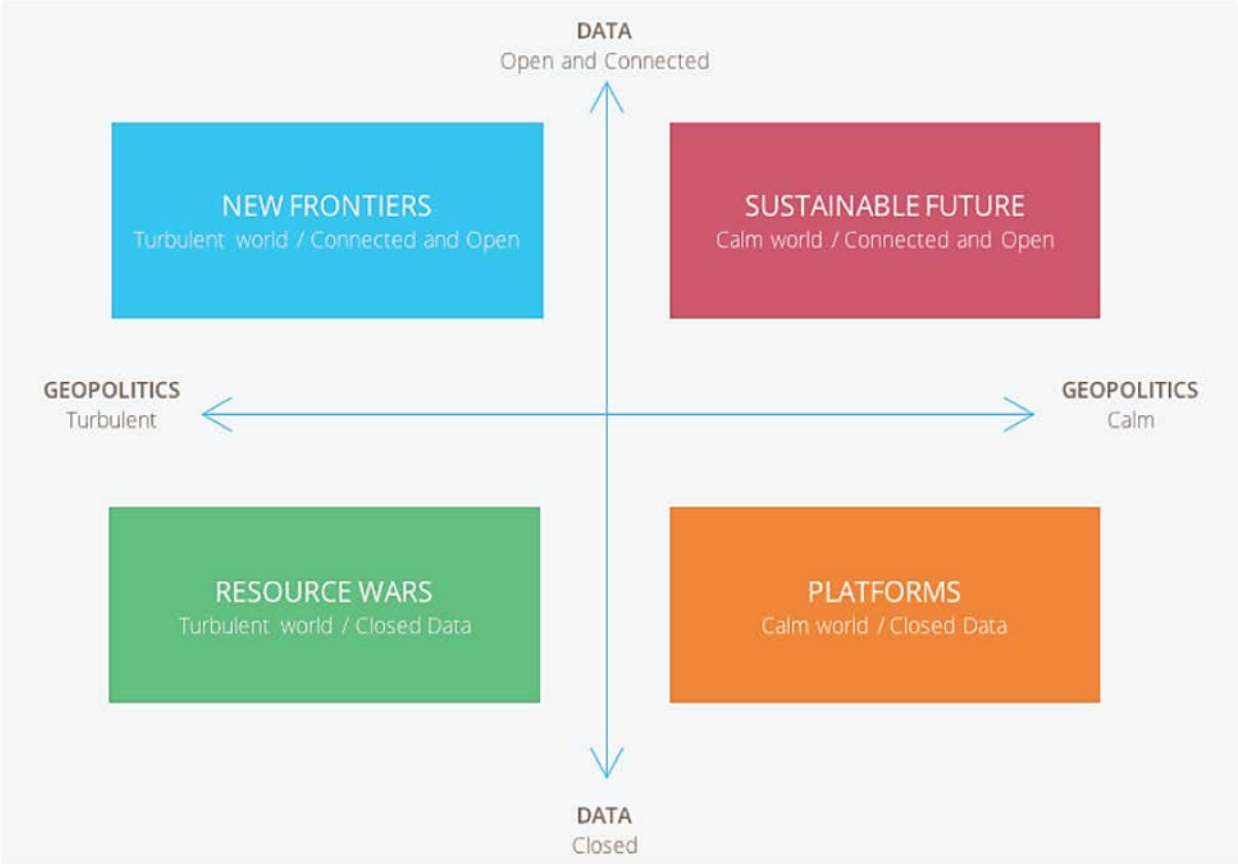
   The identification process can vary depending on the nature of the study, but the literature refers to the Delphi method, SWOT (strength, weakness, opportunity, threat) analysis, STEEP (society, technology, environment, economy, and politics), and PESTLE (political, economic, socio/cultural, technology, legal, environment). STEEP and PESTLE are methods for structuring the identification of factors affecting an organization by considering each of the categories encapsulated in the name.
2. **Shortlist critical uncertainties.**

   Having established a possibly large set of key drivers or risk factors, the next stage of the process focuses on determining where there is the area of greatest uncertainty, which will give focus to the later scenarios. One common approach is to narrow the focus to two critical uncertainties, which are represented in a 2 by 2 matrix. For example, in 2018, IATA with the School of International Futures (SOIF), conducted a study, *Future of the Airline Industry 2035*, using STEEP analysis to identify up to 50 drivers of change (IATA and SOIF, 2018). From this, the study team amalgamated and shortlisted critical uncertainties—geopolitics (from calm to turbulent) and data (from open to closed)—into the 2 by 2 matrix shown in Figure 28. This exercise can lead naturally to the development of scenarios based on the four quadrants of the matrix. It should be noted that while this narrowing to two critical uncertainties is common, it is not the only approach. For example, this approach was not used by Shell (Wilkinson and Kupers 2013).
3. **Develop a set of plausible scenarios.**

   The next step is to generate a small number of scenarios based on the information from the previous tasks. The number of scenarios should be small to prevent the exercise from becoming overly complex and unstructured. A maximum of four scenarios is typically recommended (based on the 2×2 matrix), but the Shell team initially developed three during their exercises: one status quo scenario plus two divergent alternative scenarios (Chermack 2011). In the 1980s, the Shell team switched to two scenarios as a way of stopping managers from focusing on the "middle way" (Wack 1995). The literature urges avoiding the trap of developing best-case, worst-case, and most likely scenarios or scenarios that reflect purely good or bad outcomes. The scenarios should be plausible but not necessarily probable and relevant to the decision-maker while challenging conventional (or business-as-usual) thinking. They should be more than an "obvious" variation in the business conditions—for example, not simply a $\pm$ 20% variation in traffic or oil prices.

   There is an emphasis on developing a compelling and realistic narrative with each scenario in order to engage the decision-maker. Chermack (2011) discusses the need for "storytelling" to describe how the scenario develops and how key elements of the scenario work together (although the 2×2 matrix focuses on two critical uncertainties, other elements identified

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Figure 28. A 2 by 2 matrix of critical uncertainties (IATA and SOIF 2018).

previously can also be included in the scenarios). While avoiding being overly technical and complex, the scenario needs to provide sufficient detail and richness so that it can generate a meaningful discussion on actions that can be taken. This can include more quantitatively output such as scenario forecasts.

4. **Identify actions and strategies that can be deployed.**

   The process by which the scenarios are used to stimulate dialogue around actions and strategy can vary depending on the nature of the exercise. Scenario planning aimed at overall corporate strategy might be conducted differently from one focused on a specific project or region. Nevertheless, it is generally the case that the scenarios are used as the basis for more interactive discussion (for example, during workshops) to challenge current assumptions and practices and often to develop options for the organization to navigate these scenarios (exploiting opportunities or avoiding/mitigating adversity).

5. **Test these actions and strategies against the scenarios and refine them if necessary.**

   Sometimes this step is combined with step 4 and may involve several iterations between Steps 4 and 5. The aim is to test and challenge either the current strategic approach or the actions developed in step 4. A number of methods have been developed for this purpose:

   – Wind Tunneling (Chermack 2011). In design and engineering, wind tunnels are used to test the aerodynamic performance of an object (e.g., a wing, airplane, car, or building). In the context of scenario planning, the scenario is the wind tunnel, and it is the strategic approach or actions that are being tested. The aim is to test decisions for robustness and expose opportunities, risks, and possible improvements to the strategic approach. This is generally conducted through a series of workshops or other interactive methods.

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– Scenario Immersion (Ralston and Wilson 2006). Scenario immersion is an approach that is similar to wind tunneling and involves structured brainstorming about the implications of a set of scenarios for the organization. In scenario immersion, the scenarios are presented to as many organizational members as possible in a workshop format. Each member of the workshop group is asked to act as a decision-maker and identify strategic responses to those threats and opportunities.

The aim with scenario planning is not to produce more accurate forecasts of the future . . . rather it is presented as a learning process that challenges conventional thinking and focuses attention on how the future could be very different to the past.

As indicated previously, the aim of scenario planning is not to produce more accurate forecasts of the future, although this can happen indirectly, as with the example of Shell anticipating oil price shocks (the fact that the process can include a variety of viewpoints allows it to benefit from the "wisdom of crowds"). Rather, it is presented as a learning process that challenges conventional thinking and focuses attention on how the future could be very different from the past and the implications of that possibility. The output is a change in mindset and the enhancement of resilience and robustness. It is not a one-off exercise but rather an ongoing process of discovery and adaptation. As stated by Derbyshire (2022, 7), "plausibility-based scenario planning has a role to play in ensuring that what is 'out of sight' is not also 'out of mind.'"

War Gaming

A somewhat related methodology to scenario planning referenced in the strategic management literature is business war gaming. As the name suggests, this approach draws from military table-top war simulation exercises. In a military context, these are used to examine war-fighting concepts, train and educate commanders and analysts, explore scenarios, and assess how force planning and posture choices affect campaign outcomes (Oriesek and Schwarz 2008).

In a business context, war gaming is used to develop and test strategies in a competitive business environment. A business war game is usually set up as a role-playing simulation of a dynamic business situation. It involves competing teams, one representing the organization of interest and others cast in the role of certain stakeholders (e.g., a competitor, regulator, consumer). The aim is to play out the possible actions and counteractions in a given scenario (a new corporate strategy, launch of a new product, etc.). The focus is the dynamics between the players of interest under a specified set of conditions. As such, there is less focus on uncertainty and less of a wider perspective than in scenario planning. Some have suggested combining scenario planning and war gaming (Schwarz, Ram, and Rohrbeck 2019), but this approach appears not yet to have been fully developed.

Red Teaming

Red teaming is an approach developed in the military in which a group is formed to try and assess vulnerabilities and limitations of systems or structures. It is used to reveal weaknesses in military readiness to remedy or mitigate those weaknesses. This approach is also used in cybersecurity where "white hat" hackers attempt to infiltrate the security of information technology (IT) systems. In some cases, the exercise may also involve a blue team charged with defending the system. The FAA used red teaming to test airport and airline security in the years before 9/11, and the TSA has a Red Team with the mission statement, "Measure TSA screening effectiveness against real-world, intelligence-driven threats in order to inform enterprise risk management and performance improvement" (TSA 2024).

Red teaming has also been used in business planning to test and challenge strategy. As stated by Ryan (2020):

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> When developing a business strategy, companies often base plans on a best-case scenario, assuming that everything will go to plan. Red teaming exercises help expose strategic gaps or flaws, helping companies scan the business environment for both threats and opportunities.

Red teaming emphasizes contrarian thinking and devil's advocacy, similar to the elicitation methods described in the earlier section on scenario planning.

Need for Organizational Engagement

A common theme in the literature on strategic planning under uncertainty is the need for full organizational engagement. It is not sufficient to simply identify risks or possible black swans as an exercise; this information needs to lead to meaningful organization changes and the organization's willingness to be responsive and proactive to these risks. The literature lists examples where risks or shock events were anticipated and even modeled but without leading to any real benefit.

A common theme in the literature on strategic planning under uncertainty is the need for full organizational engagement.

- Lempert, Popper, and Bankes (2002) give the example of General Ridgeway, who, as a young officer, wrote a "war game" about a surprise attack on Pearl Harbor that fellow officers refused to play as "possibly too implausible."
- Wucker (2016) gives the example of a detailed disaster plan provided by FEMA to Louisiana state officials in January 2005 based on a simulation of a Category 3 hurricane ("Hurricane Pam") hitting southern Louisiana. When Hurricane Katrina hit in August 2005, most of the recommendations in that plan were not followed, and the plan was largely ignored.
- The UK government conducted major simulation exercises on a pandemic outbreak in 2007 (Winter Willow) and 2016 (Exercise Sygnus). However, a government inquiry on the government response to the COVID-19 pandemic found the focus of these exercises on an influenza pandemic left the country little prepared for coronavirus such as COVID-19. As noted by a previous Chief Medical Officer:

  > Quite simply, we were in groupthink. Our infectious disease experts really did not believe that SARS, or another SARS, would get from Asia to us. It is a form of British exceptionalism. . . .
  >
  > We need to open up and get some more challenge into our thinking about what we are planning for. . . . In thinking through what could happen, it would be well worth bringing in people from Asia and Africa to think about that as well, to broaden our experience and the voices in the room. (UK House of Commons 2021, 20)

Thus, there is a need for risk assessment to be more than a tick-box exercise and instead, more of a form of management realignment. The use of scenario planning and "strategy as practice" are put forward as ways to achieve this (Linden 2021), as discussed later in this document.

Systematic Approaches for Addressing Shock Events in Airport Planning

Within the aviation sector, no methods for mitigating shock events specifically were uncovered during the research. However, approaches have been developed that address uncertainty generally which can apply to shock events. Much of these have been previously documented in *ACRP Report 76* (Kincaid et al. 2012) and other publications and are summarized here, along with any updates to these approaches or new approaches that emerged.

A number of practitioners and researchers have proposed approaches to building more robustness to uncertainty in airport planning (or infrastructure planning generally). These include the following.

- Dynamic strategic planning (de Neufville and Odoni 2013).
- Flexible strategic planning (Burghouwt 2007).
- Dynamic adaptive planning (Walker et al. 2019).

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Although these approaches differ in methodology, the aim of each is to ensure that the airport's planning (strategic planning as well as master planning) is flexible and adaptable to changes in market, political, social, regulatory, and technical conditions. Rather than focusing largely on a single forecast, these approaches consider a wider range of future scenarios and contingencies that can be built into the airport strategy and master planning. For example. Walker et al. (2019) provide an example of an application to Amsterdam Schiphol Airport (AMS), as shown in Figure 29.

While drawing on real-life experience, these approaches are largely theoretical and there is little evidence of them being formally applied by airports, although elements have been used in practice.

ACRP Report 76 (Kincaid et al. 2012) examined the issues of uncertainty generally in airport forecasting and planning. It put forward a systems analysis framework that draws on aspects of

Vulnerabilities and opportunities	Hedging (H) and Shaping (SH) actions
Necessary condition for success: retain market share	
Demand for air traffic grows faster than forecast	H: Prepare Lelystad Airport to receive charter flights
Demand for air traffic grows slower than forecast	SH: Advertise for flying from Schiphol
Collapse or departure of the hub carrier (KLM) from Schiphol	H: Prepare to adapt Schiphol to be an O/D airport H: Diversify the carriers serving Schiphol SH: Develop a close working relation with KLM
Rise of long-haul low-cost carriers	H: Design existing and new LCC terminal to allow for rapid customization to airline wishes.
Rise of self-hubbing, resulting in increasing transfers among LCC operations	H: Design a good connection between the existing terminal and the new LCC terminal, first with buses, but leave room for replacing it with a people mover
Necessary condition for success: population affected by noise and the number of noise complaints should not increase	
Maintain current trend of decrease of environmental impact of aircraft	SH: Negotiate with air traffic control on investments in new air traffic control equipment that can enable noise abatement procedures, such as the continuous descent approach SH: Invest in R&D, such as noise abatement procedures
Increase in the population density in area affected by noise	H: Test existing noise abatement procedures, such as the continuous descent approach, outside the peak periods (e.g., at the edges of the night) SH: Negotiate with surrounding communities to change their land use planning SH: Invest in R&D, such as noise abatement procedures
Change in the valuation of externalities by the public	SH: Invest in marketing of the airport to brand it as an environmentally friendly organization SH: Join efforts to establish an emission trading scheme
Necessary condition for success: Schiphol's competitive position in terms of available capacity in Europe does not decrease	
Other major airports in Europe increase capacity	No immediate action required
Development of wind conditions due to climate change	H: Have plans ready to quickly build the sixth runway, but do not build it yet. If wind conditions deteriorate even further, start construction

NOTE: This was a theoretical application rather than a study conducted for the airport.

Figure 29. Vulnerabilities and opportunities and proposed responses at AMS (Walker et al. 2019).

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flexible planning, real options, and strategic planning under uncertainty. The systems analysis framework and related methodologies are designed to assist airport decision-makers with

- Identifying and characterizing risks (threats or opportunities), including their plausibility and magnitude;
- Assessing the impact of these threats and opportunities, i.e., determining what could happen, to which air facility, and when it might occur; and
- Developing response strategies to avoid or lessen the impact of threats or foster the realization of opportunities.

The framework and methodology involved five key steps as summarized in Figure 30. The framework allows planners to consider a broad range of events and risks; forecast or model these risks; and develop approaches to avoid, mitigate, or exploit these risks. It is designed to augment (rather than replace) existing strategic planning and master planning methodologies and

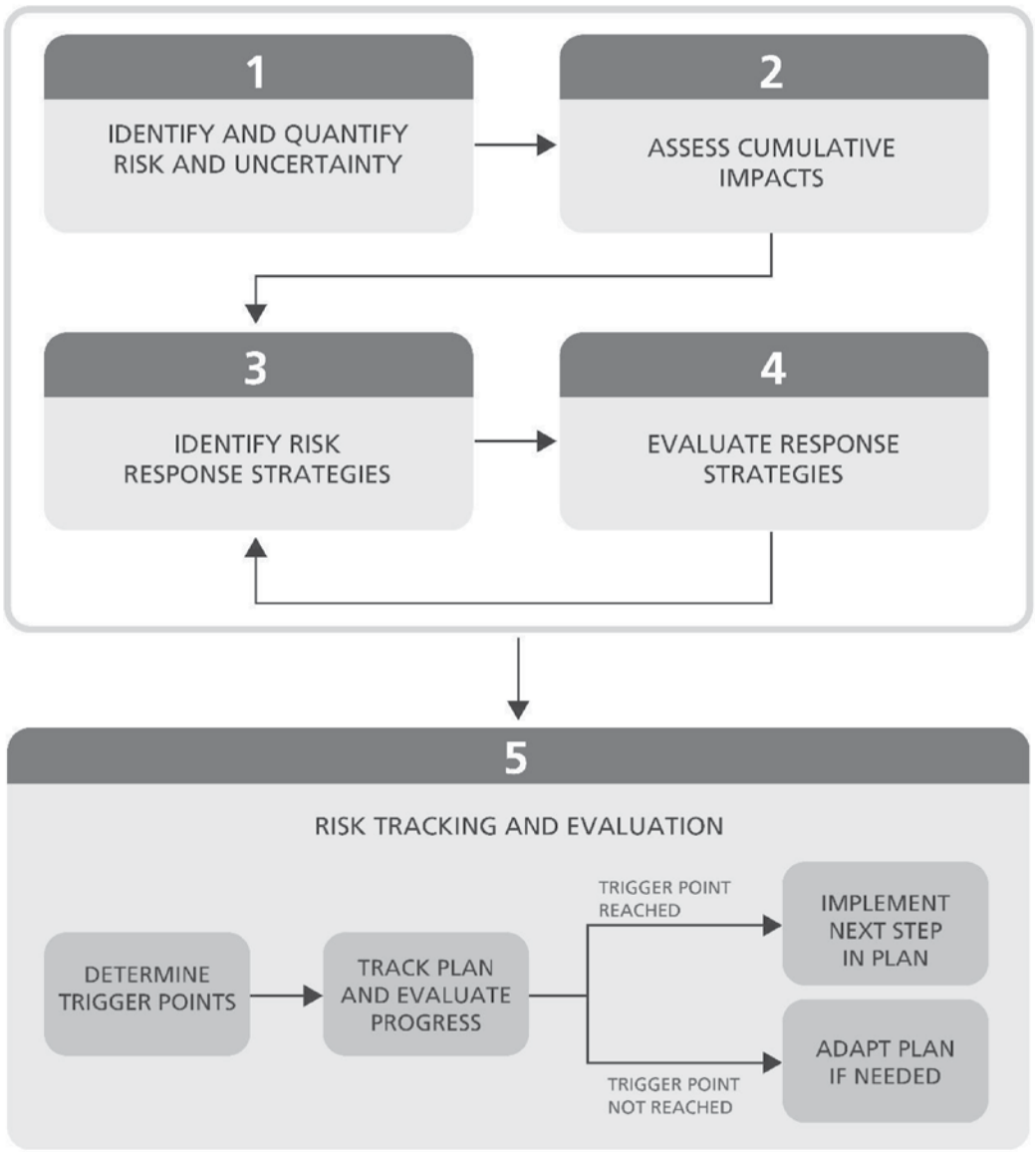


Figure 30. ACRP Report 76 systems analysis methodology overview (Kincaid et al. 2012).

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is scalable to the nature of the project and the resources of the airport. While the methodology did consider shock events, this was not a focus of the research, and the need for further research into shock events was acknowledged in the report.

The methodology set out in Part III to address shock events draws on some of the ideas and methodologies discussed in *ACRP Report 76*. As such, the shock events methodology can complement or augment the methodology set out in *ACRP Report 76* and would fit within its framework. While the shock events methodology and this guide are designed to be stand-alone, readers are encouraged to also review *ACRP Report 76* (Kincaid et al. 2012).

Practical Approaches for Airports to Address Shock Events

There are various strategies and actions airports can and do take to mitigate against uncertainty. While not specific to shock events, some of these apply to such events. Based on a review of previous research, observation of airports generally, and the experience of the research team, these strategies are summarized in Table 4 (this list is not exhaustive).

Supporting Airport Recovery with Revenue Diversification

Even before the COVID-19 pandemic, airport operators were diversifying revenue sources to fund the maintenance of airport facilities, day-to-day operations, capital improvements, and changing demands for passenger services. The pandemic intensified the need for airport operators to address severe declines in activity, fluctuating passenger demand, and uncertainty about the paths for recovery. Much has been discussed about revenue strategies in previous research [notably, *ACRP Report 121: Innovative Revenue Strategies—An Airport Guide* (Kramer et al. 2015)]; however, a few ideas stand out as particularly germane in mitigating shock events.

- Airports balance keeping service provision in-house or contracting out to maximize revenues.
  - Develop central processing in the terminal area for deliveries of food, beverages, and retail products and distribution of products to concessionaires.
  - Operate baggage handling for airlines.
  - Provide fuel storage and/or fueling services for airlines and private aircraft.
  - Operate FBO.
  - Manage gates, passenger check-in, and jet bridge operations.
- Airports engage in new activities and land uses through lease participation, direct ownership, public-private partnerships, or joint development.
  - Expand electric power generation capacity, add charging stations, and establish pricing strategies.
  - Partner with urban air mobility (UAM) companies for airport access and pretravel processing (UAM envisions a safe and efficient aviation transportation system that will use highly automated aircraft that will operate and transport passengers or cargo at lower altitudes).
  - Develop gas plazas, convenience stores, restaurants, and personal services for air travelers or members of the community.
  - Develop mineral estates or water resources if available.
  - Build clean energy capacity for airport and community use.
  - Position the airport as a testing ground for new technology and innovations, such as wildlife management, autonomous airport security, and UAM.
- Airports refine concession strategies to address what passengers want and need throughout the day in post-pandemic times.
  - Expand market research using point-of-sale data for preordering food and preselecting parking spaces to gain knowledge about passenger preferences and segments.

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Table 4. Airport-specific approaches to address uncertainty and shock events.

Strategy	Comments
Land banking – reserving or purchasing land for future development.	Mitigates or hedges against the upside risk associated with strong traffic growth. The airport authority (or government) has the flexibility to expand and take advantage of high traffic growth, but they are not committed to doing so.
Reservation of terminal space – similar to land banking, this involves setting aside space within the terminal for future use (e.g., security processes).	Mitigates against high traffic growth or changes in traffic mix (e.g., from domestic to international or O/D to transfer). Also mitigates against changes in government policy or regulation (or the overall security environment). The space can be designed so that it remains productive in the short term (e.g., using it for retail that can be removed quickly).
Development of nonaeronautical revenues and ancillary activities	Revenue diversification (discussed in the next section) can also be an effective risk-mitigation strategy. Airports can engage directly (or partner with third parties) in nonaeronautical revenues to diversify their sources of income. By relying less on aircraft operations and passenger enplanements, airports can reduce the systemic revenue uncertainty associated with the air travel industry. However, diversification can expose the airport to greater risk from other sectors of the economy.
Air service development	Increase the range of carriers and routes operating at the airport using a diversification/hedging strategy, thereby reducing exposure to particular carriers or markets.
Shock event response plans	The development of response plans is a relatively low-cost way of enhancing robustness to shock events and has parallels with the approach already used by airports to respond to emergencies and accidents.
Stakeholder consultation	Helps ensure that stakeholders understand the airport's plans and enables the airport to respond to concerns (e.g., an airline concerned that the airport is becoming too crowded). Also allows identification of additional risks (including lack of support from certain stakeholder groups).
Trigger points/thresholds – next stage of development goes ahead only if predetermined traffic levels are reached.	Addresses both upside and downside risks and can be applied to specific traffic categories—for example, • Total passengers, • Domestic or international passengers, • Total aircraft operations, and • Large aircraft operations (triggering a runway extension). As many construction projects have long lead times (due to planning, construction, etc.), the trigger should be specified to allow for this lag. For example, the trigger to expand the terminal facilities may be when passenger volumes reach 90% of existing capacity, allowing time for the additional facilities to be built before the terminal reaches full capacity. This approach applies to not only capital developments. For example, a downside trigger could be determined for certain traffic markets, so if traffic falls below that level, additional air service development work would be undertaken. The trigger does not necessarily have to be traffic-based. For example, information from airport marketing might trigger actions or capital improvements to accommodate new air service.

(continued on next page)

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Table 4. (Continued).

Strategy	Comments
Modular or incremental development – building in stages as traffic develops.	Avoids airports committing to large capacity expansion when it is uncertain whether and how the traffic will develop. At the same time, an airport can respond to strong growth by incrementally adding modules. Provides the flexibility to delay or accelerate expansion as traffic develops. Also mitigates against traffic mix changes; facilities designed to serve one traffic type can also be designed for incremental development. This option is closely linked to the trigger point concept described above.
Linear terminal design and centralized processing facilities.	Allows the greatest flexibility for airport expansion since it is the most easily expandable in different directions (especially in combination with modular design). It also allows flexibility in the face of changing traffic mix (e.g., O/D vs. connecting).
Use of inexpensive, temporary buildings	Allows the airport to service one type of traffic (e.g., LCCs) while keeping options open to serve other types (e.g., full-service or transfer). Example: AMS's LCC pier.
Non-load-bearing (or glass) walls – as with swing gates, terminal space can be converted from one use to another.	Avoids the airport being "locked in" to a narrow traffic development path, allowing greater flexibility to manage changes in traffic mix. Unlike swing gates, this is less short term in nature (not day-to-day). Also allows broader flexibility in the overall function of the terminal, e.g., converting space from domestic to international, from retail to security, etc.
Parametric design	Parametric design is the creation of a digital model of a building, such as a passenger terminal. Within that model are certain "parameters" reflecting key design attributes of the building. When one parameter is changed in the model using sophisticated algorithms and rules reset by the designer, it adjusts other attributes of the building to match. For example, changing the roof height will automatically change the walls, pillars, windows, and related elements. Parametric design allows for great flexibility as it is possible to quickly design a wide range of variants and test their robustness to various scenarios and objectives.
Common-use facilities and equipment – such as common-use terminal equipment (CUTE), common-use self-service (CUSS), common gates, lounges, and terminal space.	Mitigates against changes in the mix of traffic and carriers operating at the airport. Also has the benefit of reducing the overall space requirements of the terminal.
Swing gates or spaces – can be converted from domestic-to-international traffic (or between types of international traffic) on a day-to-day basis.	Mitigates against changes in traffic mix. Allows a very fine level of control, as it can allow adjustment to changes in traffic during the day as well as long-term developments. Can also reduce overall space requirements as domestic and international traffic often peaks at different times of the day.
Multiple apron ramp system (MARS) gates – the same gate can service one large aircraft or two smaller ones.	Mitigates against the uncertainty around the future mix of traffic (narrow-body vs. wide-body) and reduces the overall space requirements.
Tug-and-cart baggage systems	Provides much greater flexibility to make changes to the operation of the baggage system than does a fixed system. Mitigates against upside and downside traffic growth, traffic mix changes, and checked-baggage trends (e.g., less baggage due to checked-baggage charges).

NOTE: Some of the information originally appeared in *ACRP Report 76* (Kincaid et al. 2012), although additions and updates have been made.

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  - Develop gate delivery services for meals using centralized kitchens and ordering applications.
  - Add vending machines for electronics and other e-commerce purchases.
  - Use market data about time-of-day purchases and passenger segments to tailor the concession mix.
  - Experiment with quick-change stands and pop-up-stations to target upcoming flight destinations and seasons.
  - Build flexibility into contracts and leases to adjust commercial terms in the event of a pandemic or other business-stopping situations.
- Airports invest in automation to optimize contactless services and pricing strategies.
  - Automate pretravel check-in and baggage handling with available customer service options for valet assistance.
  - Use parking management systems to optimize revenue.
  - Install autonomous airport security systems for airport perimeters and passenger clearance.

Planning and Design Techniques

Advancements in computing technology can aid in ensuring facility design provides flexibility and robustness to uncertainty. For example, the use of parametric design for buildings and structures allows for more design options to be considered and tested. Parametric design is the creation of a digital model of a building, such as a passenger terminal. Within that model are certain "parameters" reflecting key design attributes of the building. When one parameter is changed in the model using sophisticated algorithms and rules reset by the designer, it adjusts other attributes of the building to match. For example, changing the roof height will automatically change the walls, pillars, windows, and related elements. Parametric design allows for great flexibility as it is possible to quickly design a wide range of variants and test their robustness to various scenarios and objectives.

Technical Solutions

On the more technical side, the use of swing gates or multiple apron ramp system (MARS) gates not only mitigates the uncertainty around the future mix of traffic but can also reduce the overall space requirements if the peaks of different traffic segments occur at different times of the day. Figure 31 shows an example of a swing gate. The swing gate in this example can swing between three types of traffic: domestic, U.S., and international, as is required at some Canadian airports. Similarly, the MARS gate allows the same gate to handle two narrow-body aircraft or one wide-body aircraft, enhancing flexibility in the face of uncertain traffic and fleet development.

Developing robustness to uncertainty can also be applied to individual elements of the airport system. For example, *ACRP Research Report 225: Rethinking Airport Parking Facilities to Protect and Enhance Non-Aeronautical Revenue* (Inter*VISTAS* 2021) put forward approaches for future-proofing parking developments at airports so they can adapt to changes in use or be converted to other uses. For example, sparse column grids and exterior ramps enable greater flexibility for how the airport can use the space as needs change in the future.

Insurance

Insurance is another means by which airports can mitigate shock events by transferring the risk to another party (at least partially). Many airports have business continuity or business interruption insurance. These typically insure airports against a loss of business due to facilities being damaged or out of action (e.g., due to a fire or hurricane). However, in the case of

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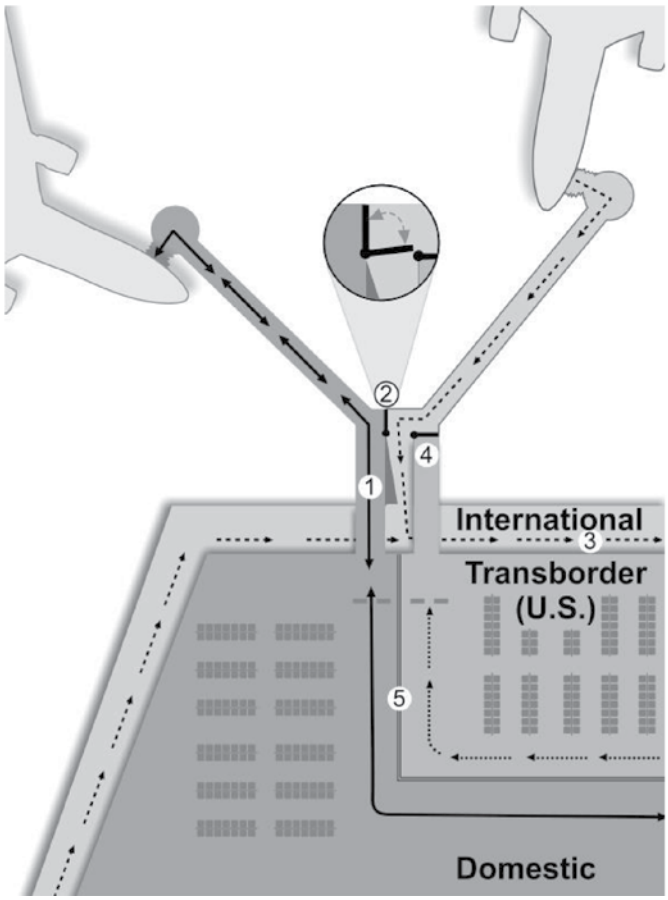


Figure 31. Examples of swing gates (Kincaid et al. 2012).

COVID-19 there was no property damage and some of the airports interviewed during the project indicated that the insurance had exclusions for pandemics, communicable diseases, and "acts of God." Thus, it is generally very difficult for airports to obtain insurance (or affordable insurance) for strategic risks such as a pandemic or loss of an air carrier.

It can be viewed that the federal government acted as insurer of last resort during the COVID-19 pandemic, through payments made to airports in the CARES Act and subsequent support measures.

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CHAPTER 9

Summary of Findings

Drawing on the research detailed in Parts I and II, several findings were identified as outlined below. These findings informed the development of the shock events methodology described in Part III.

Current Approaches to Shock Events

- Shock events are often overlooked or underresearched when considering projections of the future, both in the aviation sector and in many other sectors of the economy. This can be due to a lack of information, a lack of imagination, or an unwillingness to consider the unthinkable (e.g., "That's too horrible to consider," or "I wouldn't know what to do so I'll ignore it."), or a combination of these factors.
- Shock events are often given a low probability of occurrence, which gives them a small weighting in decision-making and a small impact on the modeling (e.g., even where Monte Carlo risk analysis is used for forecasting airport traffic, the perceived low probability of a shock event has little impact on the overall forecast). However, the shock event probability is subjective because typically there are no past experiences to draw on. Shock event probability can also reflect the biases of the decision-makers as much as any objective truth ("I don't want it to happen, so I give it a low probability.").
- The literature refers to black swan events (an event that "came out of the blue"), grey swan events (some warning signs were present), and grey rhinos ("it was staring us in the face, but we ignored it"). Ultimately, these characterizations are moot and often a matter of perspective. What is a black swan to one group may be quite obvious to another. However, they do illustrate the need for an open-minded and imaginative approach to potential shock risks.
- Airport managers are not naive to the risk of shock events. However, it is generally not a major focus for managerial and operational planning. For example, the master planning process in the United States is largely guided by the base case or most likely forecast.
- Many airports engage in ERM, and all commercial airports in the United States operate to very high standards in regard to safety and security and, as required by law (14 CFR § 139.325), have an airport emergency plan to deal with accidents, bomb incidents, fires, natural disasters, etc. However, now these plans are often more focused on operations, security, or safety and less strategic in nature. Furthermore, one criticism of ERM (not just at airports) is that it often tends to focus on compliance and known or familiar risks and less on rare events. Nevertheless, this approach to emergency planning, which is in the DNA of U.S. airports, could act as a template for approaches to more strategic shock events.

Airport managers are not naive to the risk of shock events. However, it is generally not a major focus for managerial and operational planning.

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Shock Event Identification

- Good methods for shock event identification are available, but there needs to be more focus on contrarian, left-field, devil's advocate approaches. There is a need to consider the less familiar and less comfortable possibilities when identifying shock events. For example, the Delphi method, which is a well-established iterative approach to gathering information from experts and knowledgeable stakeholders, has been put forward as a means of risk identification. However, Delphi is often focused on achieving consensus. This can result in fewer mainstream views being overlooked—and perhaps not a useful approach for identifying rare shock events. Therefore, the method used for shock event identification needs to ensure that "Cassandras" are heard. In Greek mythology, Cassandra was granted the gift of prophecy but also cursed so that nobody would believe her warnings. Thus, she could neither alter events nor convince others of the validity of her predictions.
- Any identification exercise is not about identifying all possible events in all possible detail (although there can be a positive impact in this regard). Undoubtedly, some events will be missed or occur in ways that were not anticipated. Rather, shock identification should be about opening management and planning thinking to the possibility of these events. After all, shock events for airports often result in dramatic changes in traffic volume or mix, regardless of the cause.

Forecasting and Modeling

The inability to accurately predict shock events does not mean that there is no value in forecasting. What remains useful is forecasting in this context what could happen if an event did occur, often referred to as scenario forecasts.

- There are no effective methods for predicting shock events, particularly for a long-term outlook. Models are often based on some form of historical data, and many shock events have no relevant precedent. Attempts to predict the occurrence and timing of shock events, such as earthquakes or terrorist events, require impractical (or impossible) levels of information and computation and have a poor track record. Similarly, forecasting methodologies in the financial sector have poor track records.
- To quote statistician George Box, "All models are wrong, but some are useful" (Box 1976). The inability to accurately predict shock events does not mean that there is no value in forecasting. What remains useful is forecasting in this context what *could* happen *if* an event did occur, often referred to as scenario forecasts.
- Scenario forecasts are not uncommon in air traffic forecasting [several examples are given in *ACRP Report 76* (Kincaid et al. 2012)], but generally, they are an adjunct to the main forecasts that drive an airport's strategic planning and its master plan. There is a need for a wider range of scenario forecasts to be given greater attention and consideration.

Addressing Shock Events in Planning and Strategy

- Scenario planning holds promise as a way of exploring how an airport would be affected by a shock event and how it could respond. This would involve more than producing a forecast but also considering in some detail how the airport could respond, e.g., how the airport's development might change, financial impacts, or more strategic aspects. It would involve undertaking exercises with relevant parties to consider impacts and approaches.
- As a result, airport plans and strategies could be modified to ensure that there is robustness and readiness built in, where possible. The output could also include the development of response plans or contingencies (similar to emergency response plans) that the airport could draw upon if an event occurs.

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- A common theme in the literature on strategic planning under uncertainty is the need for full organizational engagement. It is not sufficient to simply identify risks or possible shock events as an exercise. This information needs to lead to meaningful organization changes and for the organization to be responsive and proactive to these risks. This should not be a one-off exercise that then sits on a shelf. It needs to be a living plan and updated frequently. Also, it must be recognized just because an airport has done the exercise, it does not mean it is now risk-proof. Constant engagement and vigilance are needed.
- While history is an imperfect guide to the future, there are lessons to be learned from past shock events. Therefore, the information on COVID-19 and other shock events in Part I will have significant value.

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PART III

The Systematic Methodology Approach

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CHAPTER 10

Introduction and Overview of the Methodology

Part III sets out a systematic methodology to understand and address shock events in airport forecasting, management, and planning. In developing this methodology, the research team drew on generalized findings from the research in Parts I and II as well as prior experience in risk identification and risk management exercises in a range of economic sectors. It builds on established methodologies, such as scenario planning, but is tailored to the circumstances and requirements of the airport sector.

The aim of the methodology is to contribute to a more resilient airport business model, by assisting the airport to be better able to have a robust response to shock events.

In developing the methodology, the research team has focused on the following objectives:

- The methodology should be readily incorporated into existing airport practices.
- The approach should be applicable and scalable to all types and sizes of airports.
- The methodology should not require specialized knowledge or training and should be accessible to a wide range of airport practitioners.
- The approach should be a genuine contribution to improving the robustness of airports and not simply be a "checkbox" exercise.

The methodology aims to contribute to a more resilient airport business model by assisting the airport to be better able to have a robust response to shock events, including the development of a ready-to-deploy portfolio of strategies. While not all shock events can be identified or foreseen, the intent is that an airport would be better positioned for these eventualities generally and can benefit from past expertise and experience drawn from the COVID-19 pandemic and other shock events both from within the airport industry as well as other sectors.

Overview of the Methodology

The proposed methodology is made up of five sequential elements as follows:

- Step A: Identify shock events of interest.
- Step B: Develop scenario forecasts that incorporate shock events.
- Step C: Scenario implementation (or stress testing).
- Step D: Develop options for improved robustness.
- Step E: Ongoing monitoring and vigilance.

These steps are described in more detail in Chapters 11 through 15. While the methodology is designed to aid airport processes, it is recognized that additional parties beyond airport management may be involved. For example, air traffic forecasts are often developed by third-party consultants, and, likely, these consultants would also be responsible for the scenario forecasts. Similarly, planning consultants may be involved in various aspects of the methodology.

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Scalability of the Methodology

Clearly, the more resources that can be dedicated to this methodology, the more comprehensive and detailed the findings will be, but even a simple approach is better than ignoring the issue entirely.

The level of detail and resources incorporated into this methodology will depend on the budget and time available for the airport. Larger commercial airports may seek to conduct an in-depth and comprehensive analysis, folding it into their planning processes, while smaller commercial and GA airports may undertake a high-level, qualitative approach. Both approaches are reasonable and add value in that the organization directly confronts the potential for shock events and implications for airport strategy and planning. Clearly, the more resources that can be dedicated to this methodology, the more comprehensive and detailed the findings will be, but even a simple approach is better than ignoring the issue entirely.

Fundamentally, the methodology poses the question: *What would happen if a shock event occurred, and what can be done to mitigate the impact of these events?* While the methodology outlined in the next sections discusses some of the more advanced methods that can be deployed to address this question, consideration is also given to simplified approaches that could be used by small airports or airports with limited time or resources.

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CHAPTER 11

Step A: Identify Shock Events of Interest

In attempting to identify shock events, it is important to have a clear idea of what is meant by the term "shock event." As discussed previously, shock event is used in this document as it provides a general idea of the concept without any implied assumptions around the level of prior knowledge (black versus grey swan) or likelihood of occurring (low probability). Planners and decision-makers are encouraged to consider what shock events to which the airport is exposed. Such shock events could cause traffic development to diverge considerably from the long-term trend, perhaps permanently.

Using History as a Guide, Not a Predictor

In considering potential shock events, it is useful to review past events as a starting point for the identification of and a guide to how the events may play out. However, care should be taken to avoid assuming that future events will exactly follow the pattern of past ones. For example, the airline hijackings of the 1960s, 1970s, and 1980s were very different from the terrorist attack of 9/11 even though the latter involved hijackings. Similarly, the impact of the SARS outbreak in 2003 was quite different from that of the COVID-19 pandemic despite both pandemics being caused by a coronavirus. As discussed below, there is a need to be imaginative when considering potential shock events. Nevertheless, examining past shock events does provide some guidance and can offer lessons learned.

In considering potential shock events, it is useful to study past events as a starting point for identification of events. However, care should be taken to avoid assuming that future events will exactly follow the pattern of past ones.

Table 5 provides a list of shock events that have affected airports in the past, based in part on the research described in earlier sections. These events have been grouped into the following categories.

- Air market – shock events arising from changes in the aviation industry or the market served by specific airports.
- Terrorism/war – events targeting or indirectly impacting the aviation industry.
- Natural events – natural disasters with short- and long-term impacts on aviation.
- Technology – technological changes within the industry or from competing industries.
- Pandemic – both regional and global outbreaks.
- Economic – events in the general economy with implications for aviation.
- Environmental – effects of climate change and climate change policy.
- Social – attitudes toward and demand for air travel.
- Political – national and international political events.

The table is not meant to be an exhaustive list of potential shock events but rather to seed the thinking about shock event identification.

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Table 5. Examples of past shock events impacting airports.

Category or Type	Examples	Commentary
Air Market		
Carrier failure or exit	St. Louis (STL) – TWA; Cleveland (CLE) – United; Wilmington (ILN) – cargo; Pittsburgh (PIT) – US Airways.	The failure or exit of a carrier can lead to a substantial decline in traffic from which the airport may not fully recover.
Carrier entry/stimulation	Bellingham (BLI) – Allegiant; Orlando Sanford (SFB) – Allegiant; Williston Basin (XWA) – various airlines.	The entry of a new carrier (often an LCC or ULCC) surging traffic can drive the need for changes to the airport development. This has to be balanced against the possibility that the carrier may reallocate capacity elsewhere.
New airport competition	Singapore (SIN) versus Dubai (DXB).	Singapore experienced a slowdown in the development of connecting traffic over the last two decades as Dubai (and other Middle Eastern hubs) emerged as a major competitor for Europe-Asia traffic.
Terrorism/War		
Terrorist attack on the aviation system	9/11 terrorist attacks (2001); Liquid bomb plot (2006); "Underwear bomber" (2009).	The 9/11 terrorism event resulted in a prolonged decline in traffic and significant changes to airport facilities and operations. Subsequent attempts have led to further security changes.
Terrorist attack on individual targets (airports or tourism)	Bali bombing (2002); Glasgow airport ramming attack (2007).	The Bali bombing, which did not target the airport but rather the tourism area, depressed tourism at this resort for several years (and with it, traffic at local airports). The Glasgow attack had only a short-term impact on traffic, although it did lead to additional security requirements in the UK.
Regional conflict	Kuwait (1991); Afghanistan (2001); Iraq (2003); Ukraine (2022).	These wars have led to depressed traffic levels (concerns about reprisal attacks) and in some cases, increased fuel prices. While their effects have been relatively short-lived, there is the potential for them to trigger other market effects (e.g., air carrier distress).
Natural Events		
Hurricane	Hurricane Katrina (2005).	While most hurricanes have minor long-term effects on airports, Katrina precipitated a long-term decline in traffic due in large part to factors outside the airport's control.
Tsunami/earthquake	Indian Ocean (2004); Sendai, Japan (2011); Christchurch, New Zealand (2011).	The destruction of travel and tourism infrastructure can lead to a prolonged decline in air traffic.
Volcano eruption	Eyjafjallajökull, Iceland (2010).	The volcanic eruption in Iceland closed much of European air space for nearly 2 weeks due to concerns about volcanic ash damaging aircraft (the largest air traffic shutdown in Europe since the Second World War). Volcanic eruptions can potentially generate ash clouds for several years.
Technology		
Fleet grounding	DC-10 (1979); Boeing 737 Max (2019).	The Boeing 737 Max aircraft was grounded by the FAA between March 2019 and December 2019 due to safety concerns. Exposed airlines had to cancel large numbers of flights.
Electric aircraft	In development.	The use of electric aircraft is anticipated before the end of the decade. While some deployments will be replacements of conventional aircraft, the economics of these aircraft could result in new route opportunities and changes to airline networks.

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Table 5. (Continued).

Category or Type	Examples	Commentary
New mobility models	Urban Air Mobility.	UAM is the use of small, highly automated aircraft to carry passengers or cargo at lower altitudes. This has the potential to affect demand profiles and route networks at airports.
Alternative/competing industries	Videoconferencing.	The move to videoconferencing and online working during the pandemic may have impacted some elements of business air travel, although it is not expected to be a full replacement.
Pandemic		
Global pandemic	Spanish flu (1919); Swine flu (2009); COVID-19 pandemic (2020).	The dramatic and long-term impact of COVID-19 on the aviation sector is well documented in this report. While swine flu was fairly global, it did not result in restrictions on border crossings or air travel.
Regional pandemic	SARS (2003); MERS (2012); Ebola (2014).	These outbreaks caused traffic declines in the affected regions. Concerns about a more global spread could lead to wider travel restrictions.
Economic		
Severe recession	Great Recession 2008–2009.	While economic recessions are not particularly surprise events, the severity and global impacts of the 2008/09 recession were a shock and led to significant impacts in aviation through reduced demand, airline mergers, and changes to air networks.
Oil price shocks	OPEC oil embargo (1970s); Prerecession (2007/08); Russia's attack on Ukraine (2022).	The dramatic run-up in global oil prices in 2007/08 led to the introduction of fuel surcharges raising costs for passengers as well as cargo shipments and precipitating changes in global fleets toward larger, more fuel-efficient commercial aircraft.
Decline of local business/sector	Athabasca Oil Sands, Alberta, Canada (2014/15).	A rapid decline or closure of a local business sector or major employer can lead to long-lasting negative impacts on traffic. As the global price of oil fell in 2014, traffic fell at the Fort McMurray International Airport (YMM), the primary gateway to the Athabasca Oil Sands in Alberta, as development and employment were rapidly curtailed, resulting in a long-lasting decline in traffic.
Development of a new local sector	Amazon Air Hub at Cincinnati/Northern Kentucky International Airport (CVG).	A significant development or expansion of a local/regional economic sector can rapidly spur traffic growth. Increased traffic may be a long-lasting 'level shift' or may be more transitory over a shorter period.
Environmental		
Carbon pricing or cap and trade	ICAO Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA); EU Emissions Trading System.	While not a shock in the sense that their development is well documented, the application to aviation and potential for increased penalties for carbon emissions could have a significant and long-lived impact on traffic development.
Effects of climate change	Rising sea levels; increased average air temperatures; increased extreme weather events.	Impacts on airport operations could lead to changes in fleet mix and network. Airports may need to divert capital resources to mitigate the effects of climate change (e.g., building sea walls).

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Table 5. (Continued).

Category or Type	Examples	Commentary
Social		
Social attitudes toward air travel	Flight shame or *flygskam.*	Flight shaming or *flygskam* is an anti-flying social movement, aimed at reducing the environmental impact of aviation. It started in 2018 in Sweden and has gained some traction in other parts of Europe. To date, this has not been a significant movement in the United States.
Political		
Policy or political changes	2016 UK Brexit Vote.	The decision by referendum for the UK to leave the European Union impacted air carrier services that previously benefited from the unrestricted traffic rights and ownership rules within the EU. For example, some UK-based airlines had to establish new bases within the EU to comply with new ownership laws.

Methods for Identifying Shock Events

While Table 5 provides a starting point for shock event identification, there will be a need to consider the specific situation of the individual airport and benefit from the knowledge and viewpoints of the people working at the airport. Therefore, some form of elicitation process with the airport management team is recommended. Ideally, this would involve input from a range of key functional areas to ensure all perspectives are covered, including some or all of the following:

- Airport operations, security, and safety;
- Commercial and marketing;
- Finance;
- Facility development and planning;
- Community/stakeholder relations and legal; and
- Depending on the airport, representation relating to passenger, cargo, and GA traffic.

It is anticipated the exercise would be internal to the airport. While there may be some benefit from including outside stakeholders (airlines, tenants, local government, tourism/cargo/community groups, etc.), it could lead to the censoring of some ideas. For example, it may be uncomfortable to discuss airline failure in front of representatives of the airline industry. Nevertheless, the airport will want to make sure that someone from airport management can speak to the perspective of outside sectors.

The focus is not on achieving consensus but rather "casting a wide net." Therefore, an approach such as structured brainstorming is recommended.

The form of the elicitation process deployed should encourage a diversity of viewpoints and open-minded thinking. The focus is not on achieving consensus but rather on "casting a wide net." Therefore, an approach such as structured brainstorming is recommended. Structured brainstorming is a systematic process for the liberal generation of ideas from a number of participants by encouraging each of them to volunteer their creative inputs in an atmosphere that is free of criticism and judgment from other participants. The structured part ensures that the brainstorming is focused and goal-oriented.

Structured brainstorming generally works best with small groups, preferably diverse in background and experience. The ideal format is a face-to-face session since this has been found to be most effective in sparking ideas and encouraging discussion. However, if that is not possible,

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internet video calls or similar technology can be used. In situations where the group cannot make the time commitment to moderated sessions, an asynchronous process can be used where the information is elicited through online tools, e.g., short surveys administered by email to collect the information and follow-up communications to refine the final information, which can be presented as a wiki page or electronic document. Regardless of the format, the process benefits from careful and experienced facilitation. This facilitation could be done by an outside expert or by the consulting firms already involved in the project (e.g., for forecasting or planning support). A good facilitator ensures that all viewpoints are heard and makes sure to draw out ideas from quieter people whose ideas may be as good or better than those who are more vociferous.

The broad structure and characteristics of the structured brainstorming are as follows.

- **Set a clear objective and scope.**
  The participants should receive clear messaging on what shock events they should consider. The shock events should be strategic, significant, and challenging to the business rather than minor accidents or variations. At the same time, the events considered should also be plausible and relevant to the airport; expending time on shock events that are neither (e.g., an extra-terrestrial invasion) will weaken the exercise. Nevertheless, care must be taken not to dismiss ideas as implausible simply because there is a lack of historical precedence; a careful balance needs to be made.
- **Educate and prime the participants by discussing past events.**
  This can utilize the information in Table 5 and previous sections in this guide as well as research conducted before the discussion. At the same time, emphasize the need for thinking outside of these examples.
- **Structure the discussion to aid the thinking.**
  For example, the shock events could be considered in the categories set out in Table 5: air market, terrorism/war, natural events, pandemics, economic, etc. Alternatively, the standard STEEP categorization can be used. As previously discussed in Part II, STEEP is an acronym for society, technology, environment, economy, and politics and is a means of categorizing ideas in elicitation exercises. The STEEP analysis approach is similar to PEST (political, economic, socio/cultural, technology) or PESTLE (political, economic, socio/cultural, technology, legal, environment) analysis. Depending on the size of the discussion group, subgroups could be created to look at one or more of the categories. These can then be shared and discussed with the other subgroups.
- **Encourage wide-ranging and imaginative thinking.**
  Participants should feel that they can put forward ideas without judgment, and the moderator should ensure all voices are heard. At the initial stage, the bar should be set fairly high in regard to rejecting ideas. The question to be satisfied should be, "Why can this event be rejected?" rather than, "Why should it be included?"
- **Elicit ideas of the basic characteristics of the event.**
  How will the event impact the airport, such as the overall scale, duration of the event, etc.? This does not need to be a detailed or data-driven exercise but will allow a greater understanding of how the event is perceived. The discussion can also include the probability of the event, although, as discussed previously, the probability is often subjective and subject to bias, and there is a risk that low probabilities will be used to reject ideas.
- **Ensure a balance of dialogue.**
  While the initial ideas should be free from judgment, to avoid self-censorship or less vociferous participants from being "shouted down," this does not mean that all discussion and (constructive) criticism should be avoided. Recent empirical evidence suggests that, within a cooperative environment, if people do point out that some ideas have problems, it often sparks others to share ways to improve on initial ideas (Curhan, Labuzova, and Mehta 2020).

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- **Identify common themes.**

  The facilitator can point out common themes in different shock events. For example, there may be several events that result in substantial traffic declines lasting several years. The overlapping nature of these impacts can help narrow the more detailed scenario forecast in Step B, "Develop Scenario Forecasts That Incorporate Shock Events."

Simplifying the Process

The use of structured brainstorming and other elicitation techniques is designed to develop a comprehensive set of potential shock events drawing on knowledge from across the organization. However, some airports may not have the time and resources to undertake this process. In this case, airport management can still draw on the examples given in Part I and Table 5 as guides to relevant shock events. At the most basic level, the most challenging shock event that most airports could face is a large decline in traffic sustained for an extended time, as occurred for many airports during the COVID-19 pandemic. Consideration can be given to a shock event of this nature regardless of its exact cause or development. For example, a small commercial or GA airport might want to consider an unnamed shock event leading to a reduction of 50% of aircraft operations lasting for 2 years. In addition, consideration can also be given to a small number of other shock events, such as the entry of a new carrier leading to rapid traffic growth or the loss of a certain segment of traffic or certain customers.

Outcome

This step will generate a list of potential shock events that could impact the airport, with summary information on the anticipated characteristics and effects. This in itself can provide insightful information to engage and challenge airport management and planning. It can also be incorporated into a risk register if the airport maintains such a system. It will also guide the scenario forecasts in Step B.

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CHAPTER 12

Step B: Develop Scenario Forecasts That Incorporate Shock Events

Shortlisting the Shock Events

As a final part of the initial elicitation of potential shock events, discussed in Step A, or as a follow-up exercise, the participants should be asked to identify the top two to four shock events to be evaluated in more detail. The criteria for this should be those events that are considered most significant and/or most challenging for the airport. The short list of shock events should cover a meaningful range of eventualities and not simply those deemed most likely. The shock events should be chosen because they test as much as possible the robustness of the airport under combinations of different important stressors. The idea is that by focusing on these events, the airport will also be better positioned to handle other events, such as realizing that an approach to mitigating the loss of 50% of traffic can also apply to mitigating the loss of 20%, 30% or 40% of traffic.

The short list of shock events should cover a meaningful range of eventualities, and not simply those deemed most likely. The shock events should be chosen because they test as much as possible the robustness of the airport under combinations of different important stressors.

The number of shock events selected will depend on the time and resources available, but the aim is to have a fairly manageable and digestible list. The scenario planning literature generally recommends no more than four scenarios (including a baseline or business-as-usual scenario) to avoid the exercise becoming overly complex and to maintain the concentration of the participants. In some cases, as few as two scenarios may be sufficient, examining fairly extreme positive and negative shock events. In other cases, there may be a need to examine additional scenarios, such as shock events impacting certain different traffic elements such as domestic versus international, O/D versus connecting, air cargo, etc.

It should be noted that the scope of this research project is on shock events that primarily result in changes to airport traffic, which in turn will drive changes in revenues. Therefore, the scenario forecast should generate information about air traffic, which is the focus of Step B. However, this approach could also be applied to shock events that do not impact traffic, such as real estate or commercial activities.

Further Developing the Scenarios

Having identified a shortlist of shock events for further examination, each can be developed into a storyline as to how the event might develop. The idea is to provide a certain intuitive richness and sufficient detail to allow decision-makers to develop meaningful and realistic ideas about the shock event and how to address it.

Mapping methodologies can be used to aid this storyline development. One such methodology is bowtie analysis previously described in Part II. This approach maps the factors that

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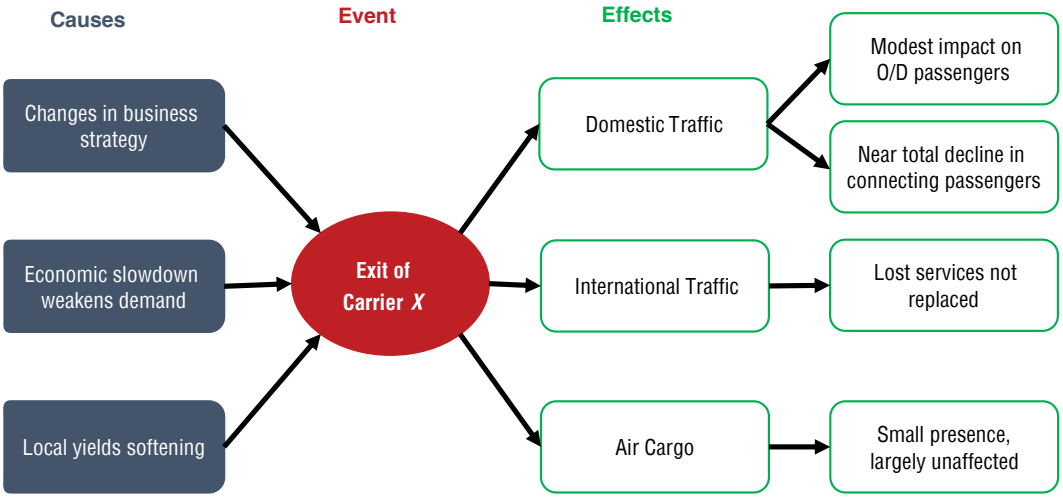


Figure 32. Example bowtie analysis for the exit of a major carrier: traffic impacts.

come together to contribute to a shock event and the various consequences after the event. The various consequences can include new risks that can emerge—for example, the airport has less revenue, which leads to a scale back of its development and refurbishment plans and results in a reputation as a dated and crowded airport. This methodology has been used to evaluate accident risks in power generation and aviation safety. Another less structured approach is the use of "mind maps" or flow diagrams, which attempt to capture the effects resulting from a shock event. Examples of a bowtie analysis and a mind map applied to the loss of a carrier are shown in Figure 32 and Figure 33, respectively.

Regardless of the approach used, the aim is to develop a compelling and plausible narrative around the shock event, describe how it comes about, and examine how it impacts traffic at the airport, which informs the scenario forecast and can be concisely explained to the decision-makers through text and diagrams.

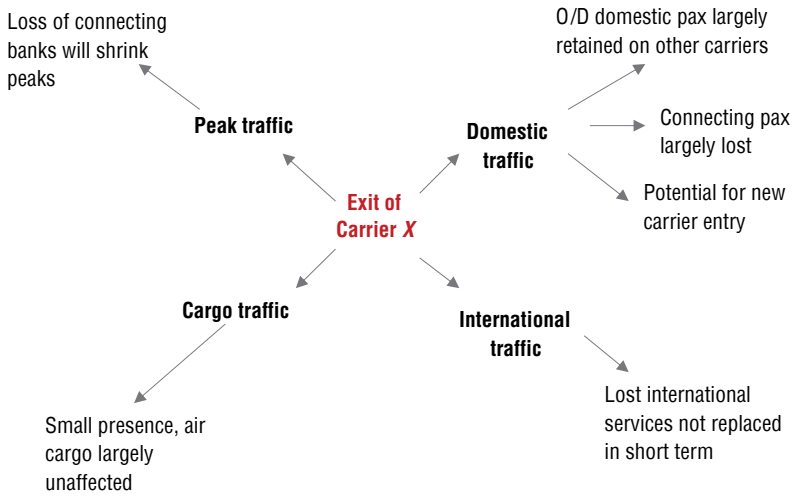


Figure 33. Example of a mind map of the exit of a major carrier: traffic impacts.

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Producing Scenario Forecasts

As well as a plausible narrative, quantification is also important to the scenarios. Quantifying the scenarios will enable the communication of shock events impacts to people who need to see numbers, such as finance and planning. As noted by DeAnne Julius, Chief Economist for Shell from 1993 to 1997, "engineers are numbers people, and if you can't quantify what you are talking about, they tend to dismiss you as interesting (at best) mystics" (Wilkinson and Kupers 2013).

In quantifying the scenario there are numerous considerations.

- **Modeling approach.** The quantification and modeling of shock events can be an add-on to the existing process of generating air traffic forecasts for planning and management purposes. Existing models can be used to aid in the scenario forecasting process and generation of output. However, the modeling approach could also be conducted as a stand-alone exercise as a deviation from current published forecasts (e.g., the FAA Terminal Area Forecasts) or deviation from current traffic levels.
- **Establishing the baseline.** As suggested above, the shock event is expected to create a divergence from some sort of business-as-usual baseline. Typically, the baseline will be the forecast used in the planning process (the base or most likely forecast). Alternatively, the shock event could simply be analyzed against current traffic volumes.
- **Timing of the event.** The timing of a shock event is impossible to predict, so an assumption about when the event occurs will need to be made. This could be now (especially if using current volumes as the baseline) or sometime in the short- to medium-term. To some degree, the timing will depend on the nature of the event. For example, the impact of some technological change may be expected to be in the medium-term given the need for product research and testing. However, as a general rule, the event should be shown occurring in the short- to medium-term. Showing an event some 10 to 15 years in the future could make it difficult for decision-makers to determine what actions should be taken now and may also give the impression that dealing with the event can be put off until later.
- **Magnitude of the impact.** In some cases, the impact on traffic volumes might be determined by examining past events as a guideline to severity and duration. But in doing so, the analyst needs to give thought to whether they need to go beyond historical precedents. As previously noted, the impact of the 2003 SARS pandemic was a weak guide to the impact of COVID-19. The brainstorming in Step A may provide information on the anticipated impact. For some events, specific analysis should be undertaken to determine magnitude. For example, if considering the loss of a major carrier, data will generally be available on the passengers and cargo carried and routes operated by that carrier. A route and network model can be used to model the potential entry of a replacement carrier(s). This modeling will need to consider the frequency and routes operated, the aircraft type(s) used (past and future), and the load factor required. Some events may impact certain traffic segments more than others (e.g., domestic versus international, leisure versus business destinations/origins); the analysis could also capture this.
- **Duration and aftermath.** The duration of the event may vary from almost immediate, such as a terrorist attack or natural disaster, to unfolding over months or years, such as an airline gradually dehubbing or climate change effects. Again, past examples can be used as a guide, but they should not blinker the analyst to other possibilities. The aftermath should also be modeled using suitable logic, such as the potential traffic recovery following a shock decline in traffic and whether there is a possible reversion to the long-term trend after a positive event. Consideration should be given to whether there will be a full recovery of traffic if it is a negative event and over what timeline. Again, the duration and recovery may vary for different traffic segments.

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- **Level of detail.** The level of detail will depend on the resources available (including the time commitment of participants) and the requirements of the project. To evaluate the financial implications of the event, information on the impacts on annual domestic and international passenger and aircraft operations may be necessary. For planning purposes, the impact on peak-hour traffic may also need to be considered. If using the airport's existing forecasting model, many of these outputs could be generated through that model.

An example scenario forecast is provided in Figure 34. This shows a fictional airport that experiences the dehubbing and exit of a network airline over a period of 2 years. Events like this have been experienced at airports such as STL, LAS (US Airways night hub), and PIT. In the example, the carrier accounts for 50% of traffic at the airport before the event. As the only hub carrier at the airport, the exit results in the loss of 90% of connecting passenger traffic, compared with a loss of only 25% of O/D traffic (some domestic routes and most international routes are lost). The event results in a 50% decline in passenger traffic after 2 years relative to preevent levels and a 54% decline relative to the business-as-usual forecast made before the event. O/D traffic takes 10 years to recover to preevent levels, while connecting traffic remains depressed. After 10 years, total traffic volumes are at 32% below preevent levels and 49% below the business-as-usual forecast. Additional detail of traffic by segment, aircraft operations, and peak can also be generated from this forecast.

Scenario forecasts do not necessarily produce more accurate forecasts but challenge planners and decision-makers as to how these outcomes can be managed and mitigated in the airport planning.

Another example of the use of scenario forecasts is provided in Figure 35, taken from *ACRP Report 76* (Kincaid et al. 2012). In this report, scenario forecasts for Bellingham International Airport (BLI) were generated based on the airport's circumstances in 2003 and 2004 when the airport was developing its master plan and before the entry of Allegiant Air, which caused traffic to surge (BLI is discussed in more detail in Chapter 5). The exercise was undertaken in 2011 by the ACRP research team (not airport management) and applied to historical data as a test case. In addition to the master plan forecast, three scenario forecasts were generated in the ACRP exercise:

- Extreme downside – exit of some existing carriers and weakened growth;
- Extreme upside – rapid growth associated with the entry of a new LCC; and

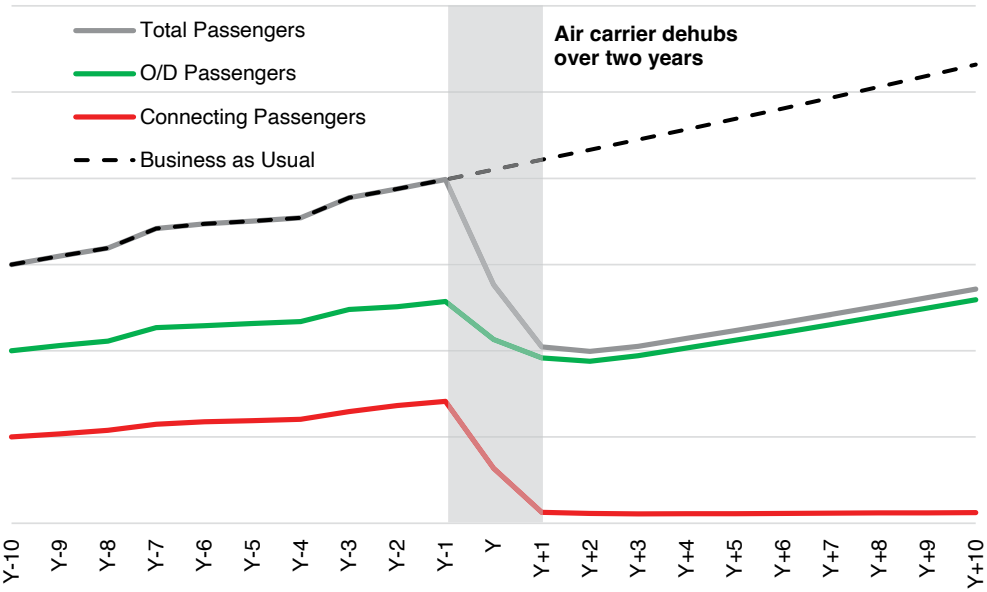
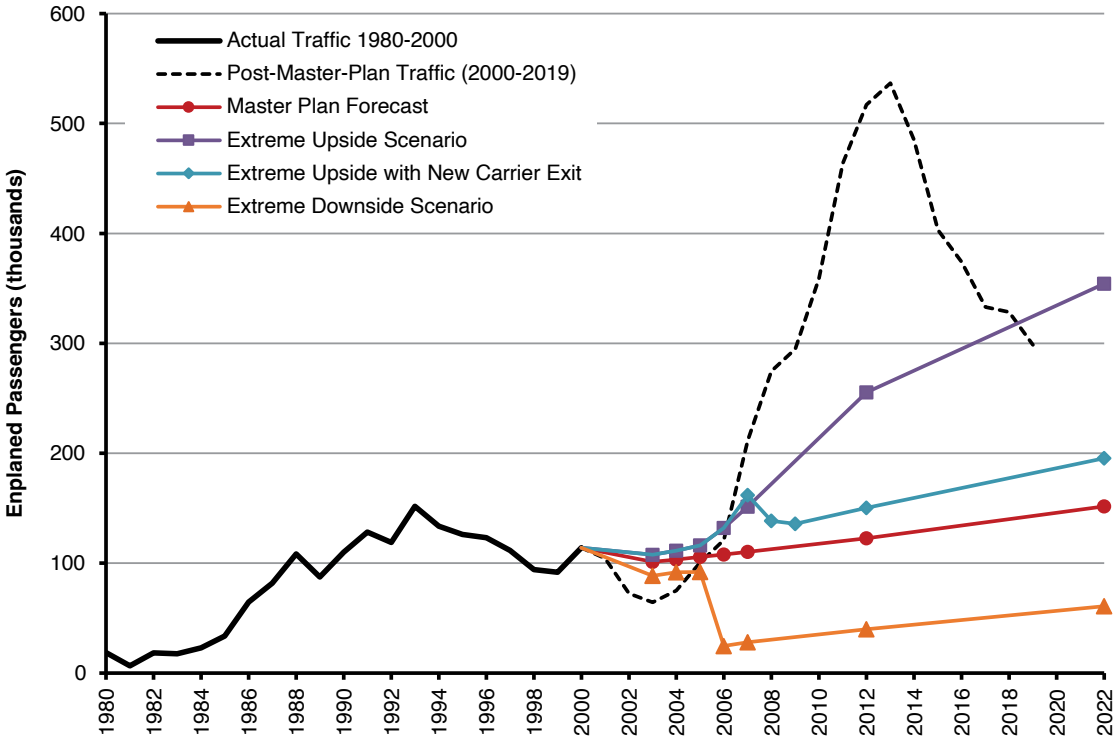


Figure 34. Example scenario forecast of carrier exit (illustrative only).

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NOTE: The actual traffic data has been updated to reflect traffic levels up to 2019 (the original chart ended the series in 2010).

Figure 35. Scenario forecasts for BLI (Kincaid et al. 2012, adapted).

- Extreme upside with new carrier exit – initial rapid growth associated with entry of a new LCC, but after 2 years the carrier scales back operations and leaves the market.

These scenario forecasts provided a wide range of outcomes with the extreme downside forecast 60% lower than the master plan forecast, and the extreme upside 2.3 times the master plan forecast. Also shown in Figure 35 are actual passenger volumes (the dotted black line). As can be seen, volumes initially exceeded the extreme upside but reduced as Allegiant scaled back services (as described in the Task 2 write-up).

It should be noted that the intention with scenario forecasts is not to necessarily produce more accurate forecasts but to challenge planners and decision-makers as to how these outcomes can be managed and mitigated in airport planning.

More Advanced Methodologies

Part II discussed the use of Monte Carlo simulation in air traffic forecasting. Monte Carlo simulation makes use of randomization and probability statistics to generate a wider range of possible traffic outcomes than a conventional forecast. As previously noted, shock events can be included in Monte Carlo simulation, but the perceived low-probability nature of shock events means that their impact may not be prominent in the output, and this could understate the significant and potentially profound effects of certain shock events.

However, there are ways to make the impacts of shock events more explicit in a Monte Carlo simulation. It involves "locking in" the event of interest so that the model always triggers the event at a given time. For example, the probability of a terrorist attack would be set at 100% in a given year. All other variables and parameters are randomized and so vary around this shock event. This is illustrated in Figure 36. The shock event is always triggered in Year 5 of the forecast,

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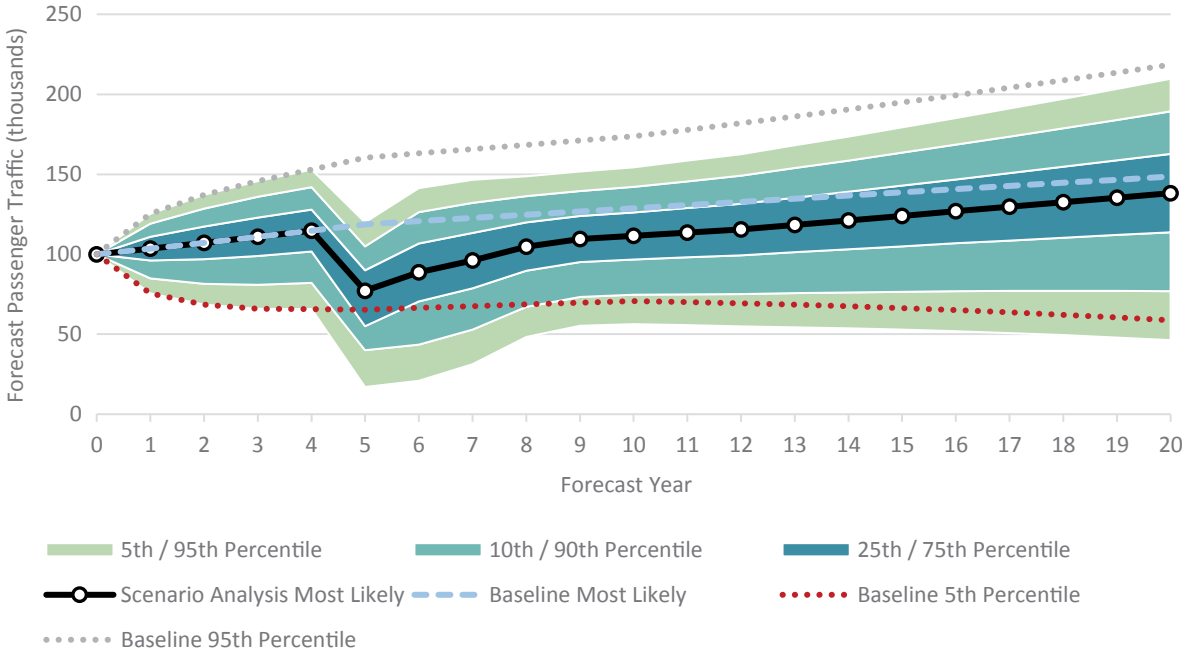


Figure 36. Scenario forecast within Monte Carlo analysis (InterVISTAS analysis).

although its magnitude is still randomized. The impact on traffic is shown as a range around the shock event—the extent of traffic decline and recovery being a function of these other factors as well as the shock event. Thus, scenario forecasts can be generated within the Monte Carlo analysis while still incorporating other uncertainties.

Further details on this approach are provided in Appendix C.

Simplifying the Process

As noted previously, the degree of complexity and detail that goes into the scenario forecasts will depend on the time and resources available and the requirements of the project. The sections above describe a number of methods for developing and modeling scenario forecasts with varying degrees of complexity. However, simple but still informative forecasts can be developed without the need for advanced methodologies. The type of forecast shown in Figure 34 can be generated using a small set of spreadsheet calculations, with the results presented in chart or table format. Similarly, if a small airport is considering a simple shock event involving the loss of half of its traffic, then a forecast based on a 50% decline in current traffic levels may be sufficient to stimulate thinking about the implications of such an event.

Outcome

This step will generate plausible and reasoned forecasts of traffic for a shortlist of critical scenarios with sufficient detail to allow decision-makers to fully evaluate the implications of these scenarios for the airport business. Although the examples above focus on passenger traffic, the scenario forecasts should give information on all relevant traffic elements, such as cargo and GA.

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CHAPTER 13

Step C: Scenario Implementation (or Stress Testing)

Evaluating Robustness

The next step in the process is to evaluate how the airport business would cope and respond to the scenarios previously developed. The scenario planning literature sometimes refers to this as "wind tunneling." In design and engineering, wind tunnels are used to test the aerodynamic performance of an object (e.g., a wing, airplane, car, or building). In the context of scenario planning, the scenario forecast is the wind tunnel, and it is the airport plans and business that are being tested, as described by van der Merwe (2008).

> Scenarios represent the different future conditions within which the strategy, business model or other decision must fit. Wind tunneling is used to test decisions for robustness and for exposing opportunities and risks. An important additional benefit of wind tunneling is that the leadership engaged in wind tunneling are continually adjusting their assumptions as they enter the different worlds described in each scenario. As leaders check their decisions or business mode in the various scenarios, they are often required to adjust their thinking based on evidence of flawed assumptions. This process is filled with critical learning opportunities in the scenario-based strategy framework and draws highly on constructivist learning principles.

Note: Constructivism is the theory that individuals construct new understandings and knowledge through experience and social discourse, rather than passively receiving information.

In the context of airports, this involves evaluating how the current airport plans and strategy would be impacted by the shock event and whether it has the tools and flexibility to mitigate or exploit the situation. Moving beyond the traffic aspects of the shock event, this can encompass a range of functional groups including (but not limited to) the following areas.

Planning

The master planning and general development of the airport are heavily guided by expectations around traffic development. In light of the traffic forecast generated for the shock event scenario, consideration can be given to the questions suggested in the following list.

- Are the current plans (or the plans being developed) able to accommodate the traffic implications of the shock event during the event and its aftermath?
- What areas of the plan may be challenged by the shock event?
- Does the plan have sufficient flexibility to allow for modification or adjustment? For example, can it be completed or adjusted incrementally to deal with substantially slower growth or accelerated to accommodate faster growth or changes in traffic mix?
- What aspects of the plan might hamper the ability to respond to the shock event?
- Does the plan allow for meaningful shedding of operating and capital costs when faced with a sustained traffic decline?

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- Does the plan include development of nonaeronautical revenue sources to provide financial resiliency in the event of significant traffic decline? Are these sources of revenue traffic-dependent, or are they more sustainable through a crisis?

Finance

The implications for the finances of the scenario forecast and measures that are in place to address impacts should also be considered. As previously documented, the COVID-19 pandemic had a major impact on both aeronautical and nonaeronautical revenues and led many airports to take cost-reduction measures. U.S. airports were able to utilize the grants and debt access available in the 2020 CARES Act and subsequent acts, but such support might not be available during and after other shock events. Issues that may need to be considered include the following:

- How will revenues and costs be impacted by the scenario forecast during the event and afterward?
- If revenue declines are expected, does the airport have sufficient reserves, access to finance, or a diversity of revenue sources that would allow it to weather the shock? What cost-reduction measures can be put in place (may need input from other departments)?
- If the airport needs to accelerate development plans or make new investments, is it in a position to do so (i.e., given current debt loads)? Are contract provisions for capital projects flexible enough to allow postponement or cancellation if there is a shock event?
- Are capital expenditure projects modular, allowing for small commitments during the project, or contracted for an entire major project?
- What level of government support can be expected, if any? Airlines and airports received direct support during the COVID-19 pandemic, but not all events may have this safety net, nor may the government be capable or willing to provide the same magnitude of support for another event.
- Does the business plan provide access to liquidity in times of crisis, e.g., are there sufficient cash on hand and reserves to comply with bond covenants?

Marketing and Commercial Activities

The impact on and role of these departments may also need to be considered. The strategies for airline marketing and route development may help mitigate some impacts of the shock event, or the event may present new opportunities, such as the ability to attract new carriers. Will marketing efforts include developing incentives to restore traffic or take advantage of new opportunities? The shock event could have implications for commercial activities such as retail, parking, real estate, and car rentals, all of which are linked to finance as well. For example, the entry of a new LCC or ULCC may change the spend rate of passengers and the retail, food, and beverage offerings they are seeking.

Operations, Safety, and Security

The implications for the safe and secure operation of the airport are always a priority. Consideration can be given to whether the shock event results in new security or health requirements or whether the advent of new aircraft technology will materially change operations (with implications for facilities and planning). There could also be human resources implications such as the need for recruitment to develop new skill sets.

Community/Stakeholder

Airport stakeholders, both inside (e.g., tenants, concessionaires, ground handlers, freight forwarders) and outside of the airport may be impacted by the traffic changes in the scenario

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forecast. Consideration needs to be given to what these impacts may be and what role the airport has in communicating with and managing these stakeholders. For example, the entry of a rapidly growing new carrier could raise activity and noise levels around the airport, leading to difficulties with the local community. Another consideration is business partners. During the COVID-19 pandemic, airports sought to provide relief and work in partnership with tenants.

Complete Coverage

Beyond the groups already mentioned, it is also important that all relevant aspects of the business are covered. While many airports are dominated by passenger air services, the impact on and role of cargo; GA; aircraft maintenance, repair, and overhaul; and other aspects of the airport business may also need to be considered.

The Engagement Process

The process by which the airport is evaluated against the scenario forecasts needs to be established. The recommended approach, drawing upon scenario planning literature, is in-person workshops whenever possible. This allows for open and free-flowing discussion of the issues and enhances engagement with the topic. As noted previously, the commitment of time by multiple participants may not be possible. In this case, the use of online video chats, surveys, email, or document exchanges will still bring about significant benefits.

A key aim of the exercise is to stress test the airport's plans. Therefore, there is a need to honestly assess the plans and not seek to protect existing ideas and work.

Before the workshop, information can be sent out to the participants to introduce the scenario planning exercise and possibly to conduct some initial analysis on the forecast data. For example, the finance group could prepare an analysis of the expected impacts of the traffic forecast on airport finances, although this is not a necessary pre-condition of the exercise.

The workshop should start with explaining the goals of the exercise and summarizing the work to date, including an overview of the scenario forecasts. Each scenario forecast is then presented in more detail followed by a discussion on the implications for the airport. Where there are participants from different departments, they can be put in the role of decision-maker for that area and consider how they could respond and what tools they currently have available to them. Key questions for participants to consider are included in the following list.

- What sort and scale of impact will the scenario forecast have on their department or decision-maker?
- What warning signs for the event might there be? Are there early signals that the airport could monitor to ensure that it is ready if the event does occur?
- What aspects are expected to hold up well against the scenario forecast?
- What aspects are expected to be most problematic or in need of attention?
- What contingencies or tools are in place that would help manage the impacts of a shock event?
- Which contingencies or tools would need to be in place to better manage the shock event, and why are they not available (or what is needed to make them available)?
- Would being in a position to better address the shock event require fundamental changes to the airport plans or strategy? If not, why not?

A key aim of the exercise is to stress test the airport's plans. Therefore, there is a need to honestly assess the plans and not seek to protect existing ideas and work. In some cases, there may be a tendency to be emotionally invested in the current plans and so be unwilling to consider

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potential weaknesses or shortcomings. The moderators need to make clear that the process is about enhancing robustness and resilience, thereby enhancing current plans, and not about tearing down or critiquing previous work.

If time and resources allow, another approach to stress testing the plans is to assign a red team or devil's advocate team whose primary goal is to identify problem areas or weaknesses in the plan when faced with the scenario forecast. As discussed in Part II, red teaming is an approach developed in military planning to identify vulnerabilities and has been used by the TSA to assess security screening. The process should be one of dialogue, challenge, willingness to critique ideas, and ultimately to devise better solutions and mitigation strategies to potential shock events. Participants should feel comfortable to engage in conversation and open to having their ideas challenged by participants. As with Step A, the moderator needs to ensure all voices are heard and that contrarian views are given a platform.

Simplifying the Process

The use of a facilitated engagement process involving multiple departments may not be possible for all airports. As noted previously, simplified approaches can be deployed instead, such as using online interactions or surveys. For smaller airports, it may be sufficient for a small group or even a single individual to list implications for the airport and share them with relevant stakeholders for confirmation (or challenge) and additional insights. Incorporating multiple viewpoints is a critical element of this process, but the amount of time and commitment from stakeholders can be reduced to ensure their involvement and to match available resources.

Outcome

By wind tunneling or stress testing the airport against the scenario forecasts, the exercise in Step C will identify areas of weakness in current plans and strategy (as well as areas of strength) and point to remedial options to be explored in Step D.

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CHAPTER 14

Step D: Develop Options for Improved Robustness

Generating Ideas for Improved Robustness

Although presented as a separate step, it is recommended that this part of the process be included as a follow-on to the engagement process in Step C. Having determined in Step C the issues arising from the scenario forecasts and areas of concern in current plans, the natural next step is to discuss how these can be addressed, including potential remedies. This should result in a more motivating and productive engagement process to focus on practical ideas to address the possible shocks. It will also reduce the time, resourcing, and scheduling required to bring the relevant participants together.

In much of the literature on strategic risk, responses are categorized into one of four categories:

- Avoid (or exploit for positive shocks),
- Transfer (or share for positive shocks),
- Mitigate (or enhance for positive shocks), and
- Accept.

These are elaborated further in Table 6, which draws on past research in *ACRP Report 76* (Kincaid et al. 2012) with additional commentary added. The participants should consider whether the overall approach to the impacts of a shock event falls into one of the four categories, which can then guide further development of the response strategies. For example, the *avoid* and *mitigate* responses can involve a mix of planning, financial, marketing, and operational responses while *transfer* and *accept* are more likely to involve more corporate (e.g., joint ventures) or financial (e.g., insurance, build up of financial reserves for contingencies). Airport-specific avoid and mitigate responses are discussed in the following subsection. In some cases, the approach may not fall neatly into one of the four categories or may span more than one. For example, an insurance policy may have a significant deductible, so this option would be a combination of transfer (for the insured amount) and accept (for the deductible).

Another consideration is the characteristics and value-added of a response option:

- **No- or low-regret options.** Decisions or actions that will yield benefits regardless of which scenario materializes and generally make good business sense. For example, an open terminal design that allows great flexibility in the use of the space will provide benefits beyond just coping with the traffic impacts of shock events. Similarly, the use of swing gates or MARS gates (described first in Chapter 8) not only provides greater flexibility in managing different traffic mixes but could also reduce overall space requirements.
- **Reactive options.** These are actions that would be taken when the shock event occurs. Often, they are specific to one event or a type of event (e.g., the loss of a major carrier, a terrorist attack, etc.). Although reactive, they can be thought out ahead of the event and thus reduce implementation time if and when needed. Airports can develop response plans for specific

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Table 6. Categories of risk and shock event responses.

Threats	Opportunities	Additional Commentary
Avoid Action is taken to eliminate the impact of a risk. For example, design and build to meet a much higher earthquake, tornado, or tsunami risk. Some threats can be avoided entirely by changing operations or eliminating practices deemed risky. These actions will often require incurring a cost, and eliminating risky practices might disappoint stakeholders or degrade the overall business case.	**Exploit** Make a proactive decision to take action and show that an opportunity is realized. See commentary.	As documented in Part I, some airports expedited facility and capital development plans during the COVID-19 pandemic to capitalize on lower construction costs during the pandemic and reduce disruptions due to the low levels of traffic. This also enabled more reliable project timelines and positioned the airport for strong growth in the recovery period.
Transfer The impact of the risk is transferred to another party that is willing to and better able to handle the risk (such as an insurance company or investors in a futures market). Services previously provided by the airport could be transferred (e.g., to the airlines) or contracted out. This may involve higher costs or payment of a fee (e.g., outsourcing to a skilled expert) or a premium (e.g., insurance).	**Share** Assign ownership of the opportunity to a third-party that is best able to capture the benefit for the operation. Examples include forming risk-sharing partnerships, teams, or joint ventures that can be established with the express purpose of managing opportunities.	Many airports have business continuity or business interruption insurance—insurance against a loss of business due to facilities being damaged or out of action (e.g., due to a fire or hurricane). However, in the case of COVID-19, there was no property damage, and some of the airports interviewed in Part I indicated that the insurance had exclusions for pandemics, communicable diseases, and "acts of God."
Mitigate Action is taken to lessen the expected impact of a risk. Mitigation generally requires positive actions, such as an investment, additional staff, or a change in procedures, and can have a resource cost. These actions should be considered new practices and controlled like any other airport operation. They may affect the airport operating budget but may be preferable to a "do-nothing" or "wait-and-see" approach (see discussion on evaluation in the next subsection).	**Enhance** Take action to increase the probability (e.g., marketing and incentives to attract a major new carrier hub) and/or impact the opportunity for the benefit of the operation. Seek to facilitate or strengthen the cause of the opportunity, and proactively target and reinforce the conditions under which it may occur.	One form of mitigation that may be effective for shock events is the development of action plans or manuals to guide decision-making during and after a shock event. This would be similar to the emergency response plans that airports develop for runway incidents, fires, and other emergencies.
Accept No action is taken. After being unable to avoid, transfer, or mitigate the threats, the operation will be left with residual risks – threats that cannot be reduced further. In active acceptance, airport management may set up a "contingency reserve fund" to account for the residual expected value of the remaining risks. A passive form of acceptance simply acknowledges the risk and moves forward with existing practices without reserves, which may seem sensible for risks with small expected values.	**Accept** Take no action when a response may be too costly to be effective or when the risk is uncontrollable and no practical action may be taken to specifically address it.	A number of the airports interviewed in Part I indicated that cash reserves that the airport had built up over previous years had aided the airport's response to the pandemic and its ability to manage its resources. Similarly, at least one transoceanic air navigation service provider had developed a traffic stabilization fund (reserve) to be used for either economic cycles or shock events.

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types of shock events similar to the emergency response plans that airports develop for runway incidents, fires, and other emergencies. These response plans are discussed in more detail in the next subsection.

- **Preemptive options.** These are options that require changes to the airport planning and strategy and may take considerable time to implement. They may also have substantial cost implications. For example, a modular terminal design that allows incremental development of the terminal in response to traffic growth and enables the pausing or acceleration of capacity growth may involve greater design and construction costs than if the terminal were to be built in one go.

To expand on the modular design, this approach may involve higher total costs as additional construction elements are required. A temporary end wall might be required at the end of the first phase of the building—a wall that would then need to be removed when the next phase is constructed. However, the present value of the construction costs of a modular design may be lower, as much of the expense is deferred to the future. Depending on interest rates, the present value computation of a modular design may be lower than building the entire terminal in one go. There are other advantages of modular design—such as the ability to incorporate new technologies and new nonaeronautical revenue opportunities—and to address future changes in regulations that would require an expensive refit of an all-in-one-go approach. The refit costs of new security regulations and required equipment after 9/11 are examples of how modular designs can better accommodate changing requirements for physical space planning compared to retrofitting existing facilities and the cost to change design plans not meant to be modular.

The no-regret and reactive options will generally be relatively straightforward and fairly low-cost to implement. The preemptive option may be more contentious with higher financial commitments, requiring further analysis and scrutiny. Such analytical approaches are set up in *ACRP Report 76*, including expected net present value and cost-benefit analysis, which weigh the cost of such mitigation measures against the benefits in conditions of uncertainty (Kincaid et al. 2012).

Airport-Specific Response Strategies

There are various strategies and actions airports can and do take to address uncertainty. While not specific to shock events, some of these apply to such events. Based on a review of previous research, observation at airports generally, and the experience of the research team, these strategies are summarized in Figure 37, ranging from strategic to technical. Further details on these approaches were discussed in Part II. The list is not exhaustive and is designed to act as a starting point for developing options. The specific circumstances of the airport and the shock risks it faces may lead to other solutions.

Preparedness and Response Plans

The development of shock event response plans has been raised in the previous sections and is discussed further in this section. The development of response plans is a relatively low-cost way of enhancing robustness to shock events and has parallels with the approach already used by airports to respond to emergencies and accidents.

The development of response plans is a relatively low-cost way of enhancing robustness to shock events and has parallels with the approach already used by airports to respond to emergencies and accidents.

Airports are already familiar with the development of emergency response plans (ERPs). Currently, these generally address safety and security issues, although there are other types of events that they have to address as well.

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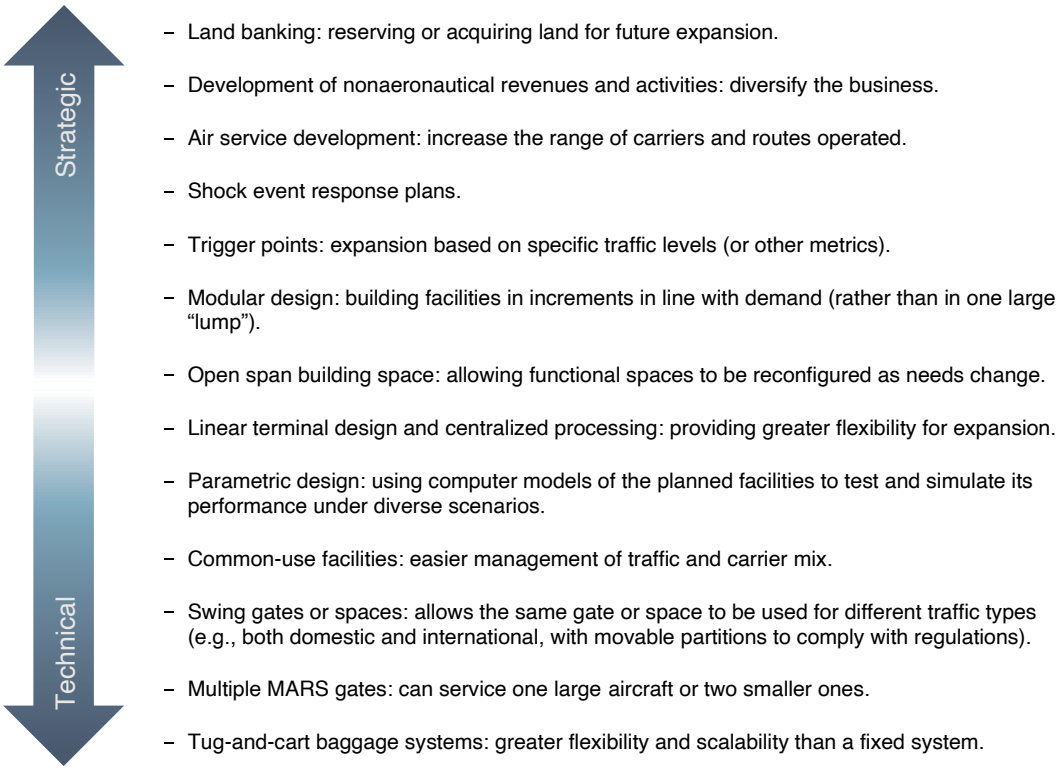


Figure 37. Airport strategies and action to address risk and shock events.

Some types of ERPs are a regulatory requirement. FAA 14 CFR § 139.325 – Airport emergency plan states the following.

§ 139.325 Airport emergency plan.

(a) In a manner authorized by the Administrator, each certificate holder must develop and maintain an airport emergency plan designed to minimize the possibility and extent of personal injury and property damage on the airport in an emergency. The plan must–
  (1) Include procedures for prompt response to all emergencies listed in paragraph (b) of this section, including a communications network;
  (2) Contain sufficient detail to provide adequate guidance to each person who must implement these procedures; and
  (3) To the extent practicable, provide for an emergency response for the largest air carrier aircraft in the Index group required under § 139.315.

(b) The plan required by this section must contain instructions for response to–
  (1) Aircraft incidents and accidents;
  (2) Bomb incidents, including designation of parking areas for the aircraft involved;
  (3) Structural fires;
  (4) Fires at fuel farms or fuel storage areas;
  (5) Natural disaster;
  (6) Hazardous materials/dangerous goods incidents;
  (7) Sabotage, hijack incidents, and other unlawful interference with operations;
  (8) Failure of power for movement area lighting; and
  (9) Water rescue situations, as appropriate.

. . .

(e) The plan required by this section must contain procedures for notifying the facilities, agencies, and personnel who have responsibilities under the plan of the location of an aircraft accident, the number of persons involved in that accident, or any other information necessary to carry out their responsibilities, as soon as that information becomes available.

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The development of an ERP involves identifying the resources that will be needed, where these can be obtained, the staff needed, and clearly stating the duties and responsibilities of the individuals (and organizations) in the response team. It also identifies actions and their sequence. This is a method and process that addresses certain types of shock events. It is a process that could be used to address other types of potential shock events. Item b (5), natural disaster, which is listed in the regulation, is a type of shock event.

As currently constructed, most ERPs focus on operational and human resource issues and tend to be largely related to the immediate response required in an emergency. These have been very successful in preparing airports for various types of emergencies and have been invaluable in practice. While they are focused on an immediate response to certain types of shock events, they also address some medium-term issues. For example, to mitigate the harm from an emergency, the ERP development process may identify the need for human and physical capital. Management may recommend and the board approve procurement of emergency response equipment or may undertake certain changes to the physical layout of the airport. Of particular importance is identifying training and skill-maintenance needs. In some cases, additional staff positions may be identified and hired. These are resources that, hopefully, may never be used, but their positioning will be invaluable should the shock event occur.

For certain types of potential shock events, an ERP approach would be a familiar and appropriate starting point. Additional steps would be required. For example, an ERP to deal with a major earthquake that destroys key parts of the infrastructure is focused on immediate-term actions. The ERP, however, typically does not address the medium-term restoration of the infrastructure. Incorporating shock events into airport planning would need to address the medium-term needs.

Two other modes of transport guide some aspects of short- to medium-restoration of capabilities: highways and railroads. Railroads experience floods or other weather-related or natural disasters that destroy sections—sometimes large sections—of track infrastructure (earthworks, ties, rails, ballast), including bridges. This is also true of state, county, and local highway departments. A number of these organizations identify, as an advance contingency, lists of potential external contractors and suppliers and prequalify them so that they can quickly be engaged for remedial supplies and work. A complementary approach is to identify and track availability of certain premade structures. Some highway departments keep track of bridge truss sections, for example, which may have been surplus to a recent construction project or which have been fabricated but not yet deployed on new infrastructure. This provides a starting point for remedial resources. In the airport context, an example might be tracking items, such as loading bridges, that were replaced when a terminal expansion was undertaken. Such resources could be accessed quickly to allow full operations to resume.

An ERP generally does not address financial issues. A major earthquake that destroys an airport and its region may result not only in financial capital needs for restoring airport capacity but also may suppress passenger and cargo demand for a prolonged period. Hurricane Katrina in 2005 left Louis Armstrong New Orleans International Airport relatively unscathed, which allowed it to serve as critical infrastructure for the disaster response. However, the destruction of major areas of the city resulted in reduced passenger traffic for several years, as documented in Part II. Many major conventions, for example, could not be hosted, and important parts of the population relocated to other regions, in many cases for years, while housing stock was reconstructed. Other global examples are found in the earthquake in Christchurch, New Zealand, and a tsunami in Sendai, Japan.

Planning for shock events should go beyond the immediate-term scope of an ERP and address financial and human resource resiliency. Airport charging structures and policies may

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need to be addressed. Compensatory (dual till) policies allow an airport to retain net income from nonaeronautical sources to build financial reserves for future projects and contingencies. Residual pricing (single till) policies generally do not build financial reserves or only make modest provisions for building contingency financial reserves.

Just as airport ERPs are typically managed by an officer who, working with a team, is given responsibility for developing plans, so too airports might convene a team to consider potential shock events and develop plans. This might consist of the ERP team but with an expanded set of expertise. The team should include individuals from operations, planning, capital projects, finance, marketing, and communications.

The response plan itself should provide key information and guidance such as

- Persons responsible within each department;
- Actions to be taken by those departments (this can be delineated over the timeline of the event, e.g., on the outside, during its development, and in the aftermath);
- Interactions with stakeholders and customers;
- Communications with government and media; and
- Recognition that the event may occur in ways not anticipated and provide options for alterations to the plan.

Simplifying the Process

Many of the possible strategies for addressing shock events (as listed in Figure 37) can also provide general cost savings for airports. For example, using swing or MARS gates can reduce the overall number of gates and terminal capacity required. As already highlighted, shock event response plans are a relatively low-cost method for addressing shock events, and for smaller organizations, the plans can be fairly short documents.

Outcome

This step will generate possible modifications or enhancements to airport plans and strategies that enhance the robustness of the airport to shock events. Where these modifications involve significant money and resources, further analysis may be required to evaluate their cost-effectiveness. The development of response plans will provide airports with "off-the-shelf" strategies to address shock events rather than having to develop the strategies in real time.

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CHAPTER 15

Step E: Ongoing Monitoring and Vigilance

The Need for Ongoing Engagement

The exercise described in the previous steps is designed to expand understanding of potential shock events and position the airport to be more robust to such eventualities. However, it is not sufficient to simply identify shock events and identify potential remedies as a one-off (or "checkbox") exercise; the consideration of shock events needs to lead to meaningful organizational changes and for the organization to be continually responsive and proactive to these risks. The aim should be to develop a much greater awareness of the potential impact of these shock events and for that awareness to continually feed into the ongoing and day-to-day planning, financing, and management of the airport. It will also ensure that the process and awareness continue even when there is staff turnover and other organizational changes.

It is not sufficient to simply identify shock events and identify potential remedies as a one-off exercise; the consideration of shock events needs to lead to meaningful organizational changes and for the organization to be continually responsive and proactive to these risks.

To ensure that the information on shock events is current and that the potential impacts of shock events continue to receive attention, the following actions can be considered.

- **Monitoring and early warning detection.** Tracking previously identified shock events and identifying new potential shocks. Step D puts forward a response plan that identifies individuals with specific responsibilities for shock event action plans. These same individuals can have responsibility for monitoring shock events and looking for early warning signs (e.g., airline financial distress or stories of illnesses in other parts of the world). Some airports may already have risk management teams or risk registers, and tracking and identifying shock events could be folded into that process.
- **Periodic and ad hoc updates.** Updates can be provided to the wider management and planning team regularly (e.g., quarterly) to maintain awareness of shock events and ensure that information is current. This could be in the form of an update memo (which regularly keeps the potential for shock events in mind) or a presentation to management on potential shock events (the latter having the benefit of providing a forum for debate). Ad hoc updates or warnings can also be sent if there are concerns that a shock event could occur or is emerging.
- **Full reassessment.** On a less frequent basis (e.g., every year or 2 years), the scenario planning exercise in Steps A to D can be redone or at least reviewed and updated. Alternatively, this could be done before any major planning or strategic decisions are made (e.g., updating the master plan). The aim is not to rewrite the airport's plans every year or two but to ensure that the focus remains on robustness to shock events and that the plans change as conditions change. Key questions to consider include the following.
  - Are there any new potential shock events that need to be considered?
  - Are any of the previously identified shock events no longer relevant?

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- Is there a need to produce and evaluate new scenario forecasts for previously or newly identified shocks?
- Are any changes needed to the airport plans and strategy?

Disposing of and Modifying Scenarios

The purpose of the scenarios and the scenario forecasts is to engage and challenge the thinking of decision-makers. They are designed to break the focus on "most likely" outcomes and encourage "out of the box" thinking on the possibility of shock events and how to address them. However, these scenarios are not forecasts of the future and should not be treated as such. Scenarios have a limited shelf life as circumstances change, and the temptation should be avoided to cling to them due to their familiarity.

Some scenarios should routinely be disposed of or, at the very least, modified, and new scenarios developed as events and thinking change. As an example, a potential shock event might be that a major carrier fails or dehubs operations at the airport. If the carrier resolves the issue, such as by a merger and commitment of the merged carrier to retain operations or by successful restructuring, then the shock event previously identified may no longer be relevant. As an example, in 2003, Air France and KLM merged under a common holding company, and there was concern that the airline group would dehub operations at Amsterdam Schiphol Airport (AMS) or downgrade AMS to a regional, intra-Europe hub. However, the airline holding company made a formal commitment to not dehub AMS and locked this commitment in through formal agreements. Similarly, in the 1990s, there were concerns in Canada that the precarious financial position of the nation's second-largest carrier, Canadian Airlines International (CP), would result in the failure of the carrier and loss of service at key hubs and spoke destinations where CP was the major carrier. When Air Canada acquired CP and made commitments to the government that it would retain a basic level of service at previous CP destinations, the concern for a major shock event reduced.

Communicating the Findings and Plans

Not all members of the organization can be involved in the scenario planning exercises, but action plans resulting from this process should be well-known and understood within the organization. As discussed in previous tasks, understanding and acceptance of risk and shock events varies considerably from individual to individual and so there is a need for clear and effective communication before and during any event. This is not to say that all the details necessarily need to be widely disseminated in all cases; certain types of shock events should be considered but are sensitive.

Communication can serve multiple roles in creating robustness against shock events and during a shock event itself as noted in the following list.

- Explaining and building confidence in decisions made to enhance robustness and decisions made during the crisis or shock event.
- Coordinating responses within the airport organization to ensure everyone is "pulling in the right direction."
- Coordinating with key stakeholders—airlines, tenants, government agencies, local communities, etc.
- Informing customers (passengers, shippers, etc.) and the general public and avoiding unnecessary public concern.
- Protecting "brand image" and customer or investor (bondholders) confidence.
- Generating interest in related issues (and in some cases, promoting the airport's plans).

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Simplifying the Process

Repeated updates to the scenario planning and forecasting may not be possible. However, taking actions such as routinely including risk factors and shock events on the agenda of relevant executive and board meetings will help to maintain consideration of these factors.

Outcome

Ongoing monitoring and evaluation of potential shock events feed through to all aspects of the organization, ensuring that the airport is better able to have a robust response to unknown and unpredictable shock events.

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PART IV

Application of the Methodology to Example Airports

The shock event methodology previously outlined has been applied to two example, real-life airports to illustrate the application of the methodology and to one illustrative GA airport to demonstrate the methodology on a smaller scale. The two example airports are Birmingham-Shuttlesworth International Airport (Chapter 16) and San Francisco International Airport (Chapter 17).

The two airports differ in size, traffic, and carrier mix and in the way the pandemic impacted their operations. The methodology was applied to a period in the past and as such, the project team has the benefit of hindsight. To the extent possible, the project team tried to work with information that would have been available to the airport management at the time. For example, it is not realistic to expect that a decision-maker could foresee the COVID-19 pandemic as it transpired. However, the possibility of some form of regional or short-term pandemic is more realistic given past examples with SARS and the swine flu.

The purpose of these examples is not to critique the forecasting or planning at these airports but rather to illustrate how the methodology can be applied and how it might affect decision-making. Draft versions of the methodology applications were provided to the airports' management; their feedback was incorporated into the final versions in this guide. However, most importantly, the approach, workings, and conclusions *remain those of the ACRP research team alone* and *do not* purport to reflect the opinions of the airport management.

While these two examples involve commercial airports, the majority of all airports in the United States are small, private-use facilities and only approximately one-quarter of all airports in the United States are under the NPIAS, of which the majority are commercial service airports (FAA 2022). Smaller, GA airports, either publicly or privately owned, may not have the resources available to conduct the full methodology as described in this guide. Therefore, an example of how the methodology can be employed at a small scale with an illustrative GA airport is provided in Chapter 18.

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CHAPTER 16

Birmingham-Shuttlesworth International Airport

Introduction

Birmingham-Shuttlesworth International Airport (BHM) was previously profiled in Chapter 4. BHM is a small hub airport operated by Birmingham Airport Authority, 5 miles northeast of central Birmingham, Alabama. It is the largest commercial service airport in Alabama, handling 1.5 million enplanements in 2019.

The methodology was applied to the airport during the period its airport master plan was being developed; the master plan was finalized in July 2018 (Birmingham Airport Authority 2018). The master plan set out the forecasts of traffic development to 2035 and the facility development that would be needed to accommodate that traffic and modernize the airport in this period and beyond. The forecasts for the master plan were developed in 2014, with forecasts of air passengers, aircraft movements, based aircraft, and air cargo tonnage generated for the period 2014 to 2035. The passenger forecasts are shown in Figure 38. To account for the inherent uncertainty of aviation demand forecasting, the forecast team generated three forecasts that captured different economic and airline industry conditions using regression modeling:

- **Moderate:** Employment and regional GDP to grow at a similar rate to recent trends, with regional GDP growing at 1.3% to 1.5% per annum (in real terms). Airline fares at BHM are assumed to outpace the region's economic growth, resulting in some demand suppression.
- **Optimistic:** Employment and regional GDP to grow at 50% higher than the moderate forecast and airline yields to grow at a slower rate than the national trend.
- **Aggressive:** Employment and regional GDP to grow in line with historical highs, in the range of 2.7% to 3.2% per annum for GDP growth. Airline yields to grow at a slower rate than the optimistic forecast.

Under the moderate forecast, traffic was forecast to decline by an average of –0.1% per annum from 2014 to 2035, while the optimistic scenario forecast growth of 1.2% per annum and the aggressive scenario of 2.3% per annum. Additional forecasts were also generated using a market share methodology, where traffic at BHM was forecast based on its projected share of forecast Alabama or national traffic. These market share forecasts fell within the range of the regression-based forecasts.

Figure 38 also shows the FAA Terminal Area Forecast (TAF) referenced in the master plan (published in February 2014). The TAF was close to the optimistic forecast as well as the market share forecasts (not shown) produced by the forecast team. For master planning purposes, the FAA TAF was selected as the primary forecast informing the airport master plan. This forecast growth of 1.8% over the planning period is higher than the optimistic forecast.

As the chart shows, traffic development after the forecast was produced closely matched the FAA TAF and optimistic forecast up to 2017. In 2018, traffic grew by 9.9% and was tracking

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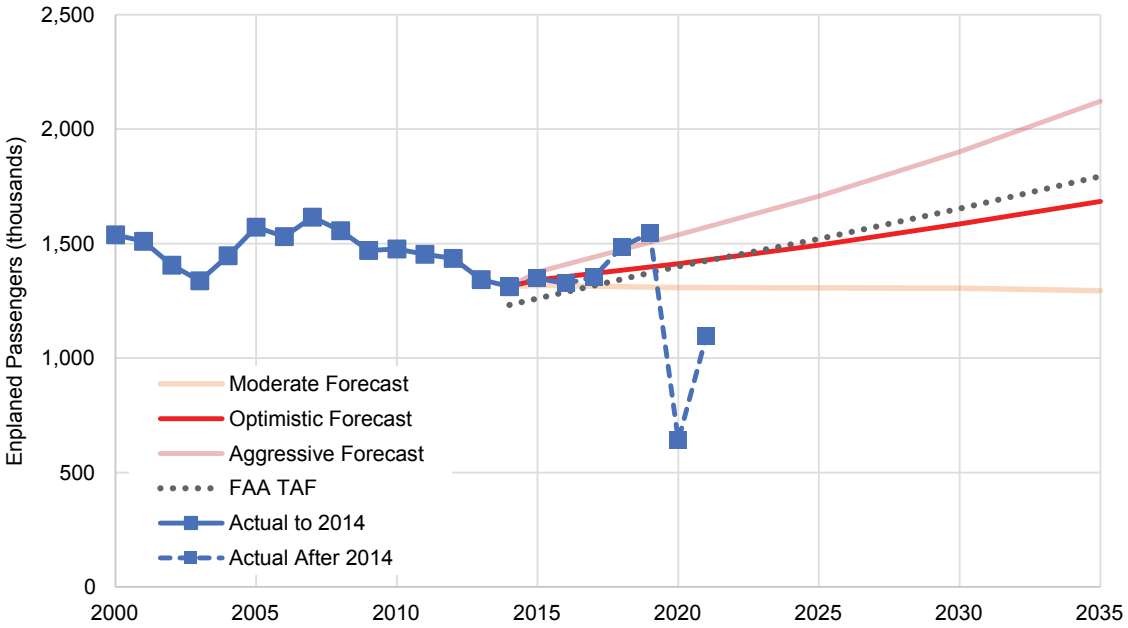


Figure 38. Forecast passenger traffic at BHM (Birmingham Airport Authority 2018).

above the aggressive forecast. This growth was due to general demand growth, the entry of Frontier Airlines (who subsequently ended service in late 2019), and capacity additions by a number of network carriers. As previously documented, the COVID-19 pandemic resulted in a 59% decline in passenger volumes in 2020 and a partial recovery in 2021.

Immediately prior to the master planning process, the airport has completed a terminal modernization program and runway extension. This put the airport in a good position to accommodate growth; therefore, the development of this master plan emphasized three areas:

- Accommodating new design standards that enhance safety and efficiency;
- Addressing operational and developmental constraints needed to support growth; and
- Creating development areas to support revenue generation.

The master plan also considered the need for a parallel runway. The airport currently has two crossing runways. The third runway would be parallel to the main runway. It was assessed that the runway would not be required before 2035, and traffic trigger levels were set for when the process of planning and building the runway should be started. The master plan set out a capital improvement program totaling $362.3 million (in 2016 dollars), with the following timeline:

- Short-term (1–5 years): $125.8 million.
- Medium-term (6–10 years): $149.9 million.
- Long-term (11–20+ years): $86.6 million.

Major capital projects included culverting or realigning a waterway (Village Creek) that ran through the airport (this accounted for over $100 million of program costs), reconfiguration of the taxiways and runway improvements (for regulatory reasons rather than capacity), road realignment, air cargo facility expansion, and site preparation and development.

Application of the Methodology

The five-step methodology was applied to BHM as described herein.

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Step A: Identify Shock Events of Interest

The identification of potential shock events relevant to BHM was made through structured brainstorming discussions within the ACRP research team as well as information on shock events from previous tasks and information obtained from the master plan document and other sources. The list of identified shock events is shown in Table 7. The list is fairly wide-ranging without any filtering applied; the next stage of the process narrows the list to those events considered most pertinent.

Step B: Develop Scenario Forecasts That Incorporate Shock Events

The events in Table 7 were shortlisted to three that were considered most relevant and challenging:

- **Loss of a major carrier.** While BHM was not highly dependent on a single carrier in 2014 (which is still the case in 2022), two carriers accounted for more than 60% of the traffic. The loss of one of those carriers would have significant implications for the airport. It was recognized that this can be linked to a recession event as an economic downturn may precipitate the carrier exit, so this scenario covers that aspect as well.
- **Entry of a new carrier.** Ensuring the airport has contingencies if more rapid growth does occur.
- **Terrorist attack.** Using this scenario forecast could cover multiple potential events where there is a sharp drop in traffic followed by recovery.

Scenario forecasts were developed for these three events, which model the deviation from the selected forecasts used in the master plan (the FAA TAF of February 2014). The key assumptions in the three scenario forecasts are summarized below.

Scenario Forecast 1: Loss of a Major Carrier

- The event is assumed to occur after 2 years (i.e., 2016). At this point, the national or regional economy is weakening, resulting in a softening demand for air travel.
- The largest carrier at BHM, Southwest Airlines, is assumed to exit at the start of 2016, due to a change in airline strategy or poor market performance.
- All of Southwest's services are terminated at that point. In 2014, Southwest Airlines was operating seven routes from BHM to Dallas Love Field, Baltimore, Orlando, Chicago Midway, Tampa, Houston Hobby, and Las Vegas.
- Immediately after the exit, some traffic is recaptured through higher-load factors on the remaining carriers where they serve the same market or can provide connecting alternatives to the lost routes. For example, United Airlines operated to Chicago O'Hare and Houston George Bush and American Airlines operated to Dallas Fort Worth and Washington National (the latter a possible alternative to Baltimore).
- Carriers serving these overlapping markets increase capacity over the next 2 years to recapture some of the demand, with capacity on those markets increasing by 25% over that period. Other routes (Orlando, Tampa, Las Vegas) are assumed to remain unserved.
- Long-term traffic growth follows the master plan forecast trend.

The purpose of this scenario forecast is to examine the impact of a negative event, so the assumptions made are fairly conservative. For example, it is possible that with the exit of Southwest, another LCC may enter the market operating additional routes or that the existing carriers will in-fill more of the lost capacity. However, it has been assumed that only modest capacity recovery will occur, reflective of a weaker economic environment.

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Table 7. Identified shock events based on the 2014 outlook.

Event	Description of Impact	Timeline and Aftermath
Major carrier exit	Loss of a major carrier. In 2014, Southwest Airlines (including AirTran Airways) accounted for 42% of passenger traffic and Delta (mainline) 20% (Birmingham Airport Authority 2018). AirTran Airways was acquired by Southwest Airlines in 2011, with the merger finalized in December 2014. The figure for Delta does not include regional services. The loss of Southwest Airlines would result in not just the loss of capacity but also connectivity, as the airline operated seven routes at the time, while the network carriers operated 2 to 3 routes to their major hubs (American Airlines added more routes in 2015).	The immediate impact of Southwest's exit would be a significant loss of traffic, particularly as the routes operated by Southwest did not fully overlap with other carriers. Some capacity may be in-filled by other carriers, but there would be uncertainty whether a full recovery would occur. While there is the potential for traffic to remain depressed for some time, it may not follow the pattern seen at airports such as STL (see Chapter 5) as BHM does not act as a connecting hub in the way that STL did (much of the traffic lost at STL with the exit of American Airlines was transfer traffic).
Entry of a new carrier	Entry of a new carrier, e.g., a ULCC such as Allegiant Air or Frontier Airlines. As documented previously, the entry of these carriers has stimulated significant volumes of incremental passenger traffic, often at a rapid rate. Frontier did in fact operate at BHM in 2018 and 2019.	The entry of a ULCC may result in rapid traffic growth, especially if there is a competitive response by incumbent carriers. The long-term success of the carrier is a risk factor with this event. The carrier could later exit due to low demand or a change in airline strategy, or an existing carrier could decide to exit the market.
Severe recession	A severe recession along the lines of the 2008–2009 Great Recession. Such a recession would lead to declining passenger demand, airline capacity cuts, and possible airline restructuring and rationalization.	Nationally the 2008–2009 Great Recession led to an 8.7% decline in traffic and a 7-year recovery. BHM experienced an 8.9% decline from 2007 to 2009, with traffic still not recovered by 2014. Another recession of this scale would result in further declines and a protracted recovery.
Loss of local industry	Loss of a major regional employer such as the Honda manufacturing plant or one of the financial or health corporations based in Birmingham. Such a loss is expected to result in a loss of corporate travel as well as discretionary travel by people previously employed at those businesses.	Short-term decline in traffic, followed by a decline in the long-term growth trend.
Terrorist attack	A terrorist attack targeting aviation in the United States, leading to a sharp decline in traffic.	BHM experienced declines in the two years following 9/11, which was also due to economic factors, and did not fully recover until 2005. Another terrorist attack would result in short-term declines in traffic due to loss of consumer confidence and government security requirements. Recovery would be similar to that after 9/11. Depending on the nature of the attack, it could also result in new or additional security requirements, affecting passenger processes and airport design.
Pandemic	BHM is impacted by a pandemic event, which may be regional, national, or global.	BHM was not affected by past pandemic events. The potential event impacting BHM would be expected to follow a similar pattern to SARS, with a short-term and large-scale decline in traffic due to government controls on air travel. The event is anticipated to last months rather than years, with a fairly rapid recovery as was seen with SARS.
Climate event or climate change	Birmingham's inland location means that it is protected from the more destructive effects of Atlantic hurricanes. However, it is subject to severe thunderstorms and tornadoes. The assumed event would be an "extreme" weather event resulting in significant damage to the airport and city.	A sharp, short-term decline in traffic due to infrastructure damage, followed by recovery over the next 2 to 3 years. An extended recovery is not anticipated such as that experienced by MSY due to the considerable population decline and loss of tourism following Hurricane Katrina.

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Scenario Forecast 2: Entry of a New Carrier

- The event is assumed to occur after 2 years (i.e., 2016). Economic conditions are generally positive.
- A new LCC starts operation at the beginning of the year, operating 166-seat aircraft, and builds up service on routes not served from BHM as well as competing on some existing routes.
- The carrier starts service in the first year to three destinations with an average frequency of four times per week, and the carrier can achieve an average load factor of 80%.
- It is assumed that 90% of its traffic is incremental (10% is taken from other services at BHM). The entry of the carrier draws a competitive response from existing carriers (fare reductions and service increases), so the overall effect has a strong stimulatory impact on passenger traffic.
- In the second year, three more destinations are added (at four times per week) and two more in the third year.
- Frequencies and destinations grow in the following years so that 5 years after entry, the carrier is operating an average of six departures a day (less than half the level of operations by Southwest Airlines in 2014).

Scenario Forecast 3: Terrorist Attack

- Terrorist attack occurs in 2016 targeting aviation, directly impacting air services at BHM.
- The government introduces strict measures to enhance aviation security, and consumer confidence is hit, resulting in a 50% decline in overall traffic volumes that year (more severe than the impact of 9/11).
- As new security measures are introduced, demand starts to return, recovering to 75% of preevent levels after one year. By the second year, traffic is close to full recovery, at 96% of pre-event levels.
- Traffic development returns to the preevent trend after 3 years.

The resulting scenario forecasts are shown in Figure 39 alongside the FAA TAF used for the master plan. The three scenario forecasts show significant deviation from the master plan forecasts

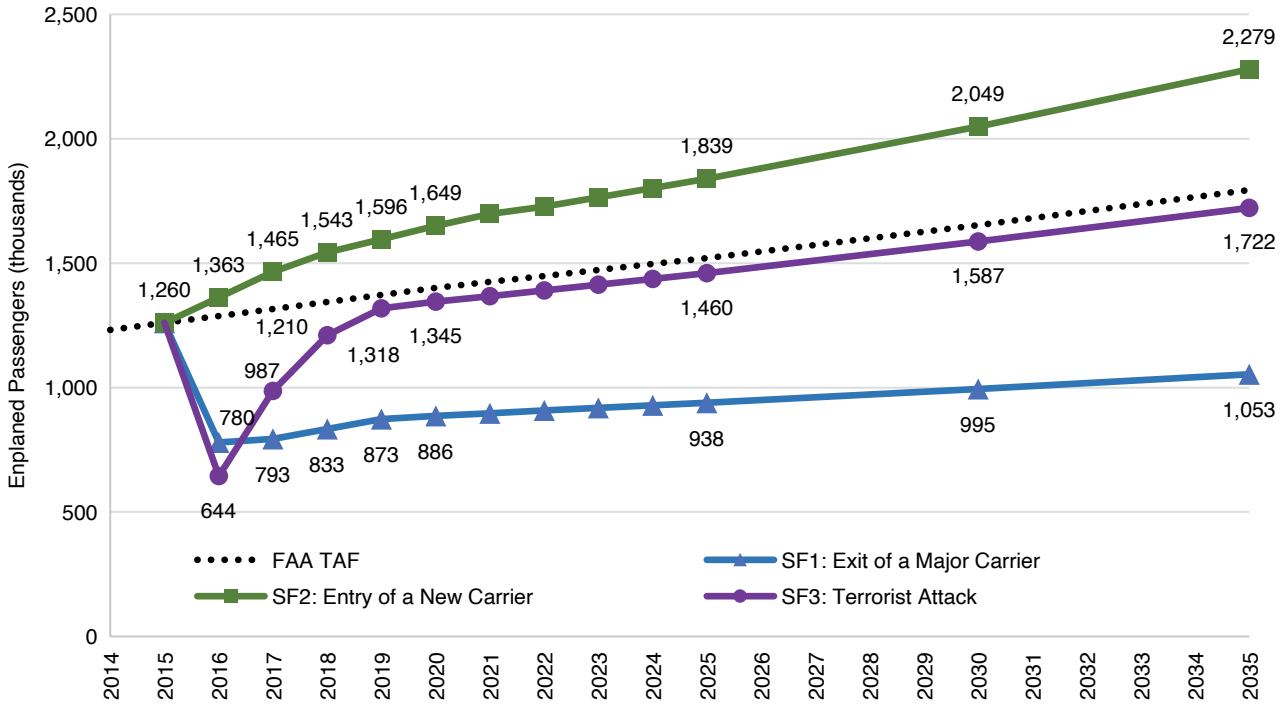


Figure 39. Scenario forecast results for BHM.

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taken from the FAA TAF. Scenario Forecast 2 (entry of new carrier) results in a 27% growth over 3 years (compared with 9% in the master plan forecast) and Scenario Forecasts 1 and 3 result in declines of 40% to 50%.

Compared with the forecasts developed in the master plan shown in Figure 39, the long-term range of the scenario forecasts is only slightly wider. The 2035 forecasts range from 1.05 to 2.28 million enplaned passengers compared with the 1.29 to 2.12 million range in the original master plan forecasts (shown in Figure 38). However, the scenario forecasts illustrate a more rapid change in traffic caused by the shock event, against which the master plan can be evaluated. Additional detail can be added to the forecasts, if required, by also producing forecasts of aircraft operations and peak-period traffic.

As mentioned previously, the purpose of these scenario forecasts is not to produce a more accurate forecast. None of these forecasts capture the traffic impacts of the COVID-19 pandemic. Instead, they provide a way of challenging and testing the master plan, as shown in the next step of the process.

Step C: Scenario Implementation

The next step of the process is to evaluate how the airport business would cope and respond to the scenarios previously developed. The idea is to stress test the master plan (and other aspects of the airport). Another elicitation exercise was undertaken with the ACRP project team to assess the impact of the scenario forecasts on five aspects of the business:

- Planning;
- Finance;
- Marketing and commercial activities;
- Operations, security, and safety; and
- Community and stakeholders.

The assessment is summarized in Table 8.

The assessment found areas of strength in the master plan and the airport's situation. Following the previous terminal and runway work, the airport was well-positioned to handle upside shocks, while the master plan did not involve a large commitment to capital expansion, mitigating downside shocks. As with most airports, fairly rapid and potentially sustained changes in traffic levels (particularly declines) would present challenges to airport planning and management. Possible measures to enhance the robustness of these shocks are explored in the next step.

Step D: Develop Options for Improved Robustness

Having determined in Step C the issues arising from the three scenario forecasts, the next step was to determine what actions could be taken to address these issues and improve the robustness of the airport plans. Although presented as a separate element, the process of brainstorming and developing possible remedies and mitigation actions was conducted by the ACRP research team at the same time as Step C.

Table 9 sets out the proposed strategies and approaches to enhance the robustness of the three scenario forecasts. The development of these options draws on the ideas set out in Part II of the guide as well as previous research in *ACRP Report 76* (Kincaid et al. 2012).

The proposed modifications to the master plan are based on an examination of how aspects of the plans would be impacted by the scenario forecasts and identifying areas where the plan could be modified or adapted to these conditions. While maintaining the overall direction of

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Table 8. Impacts of the scenario forecasts (SF), BHM.

Area	SF1-BHM: Loss of a Major Carrier	SF2-BHM: Entry of a New Carrier	SF3-BHM: Terrorist Attack and Aftermath
Planning	Given recently completed airport improvements (e.g., terminal modernization and runway extension), the 2018 master plan was largely focused on investments to achieve operational improvement, regulatory compliance, and revenue diversification. Major facility expansion was not anticipated out to 2035. Some elements of the plan could be delayed or slowed in light of reduced traffic, but other elements would still need to go ahead to meet operational and regulatory requirements. The master plan identified traffic trigger points for very long-term developments (e.g., a parallel runway).	The master plan determined that there was generally sufficient airfield and terminal capacity to handle forecast growth to 2035. Therefore, more rapid growth could largely be accommodated. (The terminal modernization was planned based on forecasts of up to 2.5 million enplanements, which is higher than the 2035 forecast in SF2.) However, there could be capacity constraints in certain areas or processes, particularly if the new carrier increases peak-period traffic volumes. Plan to manage within existing facilities, given long-term viability is uncertain.	Master plan is largely focused on operational improvement, regulatory compliance, and revenue diversification. These plans may not need to be delayed or delayed only for a short period. As with 9/11, the event could bring about additional security requirements and processes. The terminal modernization has improved security processes, and the 2018 master plan includes improvements to perimeter security.
Finance	The loss of traffic would significantly impact aeronautical and nonaeronautical revenues. Airline agreement is a mix of residual (airfield) and compensatory (terminal), so airport charges will likely rise, reducing some aeronautical revenue losses. Cost reductions will need to be sought, particularly on nonaeronautical activities. Bond covenants require a certain amount of cash on hand and reserves. The event would likely lead to a dramatic drop in revenue, and action may need to be considered to maintain the reserves in order to fulfill those covenants.	Entry of a new carrier would provide upside revenues, especially around terminal charges and nonaeronautical revenue (e.g., parking, concessions). Operating costs could also increase to manage the additional volume.	Reduced aeronautical and nonaeronautical revenues during the recovery period. Degree of government support (as occurred after 9/11) is hard to ascertain. Airlines will also bear some of the revenue risk. The impact on available reserves would need to be monitored to meet bond covenants. Increases in capital and operational costs are possible due to additional security measures.
Marketing and commercial activities	Air service development activity will need to ramp up to help replace service. May need to increase the marketing budget at a time when airport revenues are challenged. Master plan also covered cargo and commercial activity, which could be accelerated to offset passenger losses.	Passenger mix may change (e.g., more leisure travelers), which could impact the commercial revenue performance and mix.	Use incentives and other air service marketing to encourage the return of air service and to aid airline partners.
Operations, safety, and security	Declining traffic levels may lead to operational reductions (with corresponding cost savings) balanced against the need to maintain safety and service levels.	Would increase the need for greater operational capacity, especially at peak times.	Lower traffic levels may reduce operational requirements, which may be offset by additional security requirements.
Community and stakeholders	Will need to work with airport stakeholders and contractors to ensure business sustainability and achieve cost reductions. Significant loss of connectivity for the community. Work with local tourism and business representatives to develop a plan to grow air service.	Largely beneficial for community connectivity. Potential for greater aircraft noise (or perceived noise), following declines in air traffic and associated noise (as of 2014), although traffic levels would still be below historical highs.	Will need to work with airport stakeholders and contractors to ensure business sustainability and achieve cost reductions.

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Table 9. Options for improving robustness at BHM under the scenario forecasts.

Scenario Forecast (SF)	Mitigation/Robustness Options
SF1-BHM: Loss of a major carrier	**Evaluate Plan Optionality** Identify elements of the master plan that could be postponed or modified in the event of a large traffic decline in order to reduce capital and operating expenditures. This would involve separating elements of the capital development plan that are capacity-related from those that are related to maintenance or regulatory requirements. It can also identify parts of the existing facility that could be converted or shut down. This identification would need to balance cost savings with ensuring that the changes do not jeopardize the ability to recover and grow traffic in the future. **Modular Design and Trigger Points** Related to the optionality, identify where the plans for the airfield and terminal facilities can cost-effectively be made modular and incremental so investments are as closely aligned with traffic as possible and provide the flexibility to delay development if necessary. Identified trigger points, in terms of annual or peak traffic, would determine the timing of the development. Downside trigger points could also be identified to determine when plans should be slowed down or facilities converted or shut down. **Shock Event Response Plan** The response plan would identify the actions that need to be taken across the organization, should the event occur. This includes the planning items above but also covers other departments and areas: • Marketing and air service development activities to try to recover the lost traffic and revenues. • Changes to operations in order to reduce costs while maintaining service levels. • Managing the impact on tenants and stakeholders and possible contractual changes. • Managing the impacts on aeronautical and nonaeronautical/commercial revenues. • Other financial implications and how these can be mitigated.
SF2-BHM: Entry of a new carrier	**Evaluate Plan Optionality** Identify elements of the master plan that may need to be accelerated or modified as a result of higher traffic volumes. The overall facility has capacity for further growth but there may be specific areas of constraint. The evaluation should also consider additional facility requirements from the incremental traffic that are currently not within the master plan (e.g., terminal security processing, commercial facilities, employee and public car parking). These elements could be identified and characterized at a high level so that the information can be drawn upon if the event occurs. **Modular Design and Trigger Points** The use of modular design and trigger points will minimize the risk of committing to significant capital investment when it is uncertain how the traffic will develop and whether the incremental growth will be sustained. Identified trigger points, in terms of annual or peak traffic, would determine the timing of the development. **Shock Event Response Plan** The response plan would identify the actions that need to be taken across the organization should the event occur. This includes the planning items above but also covers other departments and areas: • Changes required to the airport's marketing and air service development plans in light of the new traffic and connectivity. Air service development contingencies if the carrier later exits the market. • Commercial activities that may arise as a result of the incremental traffic. • Additional operational requirements and costs resulting from the new carrier. • Managing the impact on tenants and stakeholders. • Implications for airport charges (mixed residual and compensatory) and airport revenues and costs.

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Table 9. (Continued).

Scenario Forecast (SF)	Mitigation/Robustness Options
SF3-BHM: Terrorist attack	**Evaluate Plan Optionality/Modular Design and Trigger Points** As with the other scenario forecasts, the identification of plan optionality and the use of modular design and trigger points will allow the airport to adapt the plans to changing traffic conditions. Additional consideration would also be given to the potential for new security requirements arising from the event. While impossible to predict in detail, mitigation measures include identification of terminal space that could be used for future expansion of passenger or luggage screening. This space could be initially allocated to other purposes (e.g., retail, office space) until it is required for security purposes (or in case it is never required). When required, the space can be converted to security. **Shock Event Response Plan** An event response plan can also be delivered that addresses anticipated actions and changes required in the short and long term: • Marketing and incentive plans to support airlines and recover traffic during the downturn. • Operational changes during the downturn in traffic and to cover security changes. • Communication plans with stakeholders and passengers to recover confidence in air travel. • Managing financial implications of the initial traffic decline and recovery.

the 2018 master plan, consideration is also given to how more optionality and flexibility can be added to the plan to manage shock events. Outside of the master plan, the main output is the "shock event response plans." Similar to ERPs, these involve multiple departments (e.g., operations, planning, capital projects, commercial, finance, marketing, and communications), and set out the parties involved and actions to be taken. Unlike ERPs, these plans will be generally less prescriptive, as the exact nature and timing of the event is hard to predict, and with a longer-term focus. The aim is to give the airport a head start in dealing with a shock event and set out a broad game plan or checklist for the organization while allowing sufficient flexibility to address changing conditions.

An example is given in Figure 40 with summary information for one department, terminal operations (the example is not based on BHM), although airports may prefer different formats. In the example format, a summary of key actions is provided, with linkages to other areas of the shock event response plan for more detail.

Step E: Ongoing Monitoring and Vigilance

Step E relates to the ongoing monitoring of potential shock events and updating of planning and processes where necessary. The aim is to maintain ongoing vigilance around the development of potential shock events and ensure that the airport continues to be well-positioned to manage these shocks if they occur. This can involve the following:

- **Monitoring and warning.** Individuals identified in Step D would be tasked with monitoring identified shock events and possible new events (e.g., airline financial distress, WHO advisories). This can be folded into existing risk management processes. Concern about developing events (e.g., outbreak of a novel virus in some part of the world) can be communicated to management (and if relevant, other stakeholders) and meetings undertaken to determine possible responses and preparation.

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Shock event	Loss of a major carrier
Department	Terminal operations
Officer responsible	*Name*, Director of Operations
Short term (3–6 months)	▪ Monitor peak flows and activity levels. ▪ Shut down gates X, Y, and Z and consolidate at remaining gates (subject to monitoring). ▪ Work with contractors A and B for reduced service provision (and cost reduction) – see page XX for more detail.
Medium term (6 months–3 years)	▪ Assuming depressed traffic levels, shut down Concourse X. ▪ Downside trigger for concourse closure: 1m passengers and 5 peak-hour operations (subject to management approval). ▪ See page X for further details on the required concourse changes. ▪ Work with contractors A and B for changes in scope of work. ▪ Apply staff organization plan – see page XX.
Long term (more than 3 years)	▪ Assuming downside triggers remain, conversion of Concourse X (subject to management approval). ▪ Work with planning on conversion of Concourse X to alternative uses (office, retail) – see page XX for more detailed plans.
Internal interactions	Planning – *Name*, *Name*, Senior Planner Finance – *Name*, *Name*, Director of Finance Commerce – *Name*, *Name*, Retail and food and beverage
External interactions	Contractor A (maintenance) – Contact information Contractor B (cleaning) – Contact information

Figure 40. Example summary shock event response plan for terminal operations.

- **Regular updates.** Quarterly or half-annual update memos are produced, incorporating assessments and tracking of potential shock events and other uncertainties, and circulated to the relevant airport team members.
- **Reassessment annually or biennially.** Reassess or redo Steps A to D to ensure that the airport and its plan remain well-positioned to shock events as conditions change. The aim is not to rewrite the plan every year but to allow it to respond to evolving situations and potential shock events. This assessment could be incorporated into the airport's reporting, such as annual reports.

The ultimate aim of these exercises is to enhance and maintain awareness of the potential impact of these shock events and for that awareness to continually feed into the ongoing and day-to-day planning, financing, and management of the airport.

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CHAPTER 17

San Francisco International Airport

Introduction

San Francisco International Airport (SFO) is classified as a large hub, commercial service airport under the NPIAS and is owned and operated by the City and County of San Francisco.

SFO is the second busiest airport in California (after LAX) and the primary international gateway airport for the San Francisco Bay Area. In 2019, the airport served 27.9 million enplaning passengers, of which 7.5 million (27%) were international enplanements. The airport is a hub for United Airlines for both domestic and international flights and functions as that carrier's primary transpacific gateway airport. SFO is also a hub for Alaska Airlines. SFO is located on the shores of the San Francisco Bay with all four of its runways extending into the bay on artificial land built up from the foreshore.

The methodology is based on the analysis of SFO's Draft Final Airport Development Plan (ADP), effectively its airport master plan. The current ADP was developed between late 2014 and early 2016 and sets out a comprehensive outlook for airport development to accommodate a long-term demand of approximately 35.5 million enplaning passengers.

The aviation activity forecasts were developed during the ADP's production process and were accepted by the FAA in 2014 based on a 2013 baseline. Unlike traditional master plan forecasts that forecast every year or produce outputs at 5-year intervals over a 20-year timeframe, the ADP's forecasts were developed for four timeframes: 2018, 2023, base-constrained, and high-constrained (SFO 2016, 2-1). The approach to developing the 2018 and 2023 timeframe forecasts followed a fairly standard process, incorporating regression analysis of historical determinants of air travel demand to project future O/D passenger demand and combined with projections of connecting passenger activity along with market intelligence on airline network development and fleet mixes to arrive at the passenger and aircraft movements forecasts for those two timeframes.

Unlike a traditional master plan with a central or base-case scenario with "high" and "low" alternative scenarios, the ADP has taken a different approach to its long-term demand forecast. The ADP provides a medium-term forecast for the years 2018 and 2023 and a pair of long-term constrained forecasts for design and planning purposes. Both long-term demand levels are constrained scenarios that are founded on the assumption of the maximum number of annual operations given a fixed airfield capacity and layout, with passenger levels being forecast from assumptions regarding fleet mix, gauge of aircraft, and load factors. As expanding the number of runways or taxiways is not part of the ADP's plans, the long-term traffic scenarios take a constrained approach based on airfield design and estimated operations and passenger traffic from that constraint. The base-constrained and high-constrained demand levels were designed to simulate the years 2028 and 2033, respectively.

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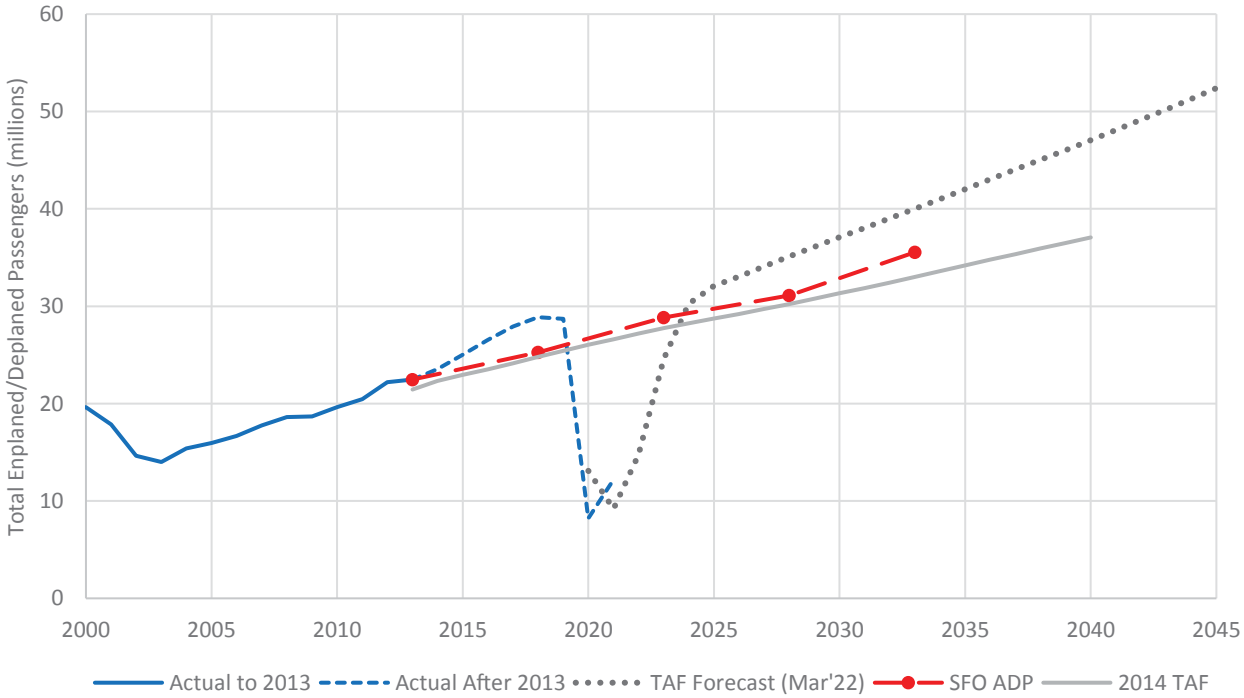
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This approach unties future forecast activity levels from specific timelines and instead links the forecast to "the maximum practical capacity of SFO's airfield." Given the approach to developing the forecasts is based on projected airside capacity, it is implicit that the designs should reflect that level of activity whenever it may happen as opposed to a more traditional forecasting/planning approach, which generally provides a timeline for the activity level. The ultimate high-constrained level of 35.5 million enplaned passengers per year could occur in 10, 20, or 50 years from the ADP's publishing, but the level of development to serve that implied level of traffic would be suitable, nonetheless. However, for planning, the base-constrained and high-constrained demand levels were simulated as approximate 2028 and 2033 time periods.

It should be noted that in the language of the ADP, the forecast projection points—2018, 2023, base-constrained (2028), and high-constrained (2033)—are denoted as **demand levels**. The base-constrained and high-constrained demand levels are not alternative scenarios but forecasts of constrained demand given assumptions at two future time points, simulated as approximately 2028 and 2033 respectively, in what is effectively a single continuous scenario. For this case study, we retain the ADP's usage of the term "demand level" when referring to the base-constrained and high-constrained forecast traffic levels.

The forecasts of passenger enplanements from the SFO ADP are depicted in Figure 41. Historical enplanement data, up to 2021 is from SFO with the ADP forecast and the FAA TAF for SFO from March 2022. As the ADP's forecast only provides data for specific years in 2018 and 2023, the base-constrained and high-constrained are plotted as traffic levels for 2028 and 2033, respectively, by design.

The forecast report considered a number of risks, described as "impact factors," which could affect activity levels at the airport over the planning horizon. These impact factors cover a number of airline and market share developments (e.g., change in the market share of LCCs), economic growth variation, and facility constraints. While it is not stated explicitly, it can be inferred that



NOTE: Historical data as reported by SFO with a minor discrepancy in the level of enplanements as reported in the TAF.

Figure 41. Forecast passenger traffic and demand levels at SFO (SFO 2016; FAA TAF).

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these impact factors could lead to the airport having traffic levels above or below the demand levels projected in the forecast.

Application of the Methodology

The shock events examined in the case studies are for illustrative purposes only and are wholly the work of the research team. SFO reviewed these potential shocks and provided feedback to the research team. The information presented here remains wholly the work of the research team and does not represent the opinions of airport management or staff.

For the case study and this guide's shock events methodology in general, it is important to keep in mind that the objective is not to devise new planning forecasts through the methodology. Instead, the objective is to examine potential shocks, unlikely as they may be, to explore how an airport's business, operations, and community may be impacted by an unprecedented turn of events. They may be worst-case or best-case alternative scenarios that should be evaluated and may not necessarily result in airport traffic, planning, or finances returning to preshock trends.

In the specific case of SFO, its ADP develops a planning forecast based on the concept of demand levels for long-term planning. This concept develops future levels of constrained traffic in the long-term (i.e., 10 to 20 years from forecast development) that could be served based on airport constraints and future demand projections.

For this case study, we examine the effects of potential shocks on airport traffic levels specifically over the medium-term. Due to the approach of the ADP's long-term traffic forecast of demand levels rather than a traditional unconstrained forecast of X traffic in Y future year, the methodology looks at what potential long-term (years 10+) impacts may be without developing specific quantified traffic levels.

Step A: Identify Shock Events of Interest

The ACRP research team developed a set of potential shock events based on their research of past shock events and information on forecast assumptions and risks (impact factors) described in SFO's ADP. The list of identified shock events is shown in Table 10 and is based on an outlook taking place in 2016 at the time of the ADP's completion.

Step B: Develop Scenario Forecasts That Incorporate Shock Events

From the shock events described in Step A, three were selected to be further analyzed through scenario forecasts. These events are projected to occur in 2014, 1 year after the last historical year of the ADP's forecast period.

- **Scenario Forecast 1: Loss of a significant carrier**
  - One of SFO's major carriers with both O/D and connecting operations at the airport is assumed to leave SFO. The carrier accounts for 8% of the airport's total capacity across a mix of both domestic and international services (but is primarily domestic). The loss of the carrier also impacts passenger traffic on connecting services by the carrier's alliance partner, particularly alliance partners on services to Asia and the Pacific.
  - The event is assumed to occur at the beginning of the second quarter (i.e., start of the Northern Summer scheduling season) resulting in a 6.5% loss in capacity in the first year and a 1.5% loss in the following year. These losses translate into a 5.5% and 4.8% reduction in traffic versus the ADP forecast for 2014 and 2015, respectively.

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Table 10. Identified shock events based on 2016 outlook for SFO.

Event	Description of Impact	Timeline and Aftermath
Loss of significant carrier	Aside from United Airlines, SFO had six carriers with significant operations with activity levels ranging from 3% to 9% of total annual airport capacity. The exit of one of these major carriers would have a substantial impact on available capacity, though it could be expected that some of that capacity would be backfilled by the remaining carriers. A secondary factor where an existing carrier leaves SFO, and instead bases at another airport in the Bay Area (e.g., OAK or San Jose Mineta International Airport), would likely have an additional negative impact as it would represent a shift in the market demand with less ability for existing carrier(s) to recapture demand and capacity.	Starting in the second quarter, one of the airport's major carriers accounting for 8% of total airport capacity, eliminates its capacity from the airport. The result is a loss of 6% in capacity in the first year, with a loss of 2% in capacity in the following year. It is assumed that this carrier operates both domestic and international services and impacts originating and connecting traffic, though the rate of connecting traffic is less than the overall average for the airport. The impact on passenger traffic is not likely to be 1-to-1, with existing carriers absorbing some of the lost capacity in the immediate time period. Over the medium-term (1–4 years) some portion of the exiting carrier's capacity is recovered. Over the long-term (5+ years) other carriers will likely backfill lost capacity, and fundamental demand would not likely be impacted versus the ADP's long-term demand level forecasts.
Downsizing of United hub	Specific to the operations of United Airlines, it is possible that the carrier could downsize its hub operations at SFO. In 2013, United accounted for more than 45% of total capacity at SFO and was one of the primary drivers of connecting traffic at the airport. A reduction in capacity would have an impact on both originating and connecting traffic, and a downsizing of the hub would likely be realized by reducing connecting activity at SFO.	In 2013, the last historical year for this analysis, United accounted for 45% of SFO's total scheduled capacity and an estimated 77% of connecting passengers. The most significant change in network structure is the major reduction in United's international network at SFO. Transpacific and transatlantic services are primarily impacted, but Canada and Mexico are less impacted. The airline's domestic network is also substantially reduced, with frequencies and destinations cut back and a resulting network focusing primarily on O/D markets rather than acting as a connecting hub. The immediate impacts are assumed to be as follows: The carrier's connecting activity is reduced by 60%, approximately a 45% loss in total connecting traffic at the airport. United's international capacity is reduced by 40% for an estimated loss of 15% in international passengers. United's domestic capacity is down by 50%, resulting in a 30% loss of domestic passengers. However, as the fundamental demand economics of the market are assumed to be unaffected, passenger traffic is likely to recover over the coming years as other carriers backfill the market both to meet O/D demand as well as utilize SFO for further connecting activities. Over the long run, it may result in constrained demand levels being reached later than in the baseline ADP assumptions, depending on the pace of capacity recovery.
Earthquake	The San Francisco Bay Area is an active seismic zone where resilience to earthquakes is a design and planning requirement. SFO is built primarily upon artificial landfill over an estuarine mudflat. Because of that underlying geology, the airport lands are at high risk for soil liquefaction and thus damage to built structures in the event of an earthquake. (U.S. Geological Survey and the California Geological Survey 2006) While the	A major earthquake hits the San Francisco Bay Area causing widespread damage. Traffic at the airport is halted for a short period (less than 1 week) while surrounding infrastructure, such as highways and public transit, is similarly negatively impacted, reducing access to the airport. A major earthquake would also have negative impacts on the local economy with disruption of business activities and impacts on households through damage or loss of life.

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	facilities at the airport have been designed for seismic resistance, there is a risk that a severe earthquake could lead to substantial damage of airport facilities such as terminal buildings, access roads, airside infrastructure, or utilities—any of which could impair normal operation of the airport for potentially weeks or months.	In a worst-case scenario, airport infrastructure such as terminal buildings or runways/taxiways are damaged and require repair to return them to operational status. Return to full operations in terms of capacity may take from weeks to even months, but the effects are likely to be transitory. The oldest terminal structures are most susceptible to earthquake damage as newer buildings (e.g., the International Terminal Building) are built to high seismic standards and designed to remain functional even in the event of an earthquake.
Climate change shocks	An extreme weather event, coupled with rising sea levels, leads to serious structural damage to the airfield through water overtopping and breaching the seawalls or inundation of the airside.	An extreme storm leads to a complete closure of all airport operations for a relatively short duration—a week, for example. However, the shock damage to the airside renders the portions of Runways 11/28 (both L and R) east of the interception with Runways 01/19 unusable. Taxiway L is also impacted and unsuitable for use. It is assumed that it will take 3 months for the airside infrastructure to be repaired to a mostly operational, but full return of the airside does not occur until the next year. The loss of operations of SFO's two longest runways and a major taxiway has an immediate impact on the operational capacity of the airport. The most significant impact is the inability to operate a number of long-haul flights due to the longest remaining operational runway being less than 9,000 ft. As a result, many long-haul operations are suspended with demand flowing over to other gateway airports in the United States. The impact on passenger traffic is assumed to be resolved within 1 year, though there is the possibility that some airlines may choose to move operations to another airport. The financial cost of the shock event is high.
Significant growth in Asia and Latin American markets	Economies, and thus passenger demand from Asian and Latin American markets, grow faster than expected, leading to a long-term increase in growth of O/D and connecting traffic at SFO.	Growth in the economies of Asian and Latin American markets is listed as one of the key impact factors in the ADP's forecast. Faster than expected growth in these regional economies would result in higher growth of international traffic—both O/D and connecting—at the airport. Faster growth would lead to traffic levels above the medium-term forecast and reach constrained forecast traffic levels sooner than otherwise anticipated. Such growth also carries a risk that this demand could slow or collapse in the future if a shock increase in economic growth is accompanied by a particularly harsh recession. This shock to demand would be experienced over a long period, but over the long run would result in higher traffic levels *in a traditional, unconstrained demand forecast*. Due to the constrained nature of the ADP's demand level projections, it would likely manifest as a reallocation of traffic toward these markets. Facility investments specific to international traffic may need to be accelerated. Additionally, services specifically targeting the demands and needs of travelers to and from these regions may need to be implemented as the demand profile of the airport's traffic shifts due to this shock.

(continued on next page)

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Table 10. (Continued).

Event	Description of Impact	Timeline and Aftermath
Dot-com bubble burst, Version 2	A major collapse in the Bay Area's tech economy results in reduction in employment and general economic activity in the airport's catchment area. Given the size of the tech economy and its importance to the overall economy of the San Francisco Bay Area, a shock to this industry would simulate a localized recession or downturn with potentially wide-ranging effects.	The San Francisco Bay Area and SFO's primary catchment area has the highest concentration of tech and IT companies in the United States. A major downturn in this industry would have outsized effects on the local economy and SFO's traffic. The 2000 dot-com bubble bursting was associated with a slowing of growth and then negative growth seen in the second half of 2000 and into 2001 (prior to the impacts of the 9/11 terrorist attack). For the first 8 months of 2001 (pre-9/11) monthly traffic at SFO was down an average of 8% versus the previous year. While the bubble bursting also coincided with the early 2000s recession in the United States (March through November 2001), SFO likely suffered reductions in travel demand as the tech industry shrunk with carry-over impacts to other sectors of the economy all impacting travel demand. A shock collapse or decline in the local tech industry would likely be realized not as an immediate drop in traffic but as a slower decline or reduction in growth that could last for many years or even be permanent.
Terrorist attack	A terrorist attack occurs in the United States leading to a temporary closure of national airspace and a reduction in travel demand following the attack.	Following the 9/11 terrorist attack in 2001, SFO saw annual traffic decline by 9% in 2002 along with a 30% decline in traffic in the last 4 months of 2001. The terrorism event had a significant impact at SFO and around the United States, but the impacts were also conflated with the lingering impacts of the dot-com crash on the local economy and fundamental underlying demand. A terrorist event that targets the U.S. aviation system would likely see similar—but less negative—effects on traffic at SFO, including a short, sharp downturn in traffic during the month following the event, along with a lagging short-term recovery in demand as travelers choose not to travel. Future terrorist attacks may also result in new security measures at airports or by airlines, which may also reduce demand in the short term (typically through substitution to other modes) but are not likely to see the same type of transformative impacts that post-9/11 security changes had on airport planning.
Pandemic	The airport is affected by a regional, national, or global pandemic, negatively impacting passenger demand and operational capacity.	In 2003, SFO suffered a downturn in traffic between February and July due to the SARS pandemic. The SARS outbreak primarily affected travel to and from Asia as the pandemic did not cause significant disruptions within the United States. Over the 7 months of the pandemic (February through July), international enplanements at SFO were down 16% versus 2002 and down 25% compared to 2000 levels (prerecession and 9/11), though traffic downturn drivers at SFO were conflated with the localized economic slowdown caused by the post-dot-com crash era. Domestic enplanements did see some declines but only at a fraction of the impact on international traffic. A similar SARS-type pandemic would be expected to see a decline in international traffic for a period of 3 to 9 months if the pandemic was foreign-based. As a major transpacific gateway airport for the United States, SFO is particularly exposed to pandemics or disruptions in Asia. A pandemic regionally or nationally within the United States, even of a SARS-type, would likely have a much more significant impact on traffic at SFO as it would negatively impact domestic demand (which has historically accounted for approximately 75% of SFO's passengers).

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  - By the second year, traffic levels have recovered to preshock levels through both organic growth of the market and through other carriers partially backfilling lost capacity and/or expanding into markets now left underserved or unserved.
  - Over the medium-term, it is projected that total passenger traffic would likely reconverge to preshock forecast levels as other carriers are able to backfill lost capacity. As underlying demand in the market was not impacted (e.g., through an economic shock) sufficient demand would remain in the market to incentivize other carriers to deploy capacity and fleet assets over time to meet demands.
- **Scenario Forecast 2: Downsizing of United hub**
  - United Airlines, SFO's largest carrier, engages in a strategy of downsizing its operations at SFO. The carrier made up 45% of the airport's total scheduled capacity in 2013 and approximately 78% of the airport's connecting traffic.
  - In the downsizing action, the carrier reduces its connecting activity at the airport by an assumed 60%. This results in an estimated 50% reduction in domestic connecting passengers and a 41% decrease in international connections. O/D passengers are projected to decline by a smaller amount (–24% for domestic and –8% for international, the latter having higher local O/D demand and therefore less exposed to shocks to connecting networks) as the shock forecast posits that the impact is based on reducing the hubbing activity with O/D less affected. However, domestic O/D is projected to be more negatively impacted as the airline is estimated to retain more of its international network for specific O/D demand, while the reduction in domestic capacity will negatively impact domestic O/D demand (which will likely spill to other domestic carriers).
  - The total impact in 2014 is a 25% reduction in total passenger traffic versus the ADP forecast. An additional year of declining traffic is assumed as connecting passenger volumes decline further, and other carriers do not have sufficient fleet resources in such a short timeframe to significantly backfill capacity.
  - By 2016, traffic growth is assumed to resume again both as a result of organic growth in the market (as forecast based on underlying economic demand) and as other carriers build up capacity to recover lost market share. However, in the short term, the airport will likely have a relatively lower share of connecting traffic than projected in the ADP forecast as carriers with backfilling capacity build expanded O/D and connecting networks.
  - Over the medium-term (i.e., years 4 to 10) passenger traffic is expected to see an accelerated recovery process as other carriers backfill demand through fleet acquisitions and network shifts to meet underlying demand to and from the Bay Area. It is possible that many network carriers may build up strengthened hub operations at SFO to take advantage of the airport's strategic location on the U.S. West Coast and for transpacific routes—potentially similar to LAX as an airport of many hubs. However, as fleet acquisitions and airline networks cannot instantly shift and based on the experiences seen at other airports that have seen major shocks to the hub activities of the largest carrier, it is projected that over this medium-term period, traffic would remain below ADP forecast levels as it recovers and rebuilds.
  - The base-constrained and high-constrained demand levels would likely be unchanged due to this shock as long-run constrained estimates of traffic potential at the airport. Reevaluation of these demand levels may be necessary to determine how, with a wider mix of airlines providing hub networks at SFO, fleet mix, and aircraft types may change the projections for passenger and aircraft traffic at these future demand levels.
- **Scenario Forecast 3: Climate Change Shocks—Ocean Inundation**
  - Scenario forecast 3 examines the risk of major flooding and structural damage to the airport's airside system in a major shock event due to sea level rise and increasing extreme weather activity arising from climate change. The specific damages are simulated as the closure of the

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eastern ends of Runways 10/33 (both runways) beyond the intersection of Runways 01/19. The event is assumed to happen at the beginning of August 2014.
- The entire airport is assumed to be closed for 1 week due to the damage. More extensive damage from inundation of the San Francisco Bay leads to severe structural damage to the Runways 10/33 infrastructure. A single runway is made available for use within 6 months, but full repair and return to normal operations do not occur until the following year.
- As a result of the damages, SFO is without its two longest runways, which impacts the ability of airlines to operate long-haul flights. Repair works limit the operational weight of aircraft taking off and landing such that a full return to preshock operations cannot occur until the following year. The longest operational runway at the airport during the shock is less than 9,000 ft, which limits airline operations of long-haul flights. Not until the airside repairs are fully complete will the restrictions on stage length be lifted.
- The effect on passenger volumes is assumed to be transitory and to be recovered within 2 years. While there will likely be an impact on airport finances, the airport's Lease and Use Agreement and residual rate-setting method are anticipated to reduce the financial risk to the airport itself by spreading risk to its airline users.

The scenario forecasts are depicted in Figure 42. Actual passenger traffic from 2003 to 2019 (excluding the impacts of the COVID-19 pandemic) is included as well as SFO's ADP forecast from 2013 to 2023. The scenario forecasts are all relative to the ADP's forecast, rather than actual passenger growth, which was above forecast levels. While SFO-3, showing the potential impact of major damage to the airside infrastructure, is projected to have only a relatively modest impact on traffic—compared to the other two scenario forecasts—it is possible that this scenario could bear a heavy financial burden to finance repair costs.

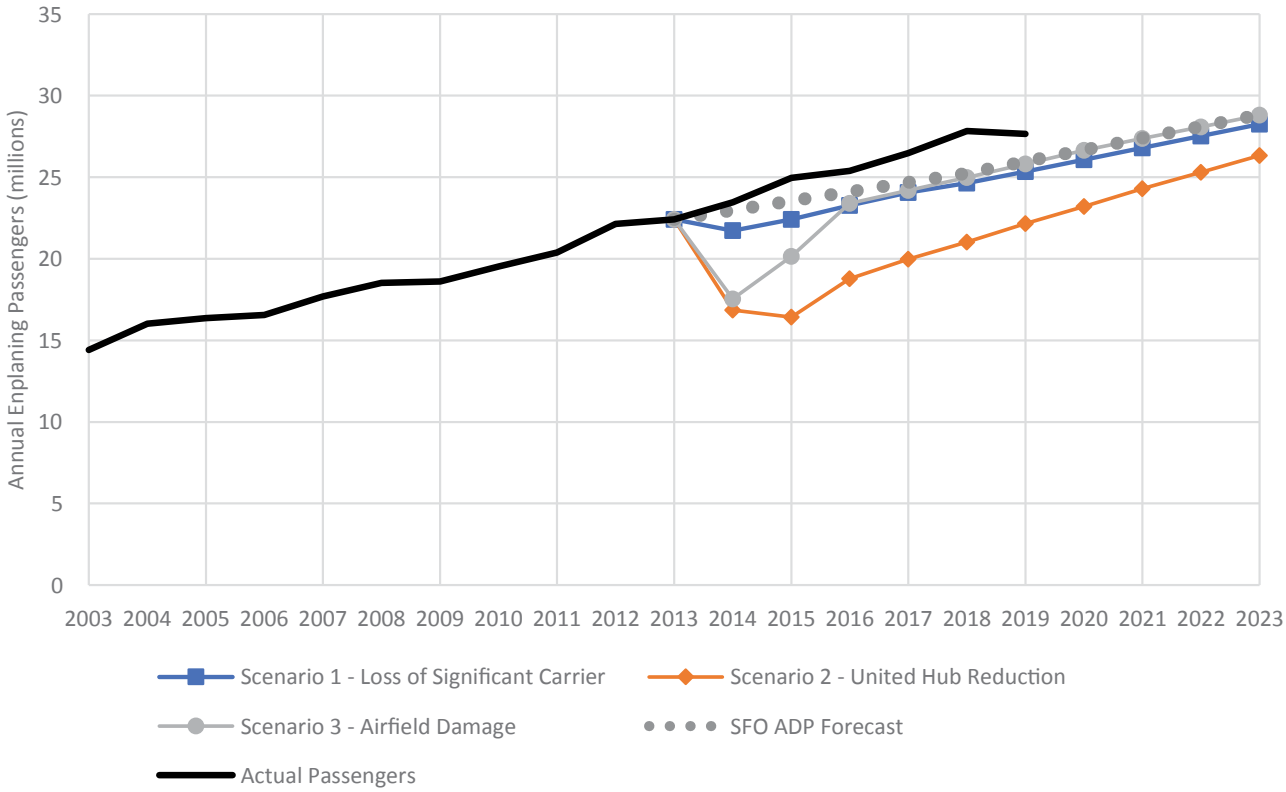


Figure 42. Comparison of scenario forecasts for SFO.

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Step C: Scenario Implementation

Following the development of the scenario forecasts, the next step in the methodology is to evaluate how the airport would cope with and respond to these scenarios. The intention is to test the airport's current plans and see how the scenarios could impact a range of aspects in airport operations including

- Planning;
- Finance;
- Marketing and communications;
- Operations, security, and safety; and
- Community and stakeholders.

The results of the implementation are summarized in Table 11.

Due to the nature of SFO ADP, in that it postulates two future capacity-constrained activity levels and designs its capital plans from that, the ADP is relatively resilient to faster or slower growth projections. By identifying priority developments, largely on a basis of need in the short term and then on a demand-basis in the long run, the ADP can accommodate changes in traffic growth patterns. The provisions of the airport's Lease and Use Agreement, specifically the residual rate-setting method with carriers and annual reallocation of gates and terminal facilities that are designed to be common use and nonexclusive, provide the airport with a notable level of risk mitigation and resiliency to cushion financial risks and operational shifts arising from shock events.

Step D: Develop Options for Improved Robustness

With key issues identified in Step C arising from the scenario forecast, the next step in the methodology is to determine what actions could be taken to address these issues and improve the robustness of airport plans. Though this is present as a separate element to Step C, the process of developing these concepts would likely be completed at the same time the impacts are discussed in a holistic process of identifying issues and mitigation strategies that can be developed simultaneously. Below are a series of common actions and themes that relate to one or more of the Step C shock events, discussing both preventive and reactive actions the airport operator may take to improve robustness of the airport.

Modularity in Design and Planning

With respect to the downside scenarios impacting traffic over a longer period of time (loss of a major carrier and United downsizes the SFO hub) incorporating modularity into the design and planning of infrastructure development would help improve the robustness of the airport if traffic demand and capacity fall sharply. Should passenger traffic take a downturn that persistently changes the short-, medium-, or long-term outlook for demand at the airport, incorporating modularity into plans and design can improve the ability of the airport to not overbuild (and overspend) facilities for a demand future that may not be realized or realized much later than expected. Rescaling projects may also be necessary, depending on the shock forecast's growth potential over the longer-term. In the case of scenario SFO-2, United downsizing its hub, there would likely be a substantive change in the peaking volumes and banking characteristics of demand at the airport. This would require reevaluating processor demands and gate/stand usage, which would then drive decisions on when subsequent phases of processor/terminal development would continue (modularity) but also if those plans need to be revised and rescaled to meet future forecast demand. The airport currently, and throughout its ADP, emphasizes

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Table 11. Impacts of the scenario forecasts (SF), SFO.

Area	SF1-SFO: Loss of Significant Carrier	SF2-SFO: United Downsizes Hub	SF3-SFO: Airfield Damage
Planning	As the impact is forecast to have a mostly transitory effect on forecast passenger traffic and only a relatively minor, long-term impact, it is likely that the planning objectives for both the Base- and High-Constrained demand levels would not need to be substantially adjusted to accommodate the change in demand. Reaching these demand levels may take longer than anticipated otherwise. As a previously major carrier, its departure would leave terminal facilities available for other airlines. Planned capital improvements to the terminal could be reprioritized to take advantage of underutilized spaces to minimize passenger experience disruption.	The major reduction in connecting activity and overall passenger traffic at the airport would likely impact the plans for terminal development, namely facilities designed for transfer passengers as well as the terminals and boarding areas [Terminal 3 (T3) E and F and International Terminal Building G] currently used by United Airlines. However, the airport's Lease and Use Agreement with airlines includes an annual process to reallocate gate and terminal usage, which are assigned on a joint, common, or preferential use agreement. This process would allow incumbents or new entrants to swiftly be reallocated to vacated gates and other terminal facilities and would mitigate many of the challenges that could be faced at an airport with exclusive-use-only gate/terminal use agreements. As T3 facilities are among the oldest at the airport, ADP plans to upgrade T3 and boarding areas E and F would need to be reevaluated to see when those capacity constraints may emerge again given the reduction in operations. There may be opportunities now to accelerate infrastructure upgrades without having such negative impacts on passenger experience as demand for terminal capacity is now lower than forecast. As one of the airport's major cargo carriers, cargo activity would also be reduced, which may impact planning of those facilities.	The ADP identified deficiencies in the airport Shoreline Protection System in all of the areas surrounding the eastern ends of Runways 10-28. The ADP has indicated that the deficiencies were identified through a separate evaluation study that made recommendations for addressing these issues. (Johnson et al. 2015) The study indicates that to protect against the impacts of sea level rise and potential storm surges, the airport's coastal protection system will need improvements with the east end of Runways 10/28 being specific areas of focus given their protrusion into the bay. We note that there do not appear to be major utility corridors or airport facilities within the estimated zone of impact for the shock event, which would mitigate some of the potential damage to only the airside infrastructure and stability of the landfill.
Finance	The airport would realize lower-than-forecast revenues in the short term due to the dip in traffic. The long-term traffic impacts are projected to be relatively minor.	A major reduction in operations by the airport's largest carrier would have direct impacts on airport finances through reduced revenues. However, as impact is projected to be largely on connecting passengers, revenue sources for originating passengers would be relatively less impacted, though there would certainly be a substantial impact on revenues regardless.	While the traffic loss due to operational restrictions is assumed to be resolved within 1 year, the cost of remediating the airside and landfill could be significant. Insurance liability and coverage would need to be examined, and additional debt may need to be issued to cover the costs to repair. However, existing plans for remediation of taxiway and runway surfaces in the affected area and Runways 10/28 could be accelerated and incorporated into the response work, removing them from future planned capital works.

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		The airport's residual rate-setting method with air carriers would reduce the financial risk on the airport itself. Fees and charges may need to be revised to meet financial obligations, but the residual method employed at SFO would share the financial risk among airlines and the airport providing mitigation against these types of shock events.	
Marketing and commercial activities	Increased air service development to attract airlines to fill lost service levels. Business-as-usual approach to reallocation of terminal facilities and gates through annual reallocation process under Lease and Use Agreement.	Air service development activity would need to be increased to find airline(s) to backfill lost capacity and connectivity. Business-as-usual approach to reallocation of terminal facilities and gates through annual reallocation process under Lease and Use Agreement.	The impact on long-haul operations would likely require negotiations with airlines—both domestic and foreign—and would result in a temporary reduction in overseas traffic impacting various formats of concessions and commercial activities at the airport.
Operations, safety, and security	Traffic declines would lead to a reduction in operations but are not expected to impact safety and security.	Significant reduction in capacity and traffic, particularly at specific terminals and airport processors in the first year, would reduce operational demands. As facility reallocation process evolves over the recovery period, potential operational challenges will need to be addressed as carriers may be shuffled across or within the terminals.	Airport operations, particularly airside operations, would be majorly impacted by this shock event. The airport would likely look to past plans regarding how airside operations were reconfigured during runway maintenance to quickly operationalize plans to keep the airport operating at as high of a capacity as possible.
Community and stakeholders	Engage with local community and stakeholder groups to identify underserved or unserved markets as part of air service development strategy.	While connecting activity is projected to be reduced, overall global connectivity is assumed to be relatively unaffected. Work with tourism; meetings, incentives, conferences, and exhibitions (MICE); and business stakeholders to identify key routes that may now be underserved or unserved to ensure connectivity to key markets. Noise emissions are likely to be reduced due to corresponding reduction in traffic, which would be seen as beneficial by local community. However, if/when traffic builds again in future, noise issues could become intensified as the community becomes accustomed to a lower volume of aircraft operations.	Airport management may need to reassure stakeholders and the traveling public that the airport is safe and that remediation work will allow for greater resilience against future flooding or water damage.

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modularity and flexibility, which are in many ways enabled through the Lease and Use Agreements provisions for annual gate and terminal facility reallocation based on the need for both incumbent and entrant airlines. Maintaining this flexibility through future Lease and Use Agreement reauthorizations would be critical to building resiliency and risk mitigation from shocks to capacity or carrier makeups.

Trigger Points in Planning

Related to modularity in design is identifying trigger points for planning activities. In the case of unexpected downturns in traffic or growth spurts, developing trigger points on capital development activities provides the opportunity to make investment decisions at the right time. SFO's ADP Implementation Plan specifically notes that the pace and sequence of planned developments are based on the forecast of traffic growth, among other factors such as finance, construction feasibility, and required sequencing of projects. The ADP notes that "sequencing construction projects solely on demand could result in an excessive number of simultaneous construction projects" beyond what is physically capable at the airport and that development projects are to be viewed as comprehensive improvements in the airport system. While the ADP does not set out specific triggers—be they passenger volumes or other operational metrics—many high-visibility projects such as the Boarding Area H development would be driven by the pace of growth in flights. The ADP does provide a "Lead Time for Decision" metric to indicate how far in advance a go/no-go decision should be made, a practice that provides a concrete expectation on infrastructure planning triggers to manage future development in the event of a major shock to demand or capacity.

Investment in Risk-Mitigation Activities

Taking preemptive actions through planning to mitigate the impacts of climate change could help improve the ability of the airport to face and manage shock events arising from climate change risks, as described in scenario SFO-3. In a similar vein to incorporating earthquake resilience (beyond local building codes) into the design of airport facilities to mitigate negative impacts should a major earthquake occur, taking steps to reduce the airport's emissions and carbon footprint to lessen the global impact of climate change and its negative consequences may pay dividends in the future. Just as regular maintenance and improvements to the airport's seawall and dikes manage the risk of flooding or water damage to mitigate the specific risk described in scenario SFO-3, other activities—like reducing carbon emissions and supporting climate change policies—work to reduce the potential risk of climate change-related shocks, like extreme weather or environment events. Since 2011 the airport has published an annual Climate Action Plan to measure emissions and to identify action strategies to reduce airport-controlled emissions and support emission reduction schemes in general, such as supporting sustainable aviation fuel (SAF) development or the transition to zero-emission vehicles (SFO 2021).

Shock Event Response Plans

During the shock event, effective and engaged communication with stakeholders will likely serve to improve the airport's ability to manage the shock event and build resiliency in recovering traffic (or managing a burst in demand). Stakeholders may need to be reassured of facility safety and operational viability (e.g., in the case of severe airside damage in scenario SFO-3), while marketing and air service development activities with stakeholder airlines, tourism partners, and convention organizers should be reviewed and new strategies formulated should there be major reductions in capacity or the airline mix at the airport (in the case of SFO-1 and SFO-2) to respond to the new market structure.

Shock event response plans must also consider the financial impacts of such shocks and plan for contingencies should the event happen. Major reductions in activity would be met with

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decreases in nonaeronautical revenues through reduced passenger volumes. Aeronautical revenues would be unaffected through the airport's residual rate-setting program, though airline stakeholders would face higher rates for a period of time—assuming costs were fixed. Risk planning and mitigation for the airport's financial needs is an important part of planning for potential shock events and should consider impacts on long-term obligations like debt servicing and future capital planning priorities as well as immediate concerns over covering operational costs during a period of revenue declines.

Step E: Ongoing Monitoring and Vigilance

The final stage of the methodology is the ongoing monitoring and vigilance of potential shock events and the updating of planning and processes, where necessary. This stage aims to make planning for future shocks, even though they may be low-probability events, an aspect of ongoing vigilance around airport development and planning so the airport can be well-positioned to manage shock events should they occur.

As a general part of risk management and to better incorporate shock events robustness into strategic planning for the airport, production of regular documents that monitor and review risks should be considered. These can take the form of memos or briefing notes and should include tracking existing risks, identifying any shock risks that may be increasing or decreasing in likelihood, and new risks that have been identified. These may be authored by a working group made up of staff from various airport departments and should be presented in a way that is readable and relevant across the airport organization. If the airport already has a regular, ongoing process to monitor and review risks—such as a continually maintained risk register—strategic shock risks should be considered for inclusion in that process.

Where applicable, warning systems may be warranted for specific risks. In this case study, the shock risk of a climate-change-related disaster was examined. Warning systems relating to factors such as sea level rise, an increasing rate of extreme weather events, or global mean temperatures can be macro-level metrics that feed into a shock warning system. These can be complemented by monitoring practices that may already be in place, such as physical sensors or regular observations of airport infrastructure (e.g., the Shoreline Protection System). Not all shock events may have such quantifiable leading indicators or warning signs, but research into the potential risk events and professional knowledge of experts within the airport organization may elucidate metrics or signs that could better warn of shock events and inform how likely a shock event may be to occur.

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CHAPTER 18

Illustrative Application of Methodology at a Small Airport

This chapter provides an example of how the methodology can be employed on a small scale with an illustration of a GA airport. While it follows the structure of the methodology, some of the steps have been simplified or combined.

Setting and Situation

This illustrative example imagines a GA airport that has no commercial service. It is an uncontrolled airport with a single runway and a small apron. The airport's users are GA aircraft including private aircraft and flight training at a local flying club but also include commercial activity such as medivac, aerial surveying, and occasional forest-fire suppression (not based at this airport). The airport's revenue sources are primarily from the sale of avgas fuel (70%) and land leases (30%). The airport does not have its own forecast, but an FAA TAF is available.

Application of the Methodology

Step A: Identification of Shock Event

For this illustrative case, only a single shock event is examined, as shown in Table 12. On a small scale, it may be sufficient to examine only a single shock event that would disrupt the airport's existing projection of traffic—either up or down—to explore the ramifications throughout the airport's business.

Steps B and C: Scenario Forecast and Scenario Implementations

Based on these high-level descriptions, the airport can now consider how these impacts would directly affect the airport's operations, finances, and planning. Quantification of impacts should be estimated whenever possible, even if it is speculative to demonstrate potential magnitudes to assess mitigation strategies.

- **Expected impact on aircraft operations.** Assume that piston-engine aircraft operations at the airport will decline by 50% once the ban is in place (47.5% decline in total operations). This may be due to some aircraft owners or operators deciding not to retrofit engines for alternative fuel sources or that conversion will take more than 2 years to complete. Turbine-powered aircraft operations are not affected, but those aircraft types make up a small proportion of total annual operations.
- **Expected recovery of aircraft operations.** Expect that within 5 years after the ban, only half of the initially lost operations will be recovered and that the airport permanently has a lower level of forecast operations than expected before the shock.

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Table 12. Illustrative case on single shock event.

Shock Event	Description of Impact	Timeline and Aftermath
Ban of 100LL avgas	A government mandate bans the sale of 100LL avgas in the near future. Ninety-five percent of all aircraft operations at the airport are piston-engine aircraft, which are currently certified for 100LL; the remaining 5% use jet fuel. As the airport only has avgas sales, this mandate will directly impact airport revenues and operations.	The 100LL ban mandate has a 2-year implementation time before going into force with alternative avgas fuels becoming more common. Over those 2 years, some local aircraft are able to be retrofitted to use non-100LL fuel types, but many users are unable to do so due to cost constraints.

- **Expected impact on airport revenues.** Before the shock, 70% of airport revenues were from fuel sales, which were only 100LL. Unless the airport makes changes to its fuel offerings, it could lose the entirety of those fuel sale revenues resulting in a 70% drop in operating revenues if no other actions are taken. If fueling is no longer considered to be a business of the airport, there may be costs associated with decommissioning the existing fueling system.

The primary outcome of this simplified exercise should be to begin to establish how an airport's activity levels and finances could be impacted by a shock. How would the airport deal with a drastic downturn in revenues (as in this illustrative scenario) or a major increase in operations in some other shock scenario? Examination of historical revenue and operations trends may allow operators to draw insights from past events or past shocks to help quantify potential impacts of the shock event under study.

Steps D and E: Options for Robustness and Monitoring

To mitigate the potential negative impacts of this shock event, the following is a series of robustness strategies that could be employed.

- **Study the potential for sale of different types of avgas other than 100LL.** Is the current fueling system capable of storing and dispensing different varieties of avgas? Could additional pumps and tanks be installed to allow for the sale of multiple types of avgas? Are there suppliers in the region for alternative aviation fuels? Could the fueling system be contracted out to a third-party operator under concession to reduce risks?
- **Diversify airport revenue sources.** The current airport derives the majority of its revenues from fuel sales, and thus, any impact to fuel sales, positive or negative, would dramatically impact revenues. Diversification of revenues, such as increasing land leases for aeronautical or nonaeronautical sources, could reduce the airport's risk exposure to changes in fuel revenues.
- **Plan alternatives for alternative fuel supply.** Could the airport consider developing plans for the provision of alternative fuels like SAF or recharging battery-electric aircraft as new revenue sources to capture emerging aircraft technology developments?
- **Engage industry and stakeholders.** Devise or outline plans for how the airport would work with industry groups and stakeholders should the shock take place. Identify who key stakeholders are (e.g., local airport tenants, Aircraft Owners and Pilots Association, and government agencies) and consider how the airport would manage the shock should it occur. Stakeholder planning could also include actions to mitigate the potential shock event, such as lobbying or activism within industry or other associations.
- **Monitor current regulatory process.** Monitoring of potential changes to regulation through industry groups is important to assess how likely such a change may be or if other related shocks could emerge that may have similar or related effects.

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PART V

Conclusions and Recommendations for Further Research

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CHAPTER 19

Conclusions

The research in this study found that shock events are often overlooked and their possibility is often ignored when considering future airport development. This is certainly not something unique to airports and relates more to the human condition. Often, we lack the information or imagination to consider such events and are biased toward what is knowable and comfortable. This can apply to nearly all fields of human endeavor—politics, economics, science, technology, and society in general. While insurance (and other forms of risk-transfer or risk-sharing) can be used to mitigate some shock events, this is generally based on known and understood risks and may not cover unknown unknowns or black swan shock events. An airport cannot obtain practical insurance that will cover the loss of a major carrier or declining traffic due to the loss of local industry.

Nevertheless, our research has found methods and strategies that can be deployed to address shock events in airport forecasting and planning. In general, it is not practical or realistic to actually predict or foresee shock events. However, just the exercise of identifying and exploring potential shock events can provide substantial benefits. Here, the role of forecasting is not about what *will* happen but rather about what *could* happen when shock events occur. From this, decision-makers can consider options to prepare for and respond to shock events, even when the exact nature of the shock event is not known.

Drawing on the research undertaken during the project, this guide sets out a systematic methodology to aid airports in understanding and addressing shock events. The methodology aims to contribute to a more resilient airport business model by assisting the airport to be better able to have a robust response to shock events and to aid in the development of a ready-to-deploy portfolio of strategies that the airport can make use of should a shock event occur (or is expected to occur). While not all shock events can be identified or foreseen, the intent is that an airport would be better positioned for these eventualities. The approach emphasizes the need to consider the less familiar and less comfortable possibilities when identifying shock events and the importance of organization-wide engagement and ongoing vigilance. Fundamentally, the methodology in this guide poses the question, *What would happen if a shock event occurred, and what can be done to mitigate the impact from these events?* Furthermore, it is not a one-size-fits-all approach for all airports and all projects.

While this guide discusses some of the more advanced methods that can be deployed to address this question, consideration is also given to simplified approaches that could be used by small airports or airports with limited time or resources. Clearly, the more resources that can be dedicated to this methodology, the more comprehensive and detailed the findings will be, but even a simple approach will provide value and is better than ignoring the issue entirely. A key lesson from the research is the need to explicitly address the potential for shock events and develop preparedness, which requires engagement throughout the organization.

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APPENDIX A

Glossary of Key Terms and Abbreviations

Aeronautical Revenue	Revenues generated by airports derived from flight operations (e.g., landing fees, terminal fees, ground handling).
Airport Master Plan	A document that describes long-term development concepts for an airport and provides the strategy for future airport development capable of serving forecast air traffic demand.
Bellyhold	Vacant space in the bellyhold of passenger aircraft used for the shipment of freight.
Black Swan	High-impact events that are deemed impossible to predict prior to them occurring. Also called an "unknown unknown."
Capital Cost	Costs and fixed expenses incurred on the purchase of land, buildings, and construction.
CARES Act	Coronavirus Aid, Relief, and Economic Security Act (2020). Specific to the aviation industry, the CARES Act provided financial support through grants to U.S. airports and airlines to help offset the negative impacts of the COVID-19 pandemic. Aviation sector grants administered by the FAA.
Connecting Passengers	Passengers who are only stopping at an airport to conduct a transfer (changing planes) or a same-plane transit as part of their itinerary's routing. Connecting passengers are counted in airport statistics as enplaning passengers, but not in Origin/Destination passenger volumes.
CRRSA	The Coronavirus Response and Relief Supplemental Appropriations Act (2020). A second set of financial support to the economy including the aviation industry. Airports were supported primarily in the payment of ongoing expenses and debt servicing. Aviation sector grants administered by the FAA.
Enplaned/Deplaned (E/D)	A measure of passenger traffic of all passengers enplaning (boarding) or deplaning (getting off) aircraft at an airport. Includes passengers where the airport is the start of the journey or final destination and passengers who are connecting (changing planes) at an airport. E/D passenger traffic by sector is based on the passenger's immediate inbound or outbound destination, not the ultimate origin or destination.

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Enplanements	The total number of passengers boarding aircraft at an airport over a specified period of time.
ERM	Enterprise Risk Management, a framework for risk management within organizations. May take many forms to identify, assess, and prepare for risks but ERM is a generalized term for risk management processes.
ERP	Emergency Response Plan, a safety and security response plan for airports. Required by regulation in the United States.
***Ex Post* Evaluation**	Evaluation of an event after it has happened. Typically concerned with analyzing what actually happened in contrast to what was planned or expected.
Full-Service Carrier	An airline that provides multiple classes of service (e.g., business class, premium economy, and economy) with fares that typically include standard options, such as complimentary checked bags or in-flight meals. In contrast to a low-cost carrier (LCC). Most network carriers operate under a full-service model but offer the option to unbundle fares to compete with LCCs.
General Aviation (GA)	GA is a civil aircraft operation such as business aviation, private aircraft operation, flight training, air ambulance, and aerial surveying.
Grey Rhino Event	Event with a high impact and high probability that should have been easily foreseen but was ignored.
GDP	Gross Domestic Product is a measure of the total output of an economy.
High-Impact, Low-Probability Event	See Shock Event.
Hub Airport/Hubbing	A hub airport is one that an airline (or many airlines) uses as a transfer point for passengers within their network. Part of an airline's hub-and-spoke model, hubs facilitate the transfer of passengers along their itineraries when a passenger's desired route is not served non-stop.
Known Unknown	An uncertainty that is deemed to have some possibility of occurring but may lack information or historical precedent.
Legacy Carrier	One of the three largest network carriers operating in the United States: American Airlines, Delta Air Lines, and United Airlines. Also refers to flag carriers and major international network carriers from other countries such as Air Canada, British Airways, Lufthansa, or Emirates.
Low-Cost Carrier (LCC)	A type of airline that operates a low-cost model, typically including low base fares, a single class of travel, unbundled fares, high aircraft utilization, and quick turnaround times. LCCs, and especially ultra-low-cost carriers (ULCCs), tend to operate on a point-to-point network model rather than a hub-and-spoke model to serve direct routes not always served by legacy network carriers. LCCs and ULCCs operating in the United States include Southwest Airlines, JetBlue, Frontier Airlines, Spirit Airlines, and Allegiant Air.

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Monte Carlo Simulation	A computerized analytical process creating thousands of simulations of a model (e.g., a traffic forecast or financial model) using randomized values of model inputs and assumptions defined by probability distributions. Often used to quantify uncertainty of a model or event.
Network Carrier	A network carrier is typically a full-service carrier that operates on a hub-and-spoke network model. The airline may include a regional feeder to serve smaller markets from their hubs. Network carriers in the United States include Alaska Airlines, American Airlines, Delta Air Lines, and United Airlines.
Nonaeronautical Revenue	Revenues generated by airports derived from sources other than flight operations (e.g., car parking, terminal concessions, land leases).
Operating Cost	Costs associated with ongoing operations of a business such as salaries, taxes, utilities, marketing, and debt servicing.
Origin/Destination (O/D)	The measure of traffic between two cities or countries where the origin is the starting point of an air journey, and the destination is the passenger's final trip destination regardless of the routing. Does not include connecting passengers. The O/D journey and sector define a passenger demand market. For example, a passenger traveling from the United States to the UK is an international O/D passenger, but that same passenger may need to make a connection at a domestic airport as part of their routing.
Real Options	A real option is the right but not the obligation to take a certain course of action. The concept of real options is based on and developed from financial options.
Risk Register	Documentation of risks and uncertainties of a business, forecast, or model that includes a quantification of estimated risk impacts and likelihood.
Scenario Analysis	A process of analyzing the impacts of future events by considering alternative possible outcomes.
Sensitivity Analysis	Varying forecasting inputs and assumptions one at a time to measure the resulting changes in outcomes.
Shock Event	Events whose likelihood, magnitude, and timing are hard to predict or anticipate and have significant impacts on business should they occur. May also be referred to as high-impact, low-probability events; black swans; or unknown unknowns.
Stakeholder	An individual or organization that has an interest in an airport. Key stakeholders for airports may include airport management, airlines, local businesses, destination management organizations and the tourism industry, and local communities.
Ultra-Low-Cost Carrier	See *Low-Cost Carrier.*
What-If Analysis	A form of scenario analysis where a single event or assumption is forecast relative to its baseline value.

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APPENDIX B

Summary of Relevant Literature Reviewed

Appendix B is available on the National Academies Press website (nap.nationalacademies.org) and can be found by searching for *ACRP Research Report 272: Incorporating Shock Events into Aviation Demand Forecasting and Airport Planning*.

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APPENDIX C

Incorporating Scenario Forecasting into Monte Carlo Simulations

An extension to the Monte Carlo simulation method is to conduct a sensitivity or what-if analysis using the simulation model. In a prior project initially developed to support a North American airport's enterprise risk management (ERM) process, members of the ACRP research team devised a system to conduct a sensitivity analysis of specific risk factor(s) in the model. The objective of the analysis was to provide insights to the airport into how specific external risks could impact airport passenger volumes and financial position. A Monte Carlo simulation forecast had been developed for this airport, including the incorporation of the airport's own risk register of strategic risks into the forecast risk register. The scenario analysis was developed to create a series of what-if scenarios such that the airport could better understand not only the expected impacts of a risk but also how the entire forecast probability range could change should a specific risk occur.

The process devised for conducting the simulation-based scenario analysis was as follows:

- Use an existing Monte Carlo forecast model as the basis for the scenario analysis tasks.
- Identify the forecast risk factor(s) to be evaluated in the scenario.
- Set the probability of occurrence for the risk factor of interest to 100% in a given year for each scenario analysis. This will then trigger the risk factor in *every scenario* at a specific year to allow for a fixed comparison to the existing forecast.
- Maintain the originally forecast statistical distribution of modeled impacts for the risk factor to be analyzed. This allows the scenario analysis to retain uncertainty on how impactful a potential future risk is, even if the scenario analysis assumes that it must occur at a given time in the future.
- Run the simulation model allowing all other risks and uncertainties to be simulated as in the original forecast model. This *ceteris paribus* approach assumes that all other uncertain and assumed factors influencing the forecast continue to operate, with the only change to the potential future world stated in the scenario analysis being the certainty of the specific risk factor in question.
- Compare the new forecast probability range (the statistical distribution derived from the aggregation of the thousands of individual Monte Carlo simulations) and most likely forecast of the scenario analysis to the preexisting forecast as well as the recovery times and paths.
- Evaluate the impact of the specific risk over the short and long term, including the change in the probability range of outcomes. Use this information to devise mitigation and avoidance strategies.
- Repeat for additional risks to be analyzed.

An illustrative example of the process and its results are provided in Figure C-1. This illustrates how scenario analysis with simulation can be used to investigate the impact of a specific risk factor on a forecast's outcome over time, which can be contrasted to the baseline forecast

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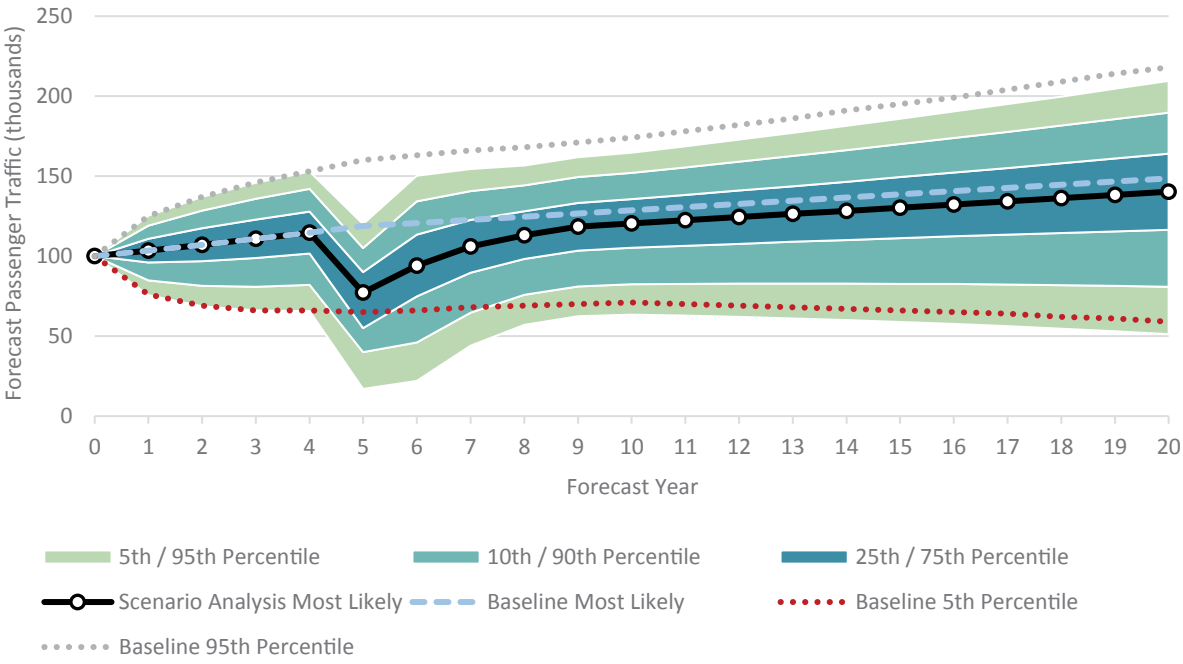


Figure C-1. Scenario analysis of individual risk factor with baseline comparison.

(shown in Figure C-1). The illustrative example assumes that in forecast Year 5, one of the risk factors will occur in 100% of all simulated scenarios but allows the magnitude of the impact to vary randomly. For this example, it is assumed that the factor has a relatively low probability of risk but with a significant potential negative impact. The expected impact is a 35% reduction in traffic in the year the risk event occurs, with traffic taking 4 years to recover and a lasting negative impact on future traffic volumes (e.g., loss of a major carrier or significant shock event with long-term effects on demand/supply). From this simulation example of a high-impact, low-probability event, conclusions can be drawn, such as there is approximately a 25% probability that traffic will not recover to preshock levels by the end of the forecast or that there is a 10% to 25% probability that traffic will return to baseline, most likely levels within 1 year of the shock event.

This approach realized several benefits for the airport and the forecasting team. First, the airport gained a greater insight into the projected quantitative impacts on passenger traffic and aircraft movements of various key strategic risks in a way that the standard stochastic risk-based forecast could not easily provide. The airport ERM team and forecast users were able to see how a given risk, even with a still-uncertain and simulated range of impacts on traffic, could impact traffic both in the short and long term, holding all other forecast factors the same as in the previous forecasting efforts. Second, by evaluating how the forecast probability range shifts in response to *a specific risk factor,* the ERM team could better understand how a given risk would impact overall and sector-specific traffic and then devise mitigation and adaptation strategies for that risk. And as the forecast is simulation-based, those mitigation and adaptation strategies can take into account the range of potential impacts and their probabilities of a given level of traffic occurring rather than some hypothetical expected value or low/high scenario with no information on probability.

Members of the ACRP research team have conducted similar simulation-based scenario analyses for other airports as part of a forecasting process for external strategic risks, uncertainties about regulatory changes, and impacts arising from major changes to an airport's airline

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market. Experience has found that this process is a valuable extension to a risk-based forecasting method to allow airport operators greater understanding when planning for potential future risk events.

A potential extension to this process would be to modify the analyzed risk factor(s) impact range, for example, to add a much higher and prolonged impact, though at a low probability in the statistical distribution of impacts, than would normally be considered in the standard forecasting exercise. Extending the scenario analysis of, for example, a pandemic event or natural disaster, to evaluate high-impact, low-probability events would be a natural evolution of this process specific to addressing the impacts of shock events.

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
GHSA	Governors Highway Safety Association
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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